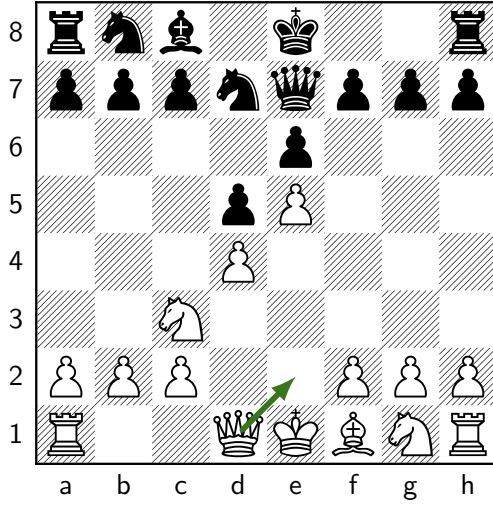


Melanie-MASTERS (human masters) — French, Classical

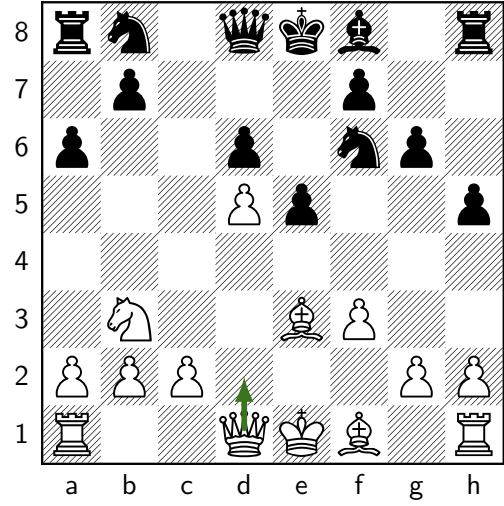
Melanie (Deep_Vision)

Abstract

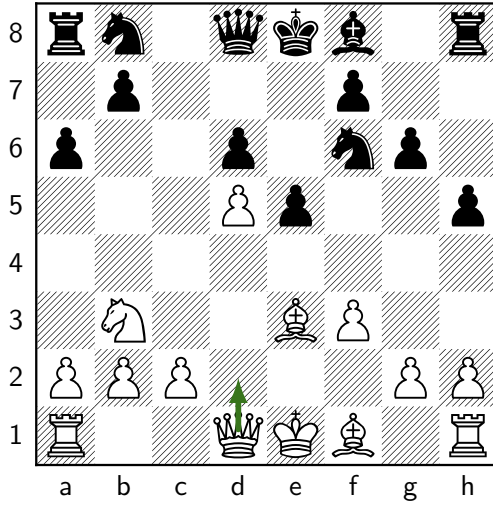
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



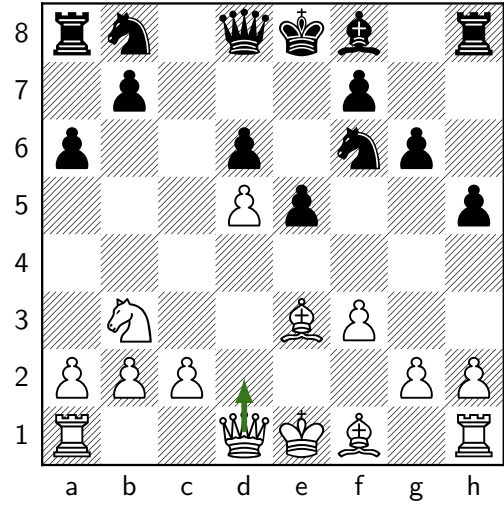
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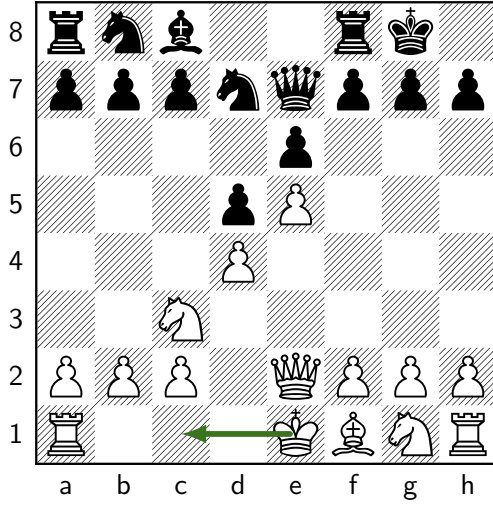


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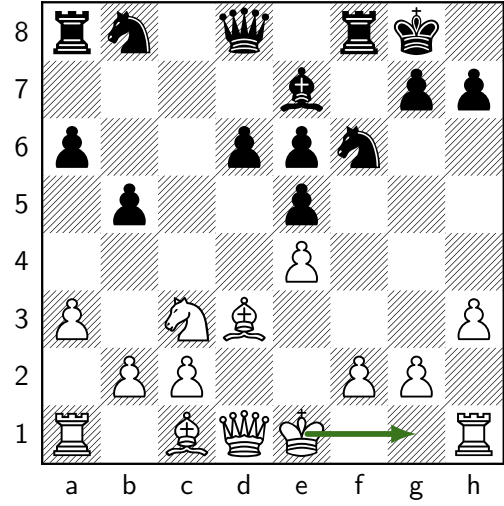


(d)

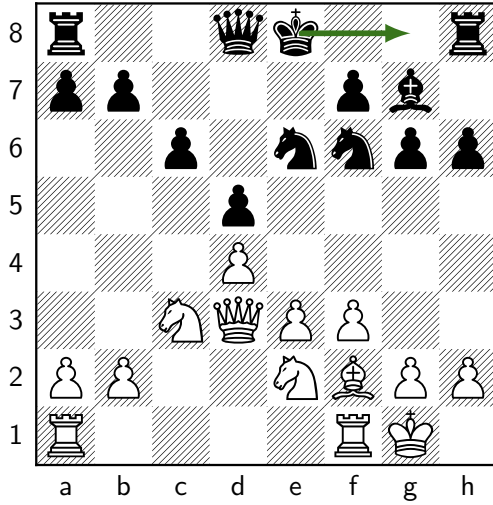
Figure 1: **Move 7, White: Qe2.** (a) Our position; Melanie plays Qe2 (d1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#3821` (ply 20) [structural distance $d=87$, QAP obj=32]: the master played Qd2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Nc3 as their Bf1, our Ph2 as their Rh1; transferred move d1e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0018#1001` (ply 20) [structural distance $d=87$, QAP obj=32]: the master played Qd2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Nc3 as their Bf1, our Ph2 as their Rh1; transferred move d1e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0025#405` (ply 20) [structural distance $d=87$, QAP obj=32]: the master played Qd2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Nc3 as their Bf1, our Ph2 as their Rh1; transferred move d1e2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qe2 (summed QAP objective 127, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



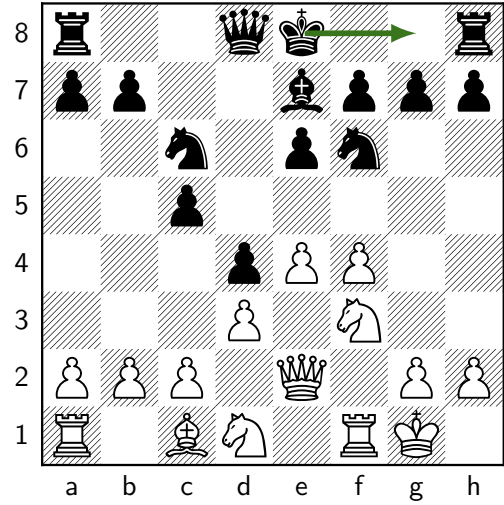
(a)



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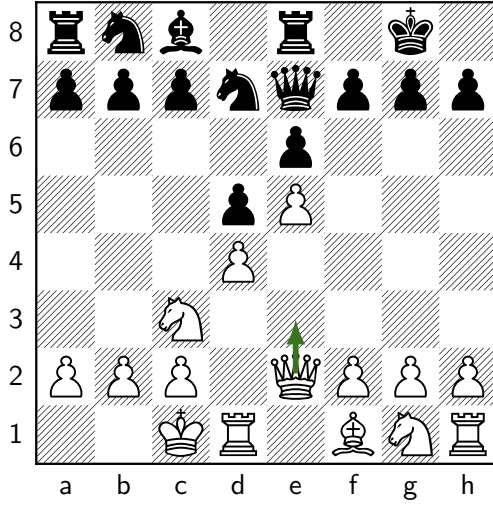


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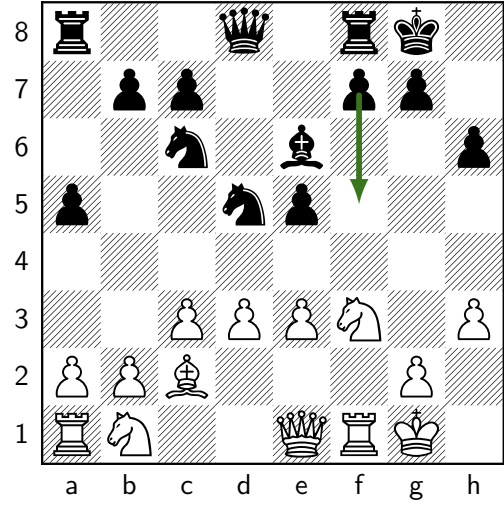


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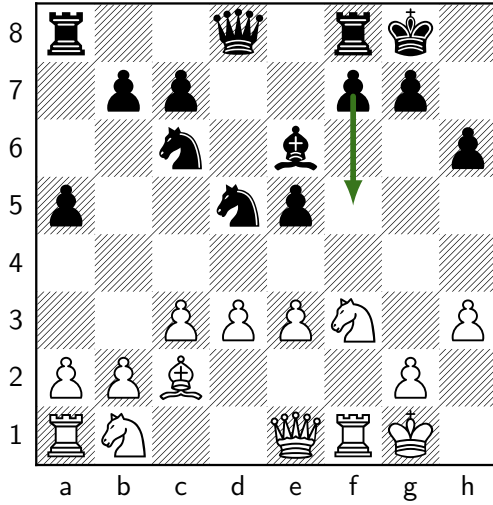
Figure 2: **Move 8, White: 0-0-0.** (a) Our position; Melanie plays 0-0-0 (e1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#1831` (ply 22) [structural distance $d=96$, QAP obj=58]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Qe2* as their *Bc1*, our *Pa2* as their *Qd1*; transferred move e1c1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0023#2374` (ply 27) [structural distance $d=81$, QAP obj=48]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Ng1* as their *Ra8*, our *Rh1* as their *Pa7*; transferred move e1c1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0018#622` (ply 21) [structural distance $d=91$, QAP obj=43]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bf1* as their *Ra8*, our *Ng1* as their *Rh8*; transferred move e1c1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to 0-0-0 (summed QAP objective 247, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



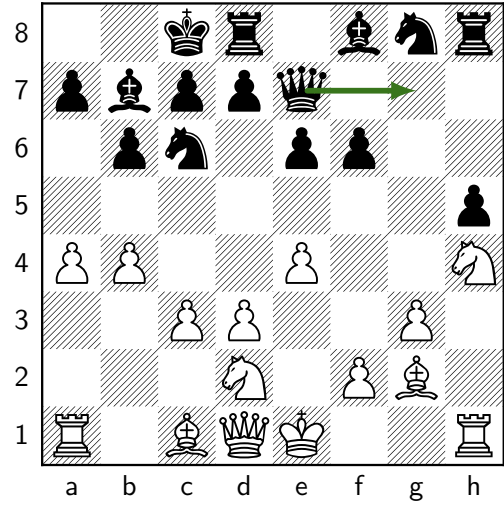
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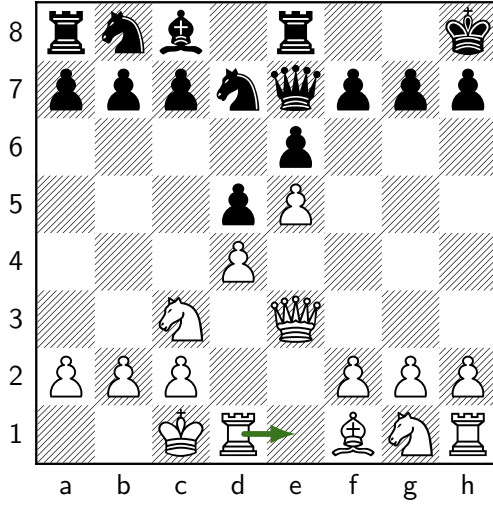


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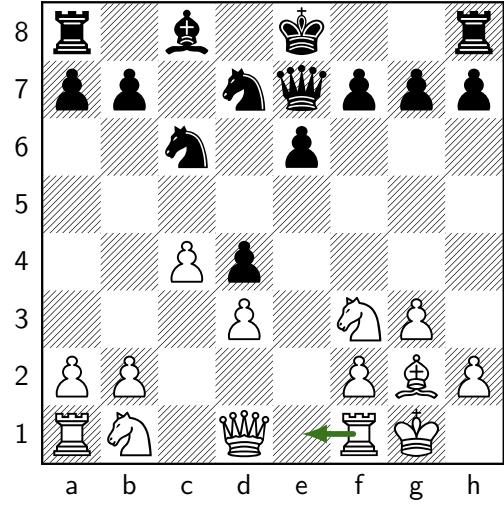


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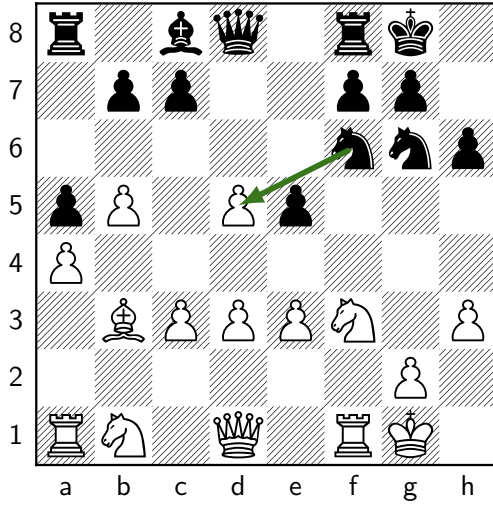
Figure 3: **Move 9, White: Qe3**. (a) Our position; Melanie plays Qe3 (e2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#2657` (ply 25) [structural distance $d=65$, QAP obj=58]: the master played f5; under μ our Qe2 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our Ng1 as their Rf8, our Rh1 as their Pb7; transferred move e2e3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0018#247` (ply 25) [structural distance $d=65$, QAP obj=58]: the master played f5; under μ our Qe2 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our Ng1 as their Rf8, our Rh1 as their Pb7; transferred move e2e3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0030#1996` (ply 21) [structural distance $d=75$, QAP obj=41]: the master played Qg7; under μ our Nc3 plays the role of that game’s Qe2 (a **cross-type** correspondence); cross-type roles: our Ng1 as their Bf8, our Pg2 as their Ng8; transferred move c3b5. Synthesis: 2 retrieved analogs transferred to Qe3 (summed QAP objective 117, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



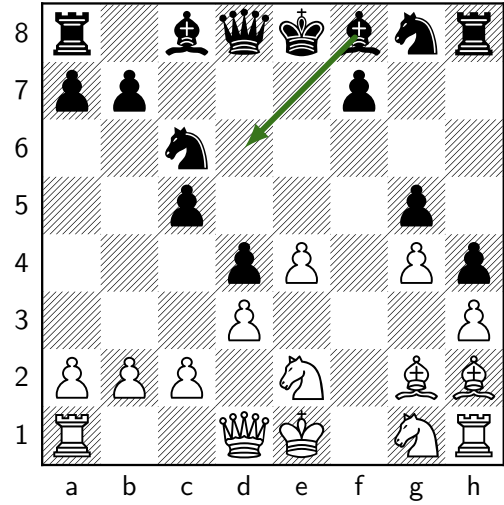
(a)



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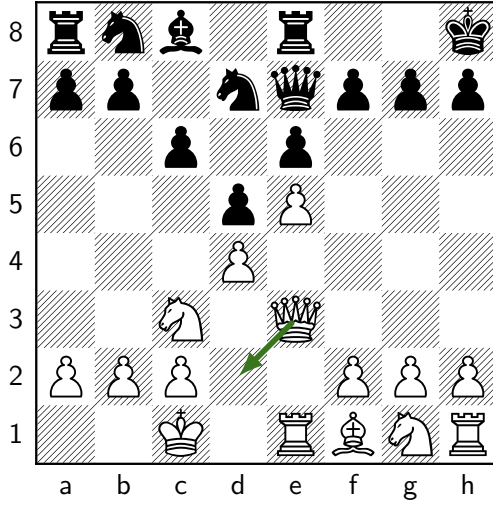


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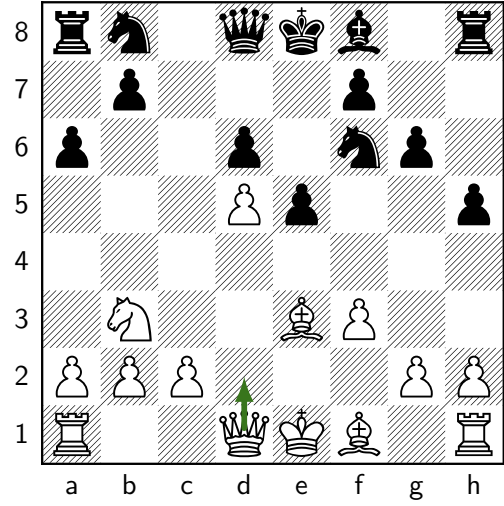


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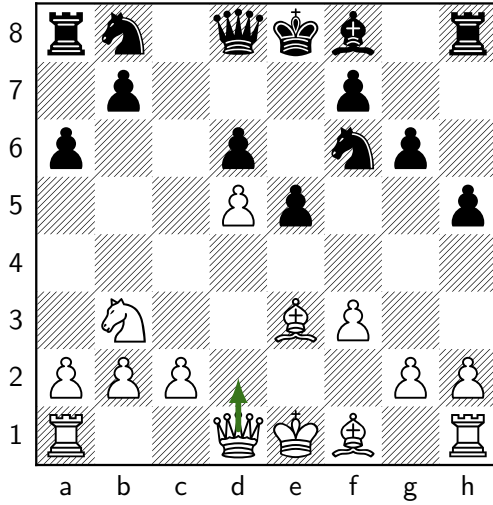
Figure 4: **Move 10, White: Re1.** (a) Our position; Melanie plays **Re1** (**d1e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0023#764` (ply 20) [structural distance $d=85$, QAP obj=26]: the master played **Re1**; under μ our *Rd1* plays the role of that game’s *Rf1*; cross-type roles: our *Bf1* as their *Ra1*, our *pc7* as their *Nb1*; transferred move **d1e1** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#3098` (ply 27) [structural distance $d=81$, QAP obj=51]: the master played **Nxd5**; under μ our *Nc3* plays the role of that game’s *Nf3*; cross-type roles: our *Ng1* as their *Ra8*, our *Pa2* as their *Bc8*; transferred move **c3e4**. (d) Retrieved from `lichess_elite_2024-09_p0023#1839` (ply 21) [structural distance $d=84$, QAP obj=33]: the master played **Bd6**; under μ our *Bf1* plays the role of that game’s *Bf1*; cross-type roles: our *nb8* as their *Ra8*, our *nd7* as their *Bc8*; transferred move **f1c4**. Synthesis: 1 retrieved analog transferred to **Re1** (summed QAP objective 26, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



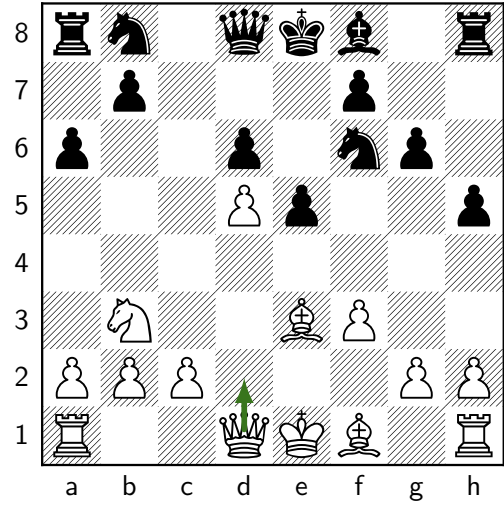
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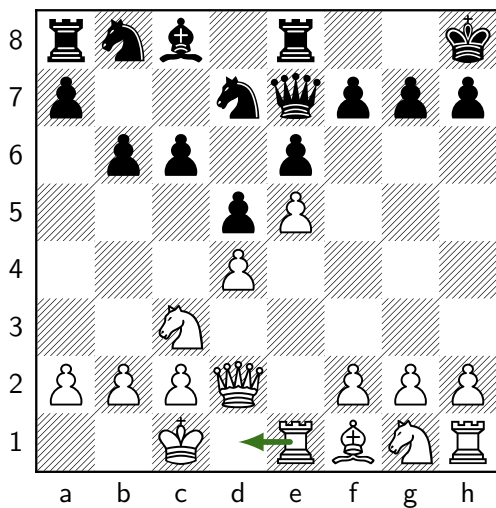


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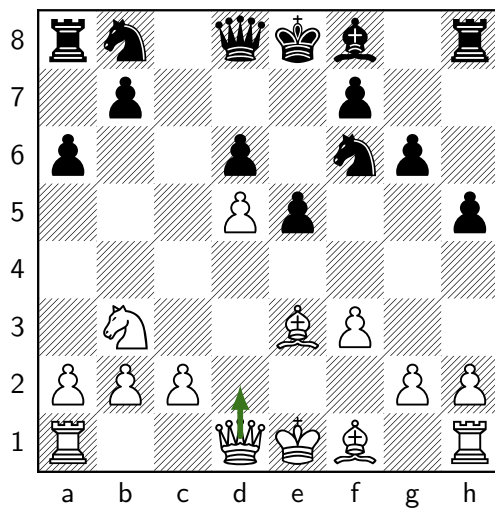


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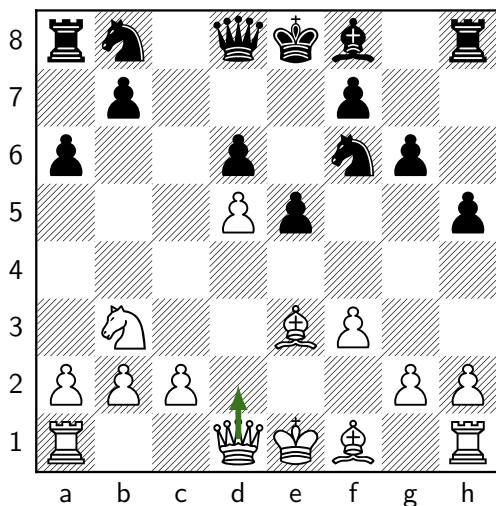
Figure 5: **Move 11, White: Qd2.** (a) Our position; Melanie plays Qd2 (e3d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#3821` (ply 20) [structural distance $d=67$, QAP obj=32]: the master played Qd2; under μ our *Qe3* plays the role of that game’s *Qd1*; cross-type roles: our *Nc3* as their *Bf1*, our *Bf1* as their *Rh1*; transferred move e3d2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0018#1001` (ply 20) [structural distance $d=67$, QAP obj=32]: the master played Qd2; under μ our *Qe3* plays the role of that game’s *Qd1*; cross-type roles: our *Nc3* as their *Bf1*, our *Bf1* as their *Rh1*; transferred move e3d2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0025#405` (ply 20) [structural distance $d=67$, QAP obj=32]: the master played Qd2; under μ our *Qe3* plays the role of that game’s *Qd1*; cross-type roles: our *Nc3* as their *Bf1*, our *Bf1* as their *Rh1*; transferred move e3d2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qd2 (summed QAP objective 129, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



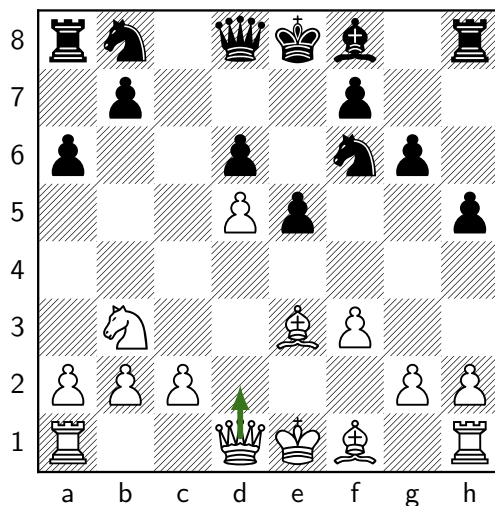
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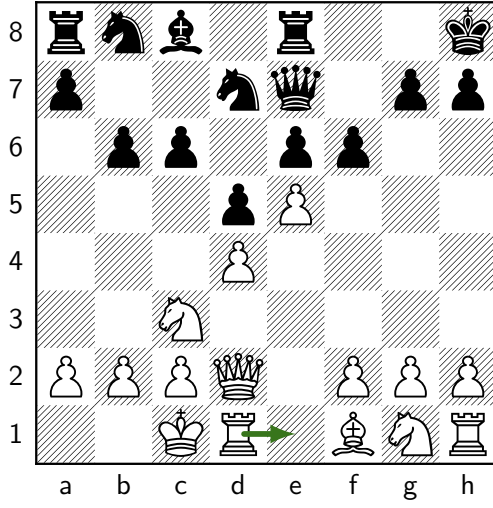


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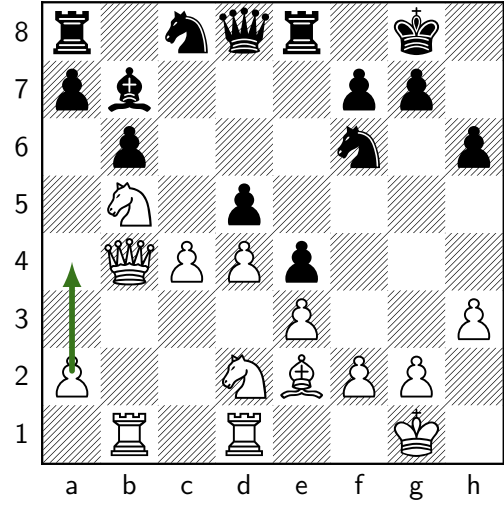


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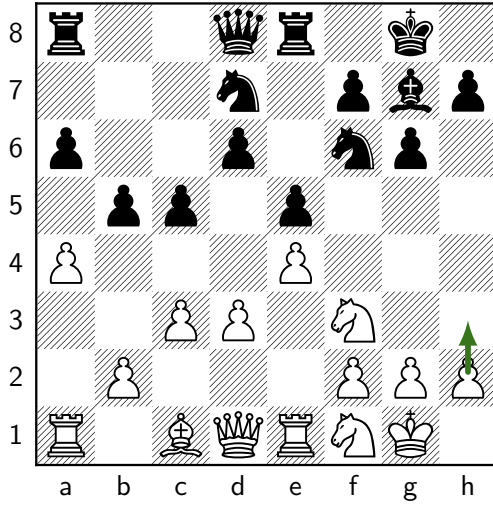
Figure 6: **Move 12, White: Rd1.** (a) Our position; Melanie plays Rd1 (**e1d1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#3821` (ply 20) [structural distance $d=82$, QAP obj=31]: the master played Qd2; under μ our *Re1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Qd2* as their *Ra1*, our *Nc3* as their *Bf1*; transferred move **e1d1 (endorses this move)**. (c) Retrieved from `lichess_elite_2024-09_p0018#1001` (ply 20) [structural distance $d=82$, QAP obj=31]: the master played Qd2; under μ our *Re1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Qd2* as their *Ra1*, our *Nc3* as their *Bf1*; transferred move **e1d1 (endorses this move)**. (d) Retrieved from `lichess_elite_2024-09_p0025#405` (ply 20) [structural distance $d=82$, QAP obj=31]: the master played Qd2; under μ our *Re1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Qd2* as their *Ra1*, our *Nc3* as their *Bf1*; transferred move **e1d1 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to Rd1 (summed QAP objective 125, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



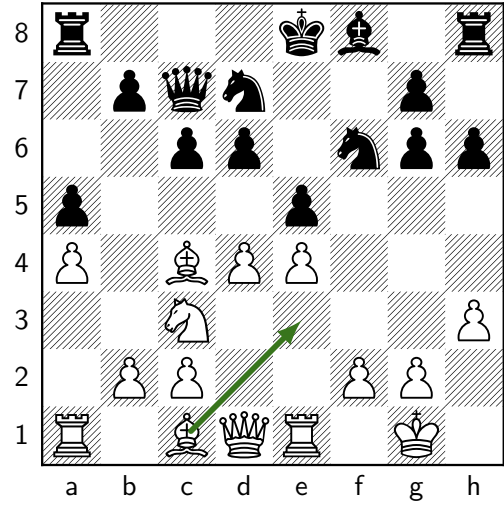
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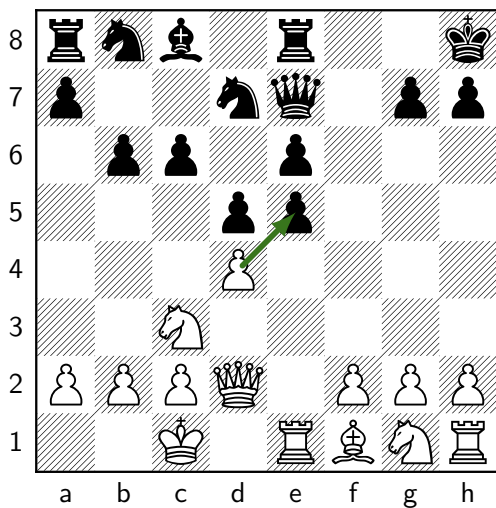


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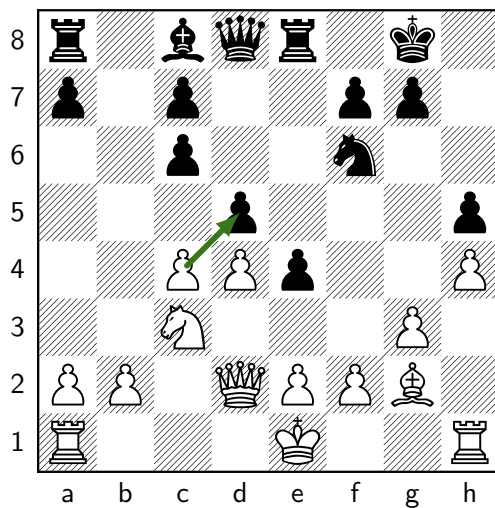


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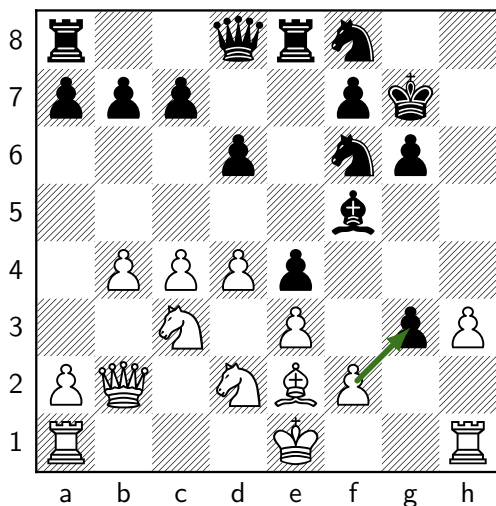
Figure 7: **Move 13, White: Re1.** (a) Our position; Melanie plays Re1 (d1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#545` (ply 38) [structural distance $d=93$, QAP obj=62]: the master played a4; under μ our *Rd1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *nb8* as their *Rb1*, our *bc8* as their *Rd1*; transferred move d1e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#625` (ply 24) [structural distance $d=81$, QAP obj=53]: the master played h3; under μ our *Rd1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *nd7* as their *Ra1*, our *Qd2* as their *Bc1*; transferred move d1e1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#2604` (ply 26) [structural distance $d=90$, QAP obj=52]: the master played Be3; under μ our *Rd1* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *qe7* as their *Ra1*, our *Bf1* as their *Qd1*; transferred move d1e1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Re1 (summed QAP objective 178, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



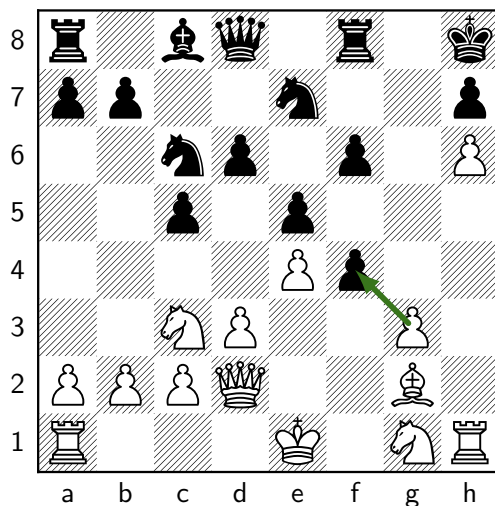
(a)



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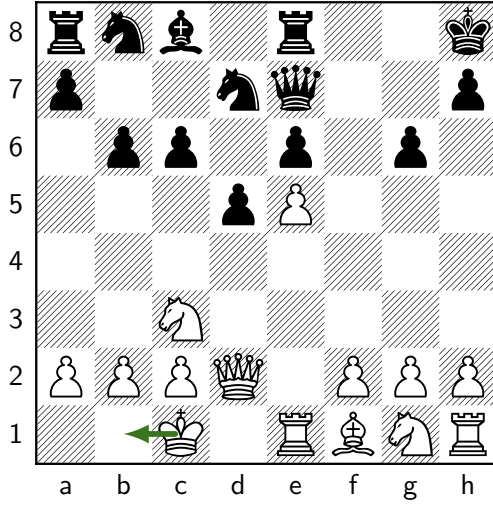


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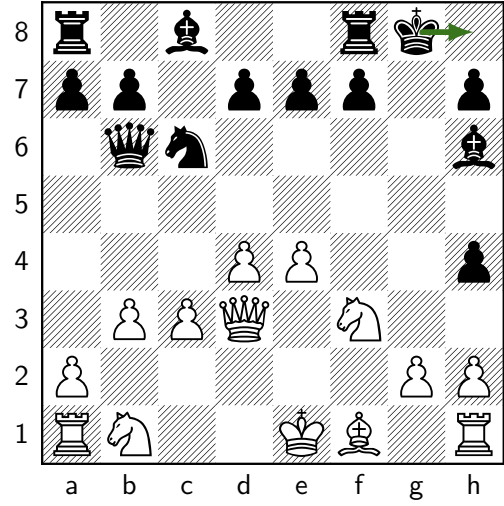


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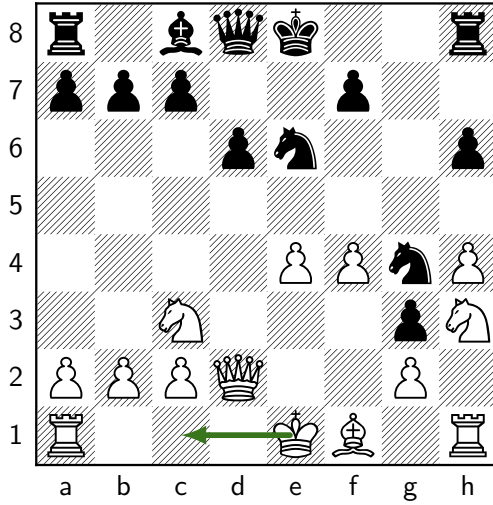
Figure 8: **Move 14, White: dxe5.** (a) Our position; Melanie plays dxe5 (d4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#1524` (ply 26) [structural distance $d=93$, QAP obj=68]: the master played `cxd5`; under μ our $Pd4$ plays the role of that game’s $Pc4$; cross-type roles: our $Pf2$ as their $Ra1$, our $Ph2$ as their $Rh1$; transferred move `d4e5` (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#1218` (ply 30) [structural distance $d=97$, QAP obj=56]: the master played `fxg3`; under μ our $Pd4$ plays the role of that game’s $Pf2$; cross-type roles: our $Qd2$ as their $Rh1$, our $pg7$ as their $Qb2$; transferred move `d4e5` (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#81` (ply 26) [structural distance $d=82$, QAP obj=52]: the master played `gxf4`; under μ our $Pd4$ plays the role of that game’s $Pg3$; cross-type roles: our $Qd2$ as their $Ra1$, our $Pc2$ as their $Ng1$; transferred move `d4e5` (**endorses this move**). Synthesis: 3 retrieved analogs transferred to `dxe5` (summed QAP objective 176, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



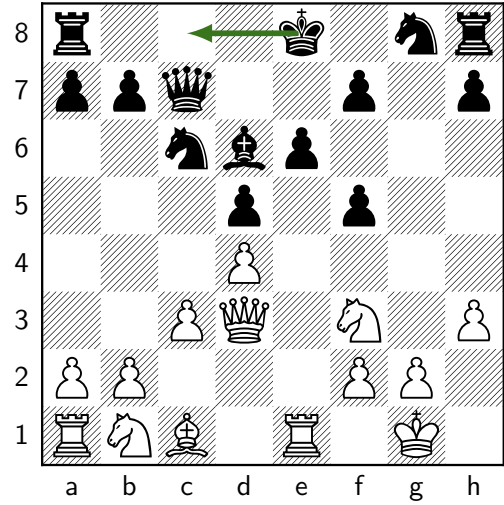
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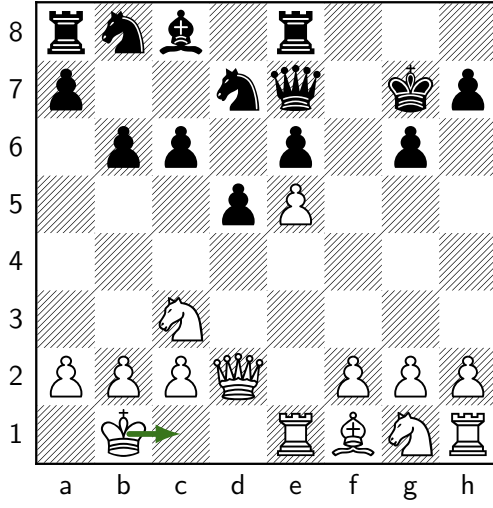


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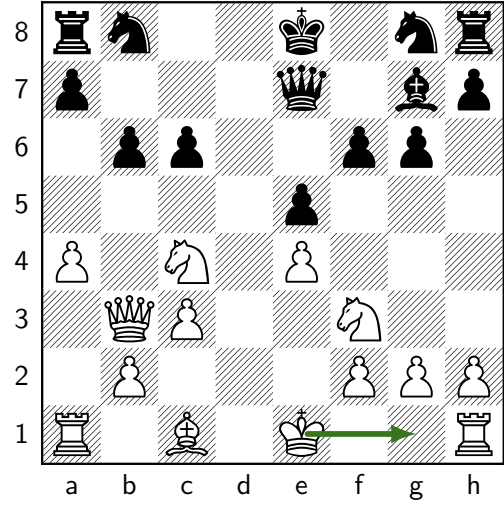


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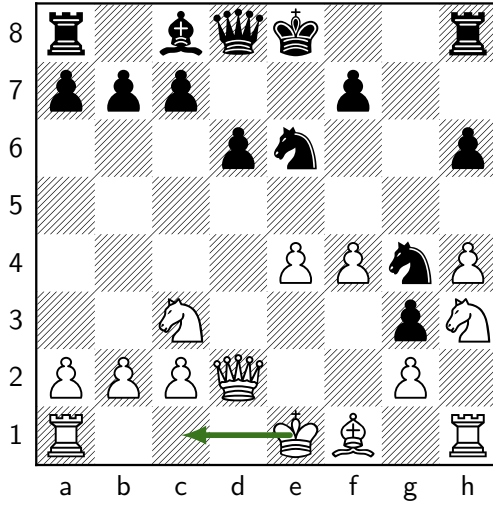
Figure 9: **Move 15, White: Kb1.** (a) Our position; Melanie plays Kb1 (c1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#2668` (ply 21) [structural distance $d=97$, QAP obj=96]: the master played Kh8; under μ our *Kc1* plays the role of that game’s *Kg1*; cross-type roles: our *Pc2* as their *Bc8*, our *pa7* as their *Rf8*; transferred move c1b1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#964` (ply 34) [structural distance $d=89$, QAP obj=55]: the master played O-O-O; under μ our *Kc1* plays the role of that game’s *Ke1*; cross-type roles: our *Ng1* as their *Ra1*, our *Pa2* as their *Bf1*; transferred move c1b1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0010#1848` (ply 21) [structural distance $d=96$, QAP obj=55]: the master played O-O-O; under μ our *Kc1* plays the role of that game’s *Ke1*; cross-type roles: our *pa7* as their *Ng8*, our *Ng1* as their *Rh8*; transferred move c1b1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kb1 (summed QAP objective 285, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



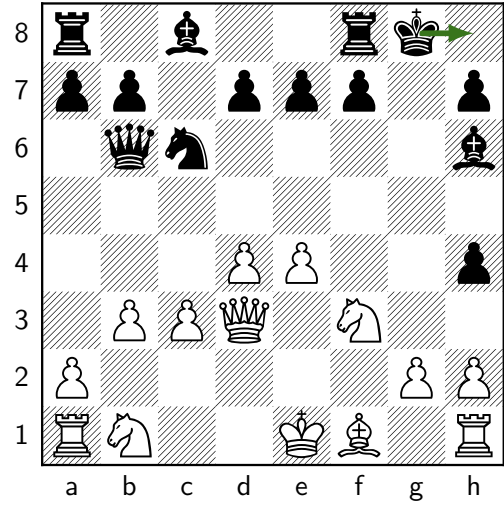
(a)



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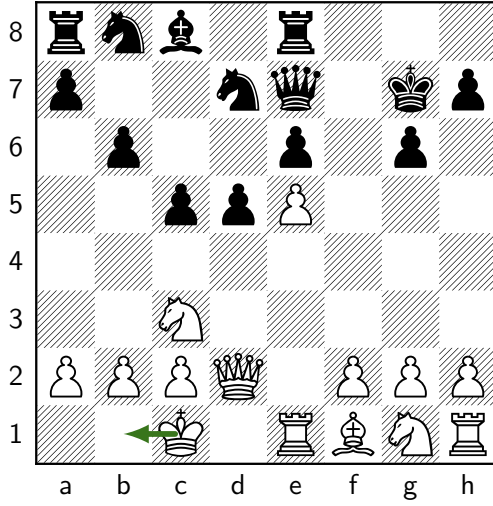


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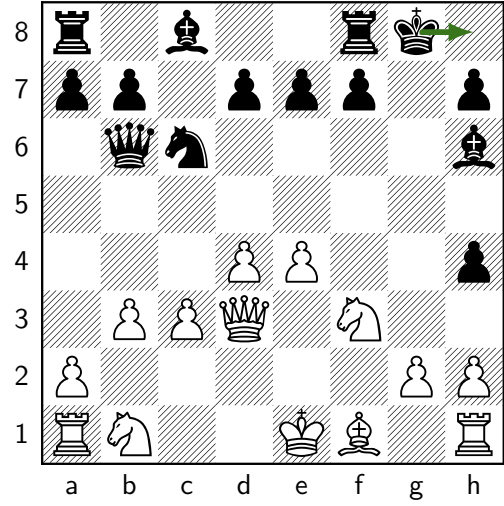


(d)

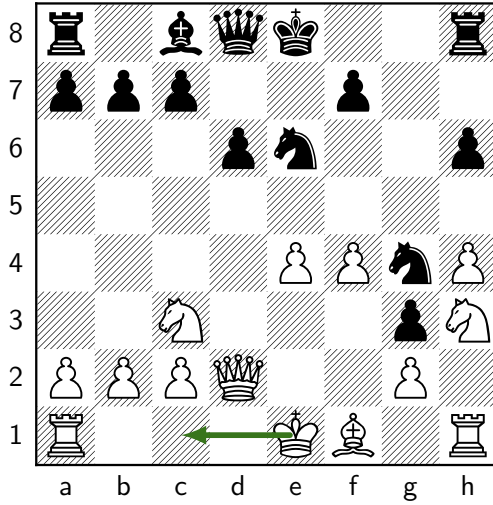
Figure 10: **Move 16, White: Kc1.** (a) Our position; Melanie plays Kc1 (b1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#521` (ply 22) [structural distance $d=88$, QAP obj=115]: the master played 0-0; under μ our *Kb1* plays the role of that game’s *Ke1*; cross-type roles: our *qe7* as their *Ra1*, our *Ng1* as their *Rh1*; transferred move b1c1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#964` (ply 34) [structural distance $d=82$, QAP obj=49]: the master played 0-0-0; under μ our *Kb1* plays the role of that game’s *Ke1*; cross-type roles: our *qe7* as their *Bf1*, our *Pc2* as their *Rh1*; transferred move b1c1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#2668` (ply 21) [structural distance $d=84$, QAP obj=98]: the master played Kh8; under μ our *Kb1* plays the role of that game’s *Kg1*; cross-type roles: our *Pa2* as their *Bc8*, our *Pg2* as their *Rf8*; transferred move b1a1. Synthesis: 2 retrieved analogs transferred to Kc1 (summed QAP objective 164, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



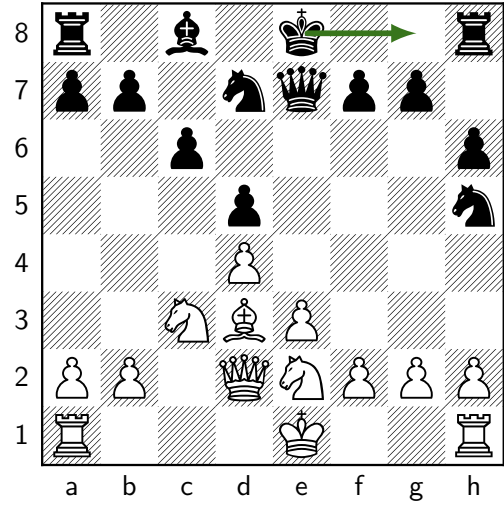
(a)



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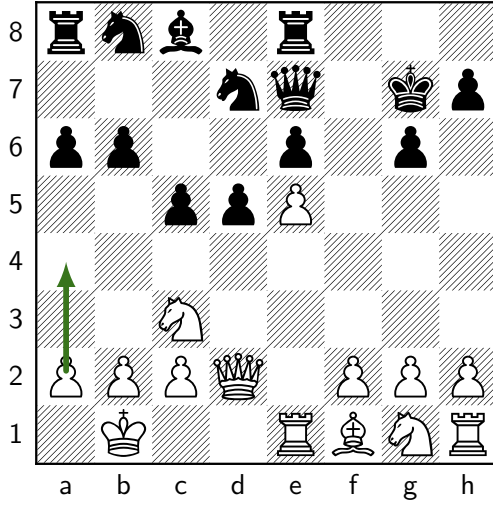


(c)

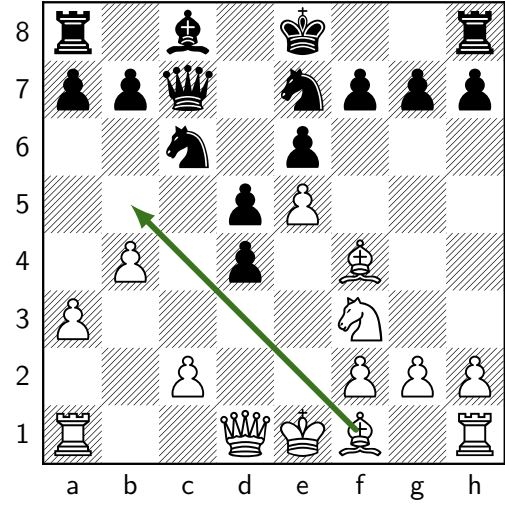


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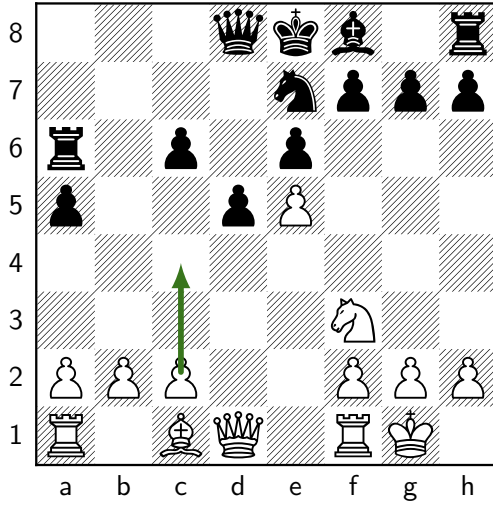
Figure 11: **Move 17, White: Kb1.** (a) Our position; Melanie plays Kb1 (c1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#2668` (ply 21) [structural distance $d=83$, QAP obj=98]: the master played Kh8; under μ our *Kc1* plays the role of that game’s *Kg1*; cross-type roles: our *Pc2* as their *Bc8*, our *pa7* as their *Rf8*; transferred move c1b1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#964` (ply 34) [structural distance $d=89$, QAP obj=58]: the master played O-O-O; under μ our *Kc1* plays the role of that game’s *Ke1*; cross-type roles: our *Pg2* as their *Bf1*, our *Ng1* as their *Pc2*; transferred move c1b1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0005#110` (ply 21) [structural distance $d=90$, QAP obj=54]: the master played O-O; under μ our *Kc1* plays the role of that game’s *Ke1*; cross-type roles: our *Nc3* as their *Ra8*, our *pa7* as their *Bc8*; transferred move c1b1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kb1 (summed QAP objective 288, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



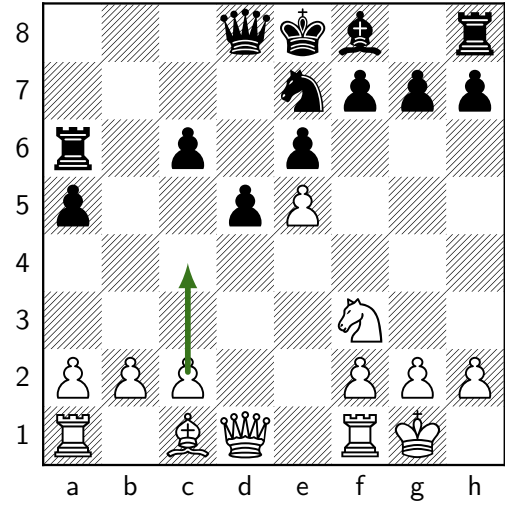
(a)



(b)

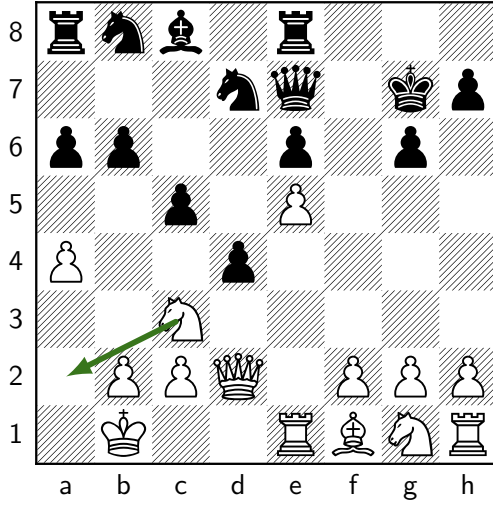


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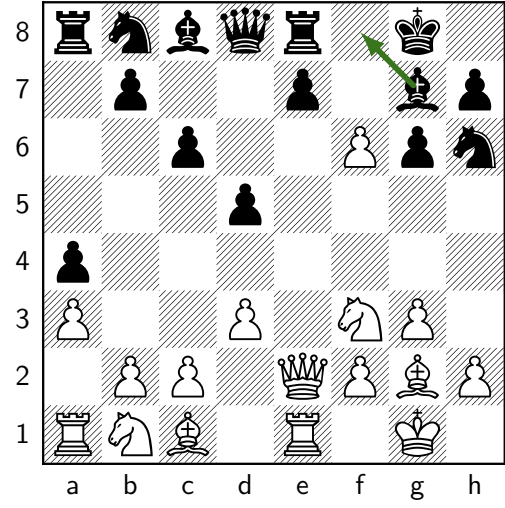


(d)

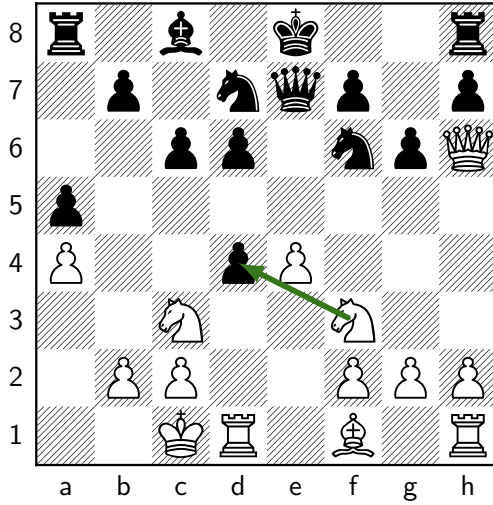
Figure 12: **Move 18, White: a4.** (a) Our position; Melanie plays a4 (a2a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#992` (ply 20) [structural distance $d=79$, QAP obj=117]: the master played Bb5; under μ our *Pa2* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *nb8* as their *Ra1*, our *nd7* as their *Qd1*; transferred move a2a4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0018#2267` (ply 22) [structural distance $d=91$, QAP obj=34]: the master played c4; under μ our *Pa2* plays the role of that game’s *Pc2*; cross-type roles: our *nb8* as their *Ra1*, our *Re1* as their *Bc1*; transferred move a2a4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#3123` (ply 22) [structural distance $d=91$, QAP obj=34]: the master played c4; under μ our *Pa2* plays the role of that game’s *Pc2*; cross-type roles: our *nb8* as their *Ra1*, our *Re1* as their *Bc1*; transferred move a2a4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to a4 (summed QAP objective 185, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



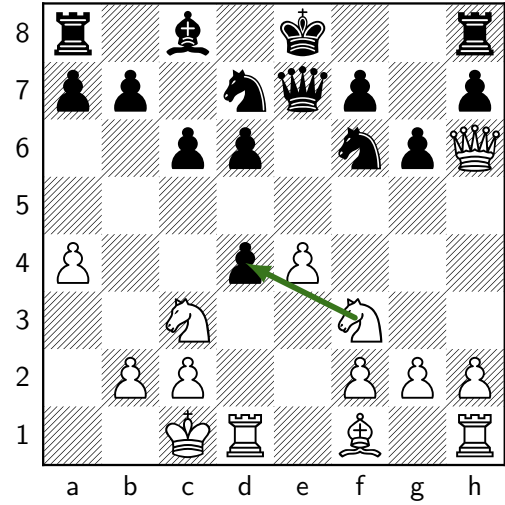
(a)



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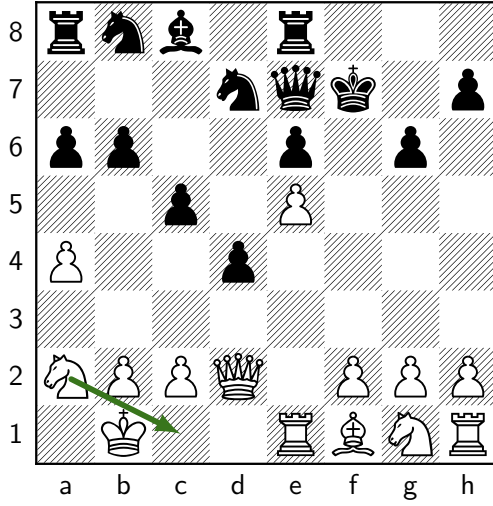


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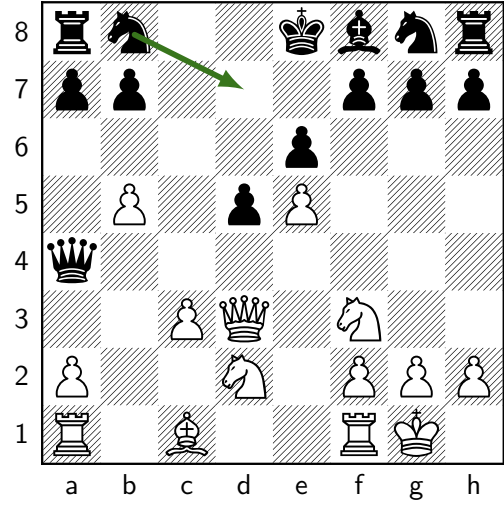


(d)

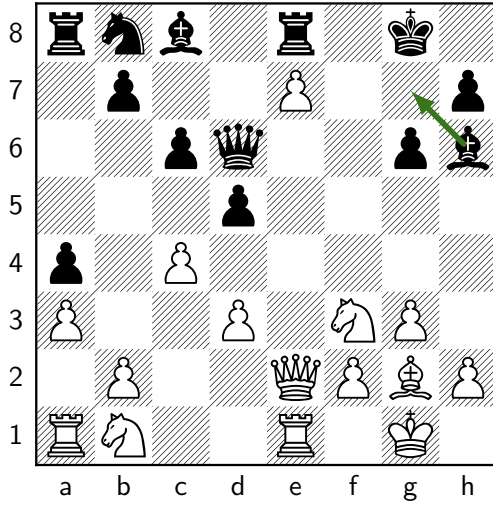
Figure 13: **Move 19, White: Na2.** (a) Our position; Melanie plays Na2 (c3a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#232` (ply 21) [structural distance $d=88$, QAP obj=30]: the master played Bf8; under μ our Nc3 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our Ng1 as their Ra8, our Pa4 as their Nb8; transferred move c3a2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#1409` (ply 22) [structural distance $d=93$, QAP obj=42]: the master played Nxd4; under μ our Qd2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Qd2 as their Rd1, our Re1 as their Bf1; transferred move d2d3. (d) Retrieved from `lichess_elite_2024-09_p0030#3189` (ply 20) [structural distance $d=87$, QAP obj=41]: the master played Nxd4; under μ our Qd2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Qd2 as their Rd1, our Re1 as their Bf1; transferred move d2f4. Synthesis: 1 retrieved analog transferred to Na2 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



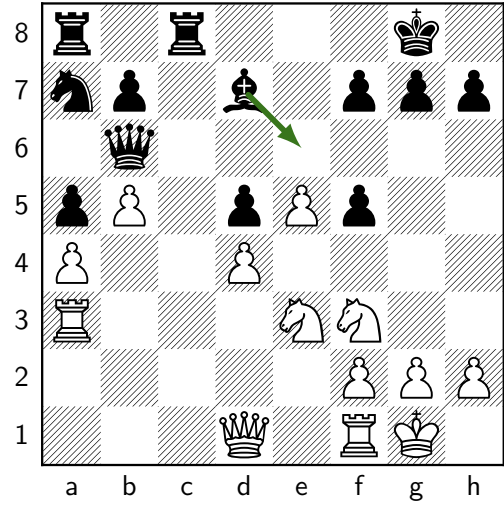
(a)



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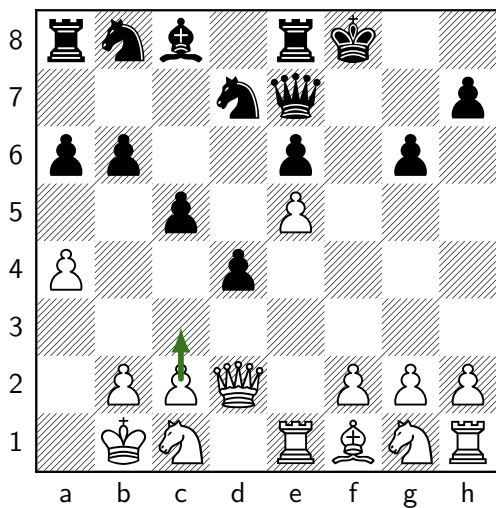


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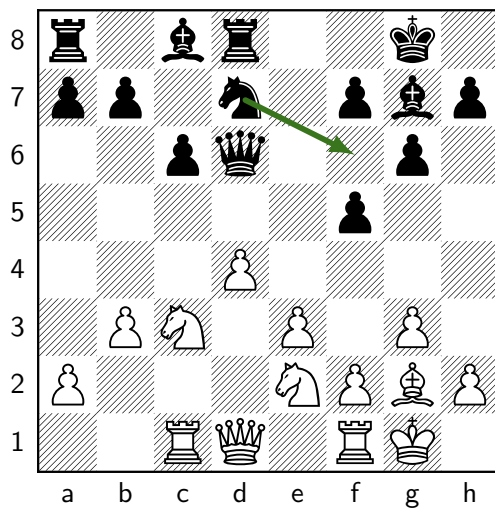


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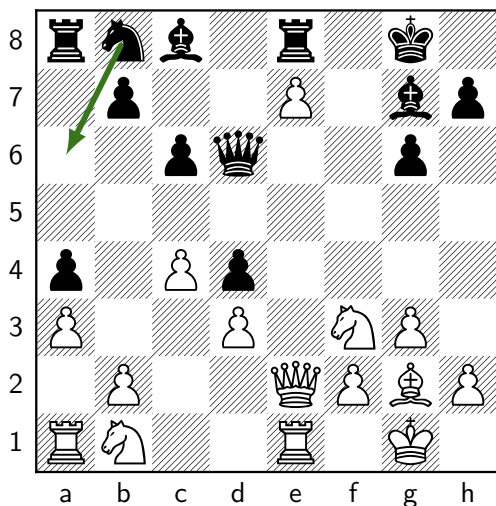
Figure 14: **Move 20, White: Nc1.** (a) Our position; Melanie plays Nc1 (a2c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#2989` (ply 23) [structural distance $d=91$, QAP obj=51]: the master played Nd7; under μ our Na2 plays the role of that game’s Nb1; cross-type roles: our Pb2 as their Bf8, our pa6 as their Ng8; transferred move a2c1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#232` (ply 27) [structural distance $d=104$, QAP obj=44]: the master played Bg7; under μ our Na2 plays the role of that game’s Bh3 (a **cross-type** correspondence); cross-type roles: our pa6 as their Ra8, our Pg2 as their Nb8; transferred move a2c1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#2205` (ply 33) [structural distance $d=99$, QAP obj=67]: the master played Be6; under μ our Qd2 plays the role of that game’s Bd2 (a **cross-type** correspondence); cross-type roles: our Ng1 as their Ra8, our Qd2 as their Na7; transferred move d2d3. Synthesis: 2 retrieved analogs transferred to Nc1 (summed QAP objective 95, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



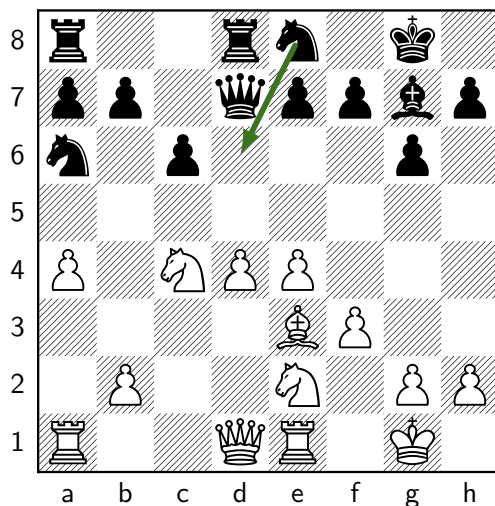
(a)



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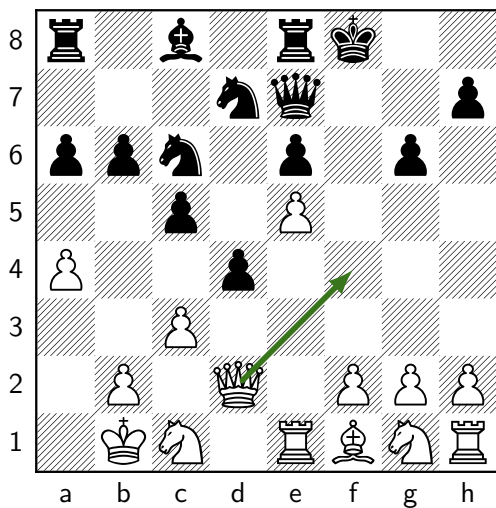


(c)

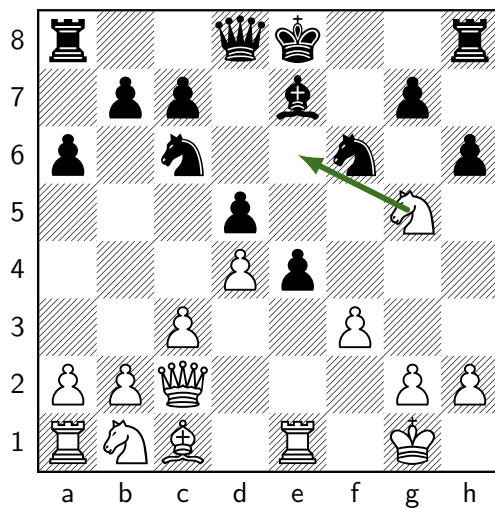


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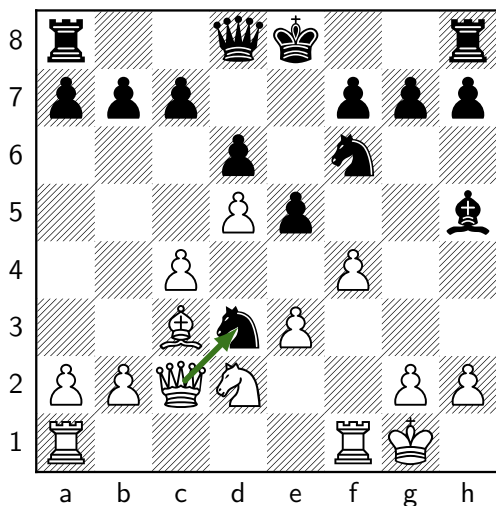
Figure 15: **Move 21, White: c3.** (a) Our position; Melanie plays c3 (c2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#3986` (ply 25) [structural distance $d=117$, QAP obj=82]: the master played Nf6; under μ our *Pc2* plays the role of that game’s *Nd2* (a **cross-type** correspondence); cross-type roles: our *Ng1* as their *Rd8*, our *Rh1* as their *Pa7*; transferred move c2c3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#232` (ply 31) [structural distance $d=98$, QAP obj=51]: the master played Na6; under μ our *Pc2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Pf2* as their *Bc8*, our *Qd2* as their *Re8*; transferred move c2c3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0024#1466` (ply 27) [structural distance $d=101$, QAP obj=41]: the master played Nd6; under μ our *Pc2* plays the role of that game’s *Ne1* (a **cross-type** correspondence); cross-type roles: our *Nc1* as their *Ra8*, our *Bf1* as their *Pa7*; transferred move c2c3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to c3 (summed QAP objective 225, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



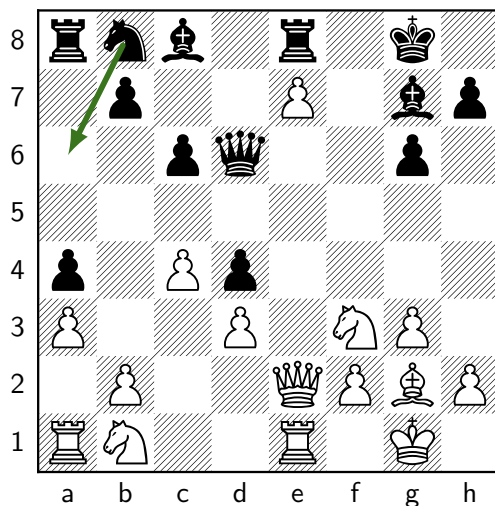
(a)



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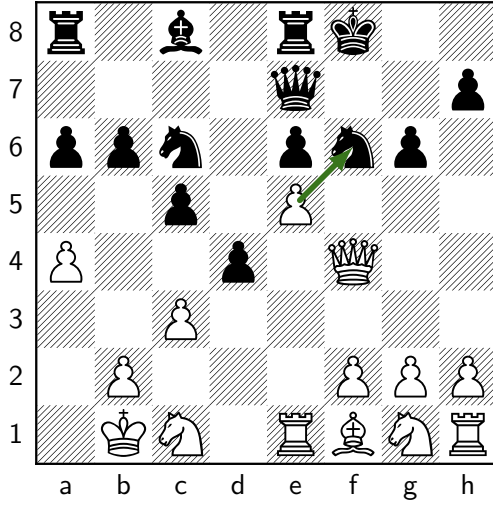


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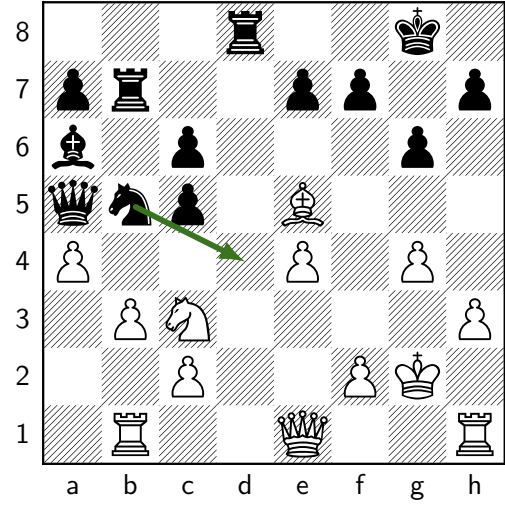


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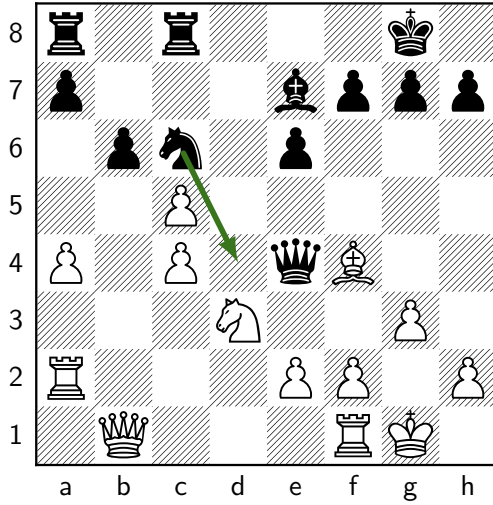
Figure 16: **Move 22, White: Qf4+ (d2f4).** (a) Our position; Melanie plays Qf4+ (d2f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#3345` (ply 26) [structural distance $d=103$, QAP obj=67]: the master played Ne6; under μ our Qd2 plays the role of that game’s Ng5 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Nb1, our Nc1 as their Bc1; transferred move d2f4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0025#197` (ply 26) [structural distance $d=106$, QAP obj=66]: the master played Qxd3; under μ our Qd2 plays the role of that game’s Qc2; cross-type roles: our pg6 as their Ra1, our qe7 as their Rf1; transferred move d2f4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#232` (ply 31) [structural distance $d=101$, QAP obj=64]: the master played Na6; under μ our Nc1 plays the role of that game’s Nb1; cross-type roles: our pg6 as their Ra8, our Pf2 as their Bc8; transferred move c1b3. Synthesis: 2 retrieved analogs transferred to Qf4+ (summed QAP objective 133, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



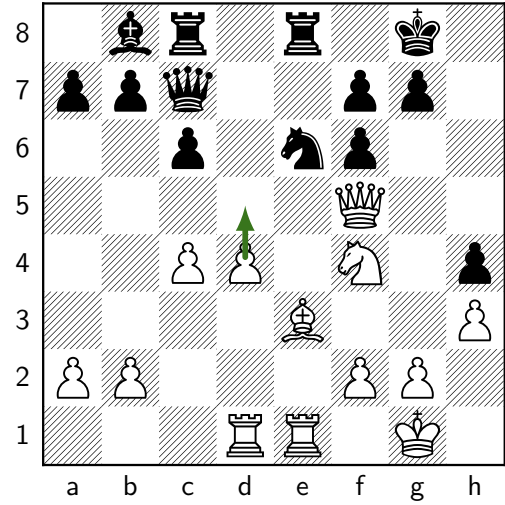
(a)



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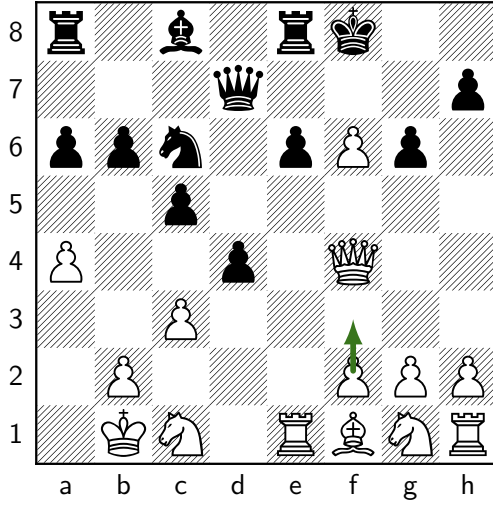


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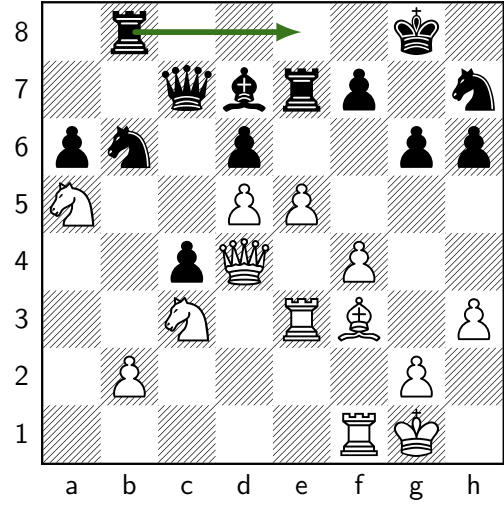


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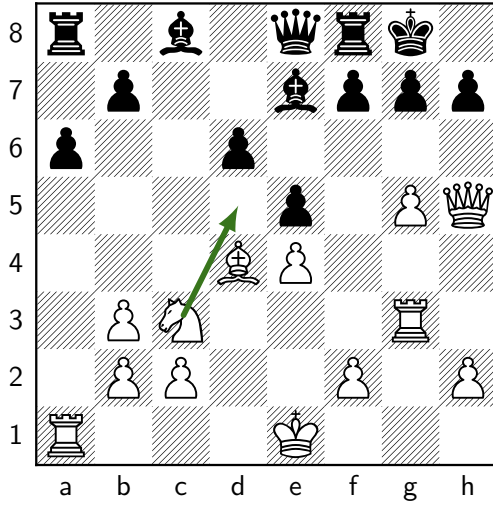
Figure 17: **Move 23, White: exf6.** (a) Our position; Melanie plays exf6 (e5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#3335` (ply 37) [structural distance $d=107$, QAP obj=79]: the master played Nd4; under μ our Pe5 plays the role of that game’s Nb4 (a **cross-type** correspondence); cross-type roles: our pa6 as their Rd8, our Nc1 as their Ba6; transferred move e5f6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#313` (ply 37) [structural distance $d=109$, QAP obj=72]: the master played Nd4; under μ our Pf2 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Ng1 as their Ra8, our Nc1 as their Rc8; transferred move f2f3. (d) Retrieved from `lichess_elite_2024-09_p0028#3087` (ply 36) [structural distance $d=103$, QAP obj=70]: the master played d5; under μ our Pb2 plays the role of that game’s Pd4; cross-type roles: our Pc3 as their Rd1, our Nc1 as their Pa2; transferred move b2b3. Synthesis: 1 retrieved analog transferred to exf6 (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



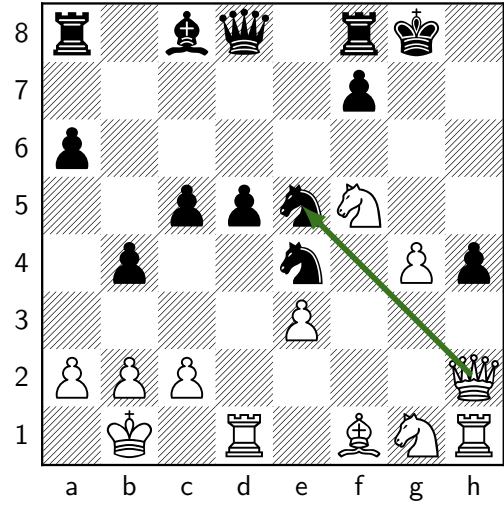
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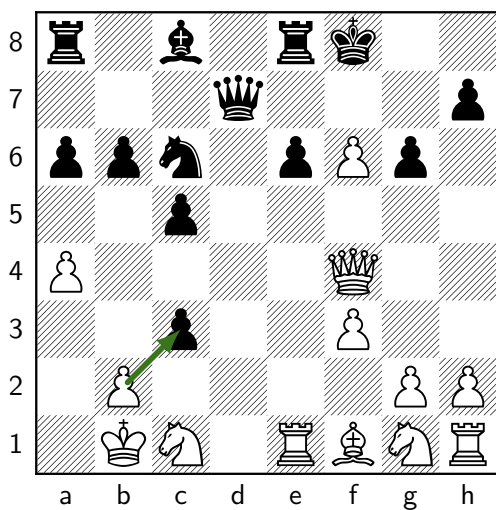


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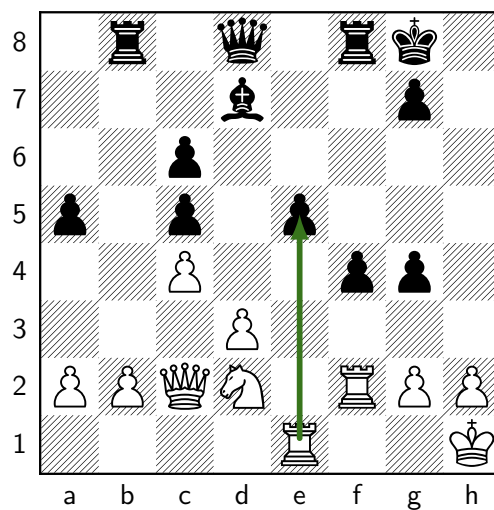


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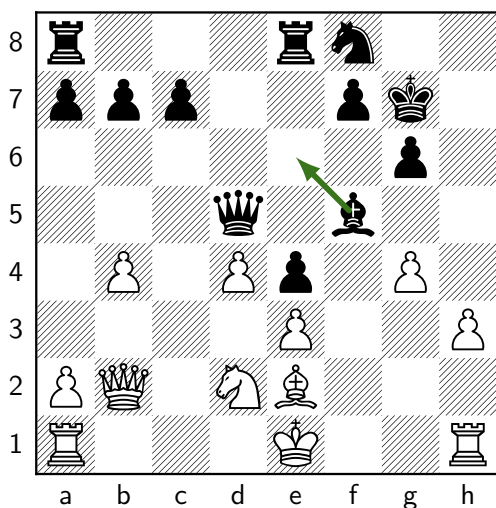
Figure 18: **Move 24, White: f3.** (a) Our position; Melanie plays f3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#3098` (ply 49) [structural distance $d=97$, QAP obj=65]: the master played Rbe8; under μ our Pf2 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our Pb2 as their Qc7, our Nc1 as their Bd7; transferred move f2f3 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#78` (ply 30) [structural distance $d=94$, QAP obj=26]: the master played Nd5; under μ our Pf2 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Ng1 as their Pc2, our Bf1 as their Pb3; transferred move f2f3 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#1674` (ply 36) [structural distance $d=90$, QAP obj=77]: the master played Qxe5; under μ our Qf4 plays the role of that game’s Qh2; cross-type roles: our Pf2 as their Ng1, our Pf6 as their Rh1; transferred move f4g5. Synthesis: 2 retrieved analogs transferred to f3 (summed QAP objective 91, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



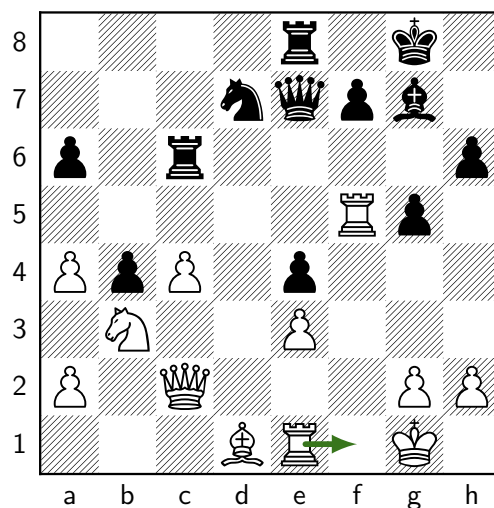
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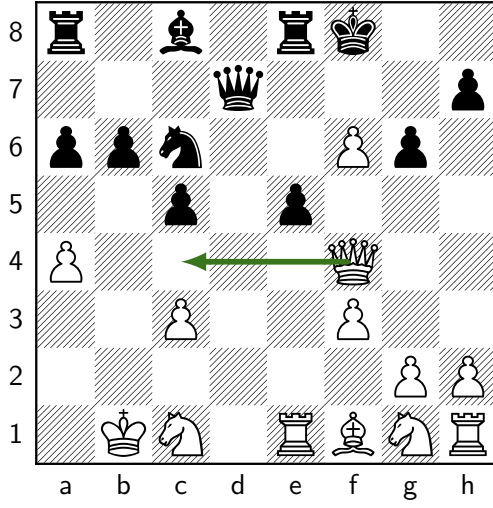


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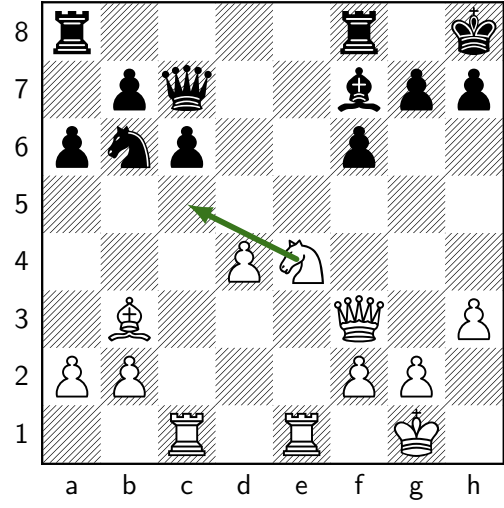


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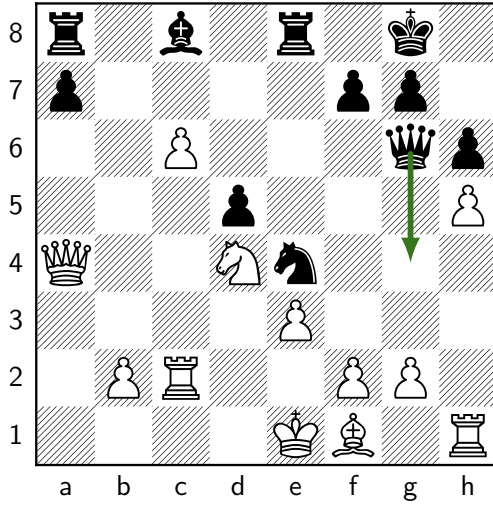
Figure 19: **Move 25, White: bxc3**. (a) Our position; Melanie plays **bxc3** (**b2c3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#3628` (ply 44) [structural distance $d=101$, QAP obj=19]: the master played **Rxe5**; under μ our *Pb2* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *nc6* as their *Pb2*, our *Nc1* as their *Qc2*; transferred move **b2c3** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#1218` (ply 37) [structural distance $d=98$, QAP obj=15]: the master played **Be6**; under μ our *Pb2* plays the role of that game’s *Bf4* (a **cross-type** correspondence); cross-type roles: our *Nc1* as their *Ra8*, our *Ng1* as their *Re8*; transferred move **b2c3** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#592` (ply 64) [structural distance $d=104$, QAP obj=37]: the master played **Ref1**; under μ our *Pg2* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *Qf4* as their *Bd1*, our *Bf1* as their *Pg2*; transferred move **g2g3**. Synthesis: 2 retrieved analogs transferred to **bxc3** (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



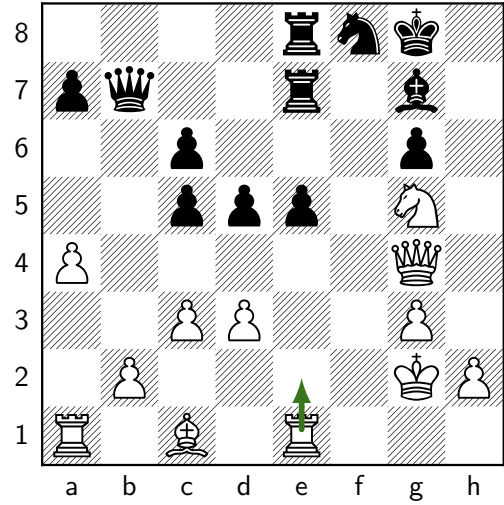
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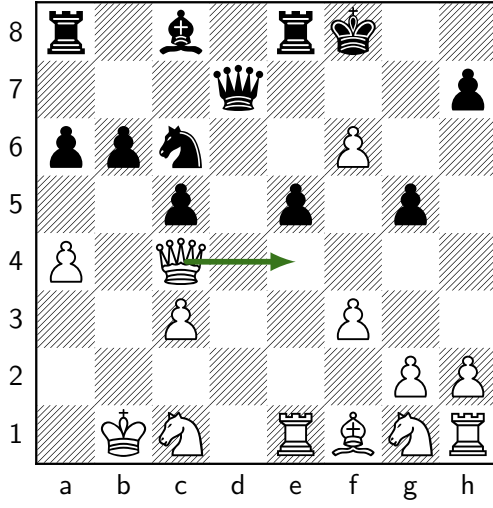


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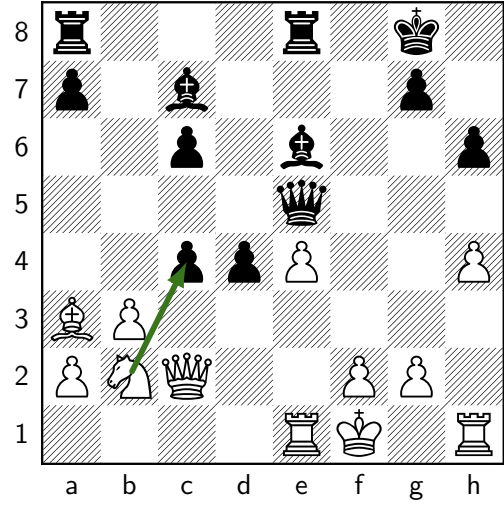


(d)

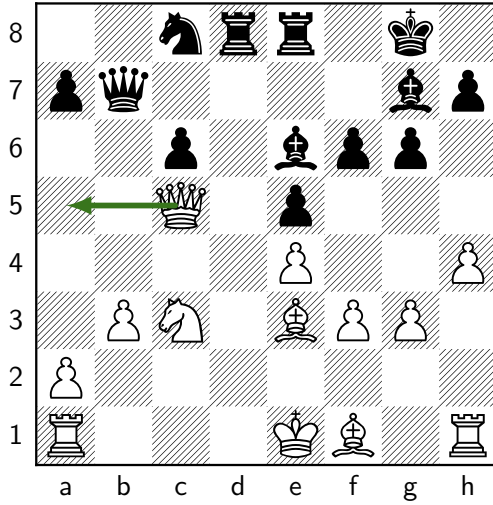
Figure 20: **Move 26, White: Qc4.** (a) Our position; Melanie plays Qc4 (f4c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#1014` (ply 42) [structural distance $d=104$, QAP obj=27]: the master played Nc5; under μ our Qf4 plays the role of that game’s Ne4 (a **cross-type** correspondence); cross-type roles: our Nc1 as their Re1, our bc8 as their Pa2; transferred move f4c4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#1623` (ply 59) [structural distance $d=90$, QAP obj=40]: the master played Qg4; under μ our Qf4 plays the role of that game’s Qg3; cross-type roles: our Ng1 as their Ra8, our Pa4 as their Bc8; transferred move f4g4. (d) Retrieved from `lichess_elite_2024-09_p0023#1109` (ply 54) [structural distance $d=98$, QAP obj=29]: the master played Re2; under μ our Re1 plays the role of that game’s Re1; cross-type roles: our Nc1 as their Ra1, our Ng1 as their Pb2; transferred move e1d1. Synthesis: 1 retrieved analog transferred to Qc4 (summed QAP objective 27, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



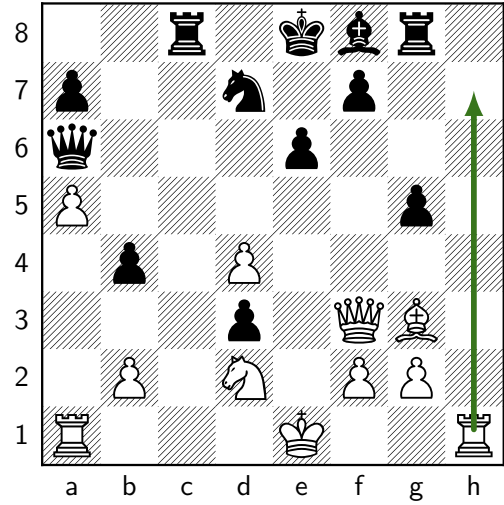
(a)



(b)

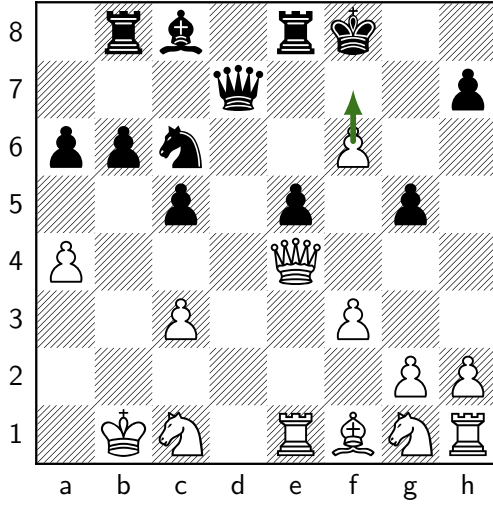


(c)

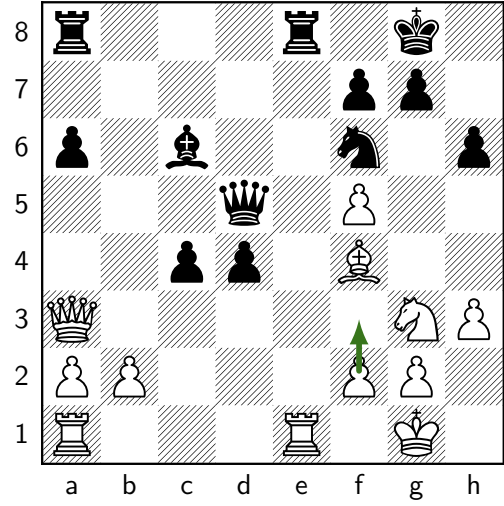


(d)

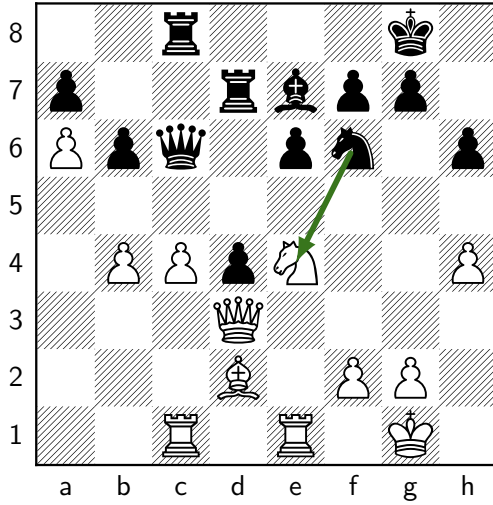
Figure 21: **Move 27, White: Qe4.** (a) Our position; Melanie plays Qe4 (c4e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0025#1086` (ply 42) [structural distance $d=120$, QAP obj=67]: the master played Nxc4; under μ our Qc4 plays the role of that game’s Nb2 (a **cross-type** correspondence); cross-type roles: our nc6 as their Pg2, our Pf3 as their Ba3; transferred move c4e4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#3178` (ply 44) [structural distance $d=109$, QAP obj=103]: the master played Qa5; under μ our Qc4 plays the role of that game’s Qc5; cross-type roles: our Nc1 as their Ra1, our Ng1 as their Bf1; transferred move c4g4. (d) Retrieved from `lichess_elite_2024-09_p0018#2849` (ply 40) [structural distance $d=92$, QAP obj=67]: the master played Rh7; under μ our Pf3 plays the role of that game’s Rh1 (a **cross-type** correspondence); cross-type roles: our qd7 as their Nd2, our Ng1 as their Pf2; transferred move f3f4. Synthesis: 1 retrieved analog transferred to Qe4 (summed QAP objective 67, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



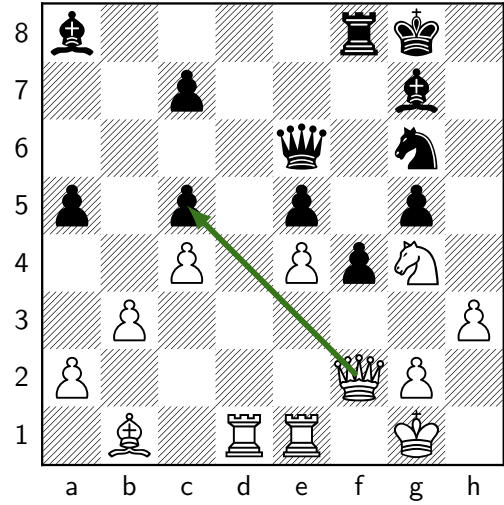
(a)



(b)

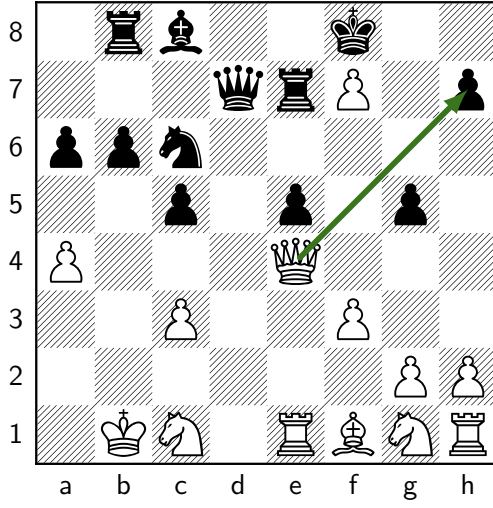


(c)

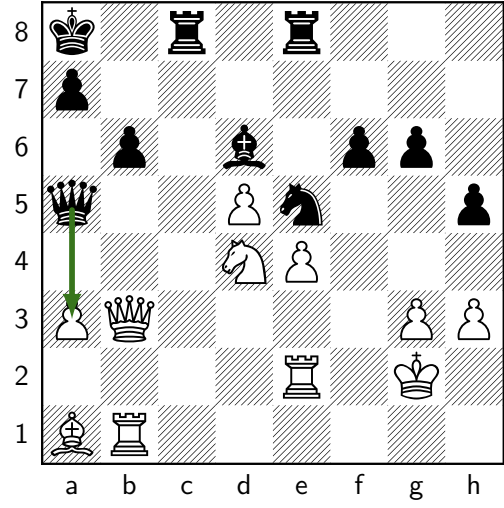


(d)

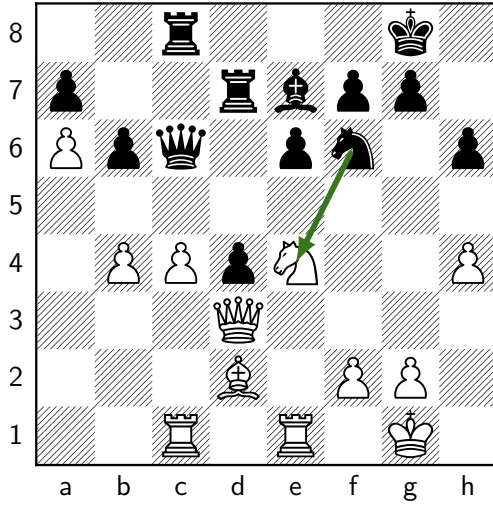
Figure 22: **Move 28, White: f7.** (a) Our position; Melanie plays f7 (f6f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#2678` (ply 44) [structural distance $d=93$, QAP obj=26]: the master played f3; under μ our *Pf6* plays the role of that game’s *Pf2*; cross-type roles: our *Nc1* as their *Ra1*, our *Qe4* as their *Re1*; transferred move f6f7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0020#377` (ply 47) [structural distance $d=97$, QAP obj=100]: the master played Nxe4; under μ our *Qe4* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *Pf3* as their *Be7*, our *qd7* as their *pa6*; transferred move e4f4. (d) Retrieved from `lichess_elite_2024-09_p0024#91` (ply 50) [structural distance $d=101$, QAP obj=53]: the master played Qxc5; under μ our *Qe4* plays the role of that game’s *Qf2*; cross-type roles: our *Pa4* as their *Bb1*, our *Bf1* as their *Re1*; transferred move e4f5. Synthesis: 1 retrieved analog transferred to f7 (summed QAP objective 26, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



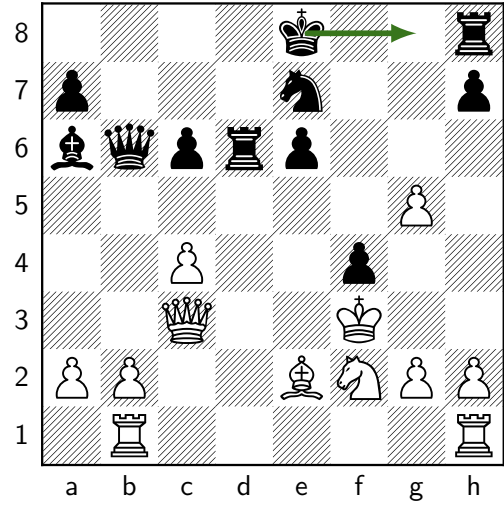
(a)



(b)

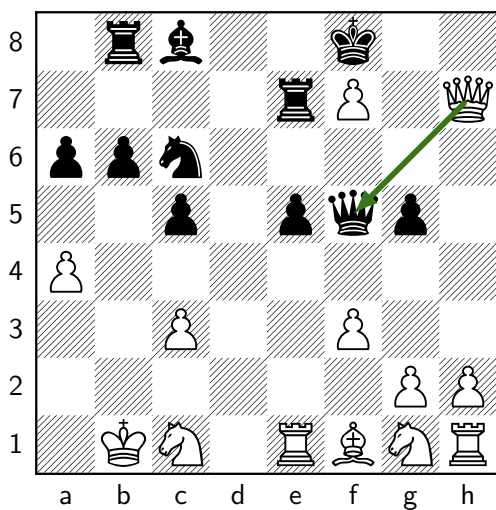


(c)

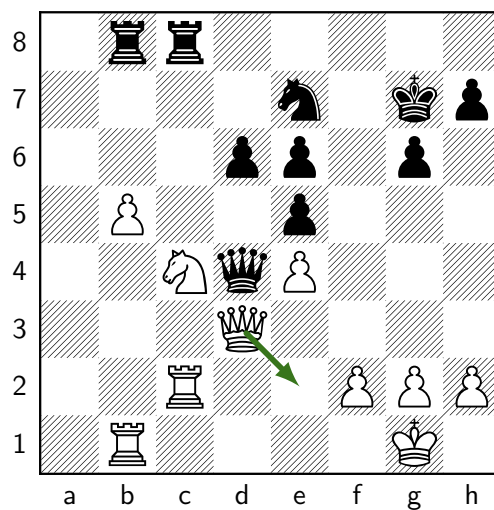


(d)

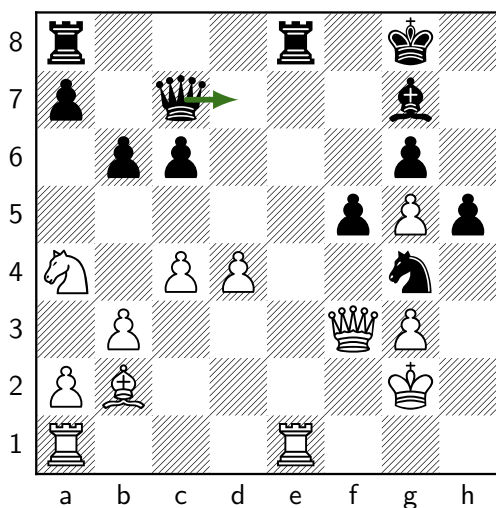
Figure 23: **Move 29, White: Qxh7.** (a) Our position; Melanie plays Qxh7 (e4h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#2497` (ply 67) [structural distance $d=103$, QAP obj=37]: the master played Qxa3; under μ our $Qe4$ plays the role of that game’s $Qa4$; cross-type roles: our $pa6$ as their $Rc8$, our $Pf3$ as their $Re8$; transferred move e4h7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0020#377` (ply 47) [structural distance $d=98$, QAP obj=100]: the master played Nxe4; under μ our $Qe4$ plays the role of that game’s $Nf3$ (a **cross-type** correspondence); cross-type roles: our $Pa4$ as their $Rd7$, our $Pc3$ as their $Be7$; transferred move e4f4. (d) Retrieved from `lichess_elite_2024-09_p0020#1571` (ply 43) [structural distance $d=104$, QAP obj=59]: the master played O-O; under μ our $Kb1$ plays the role of that game’s $Ke1$; cross-type roles: our $Pf7$ as their $Rh8$, our $Re1$ as their $Pa7$; transferred move b1a2. Synthesis: 1 retrieved analog transferred to Qxh7 (summed QAP objective 37, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



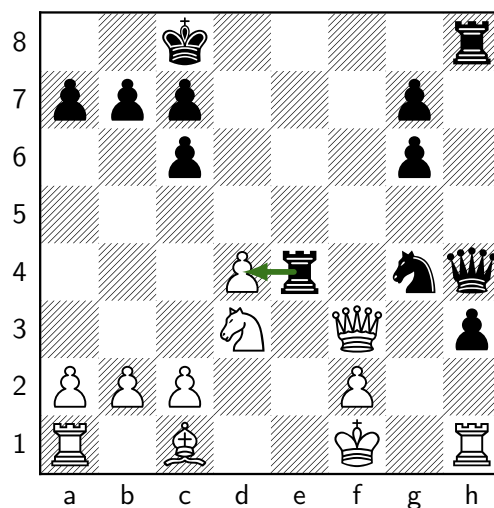
(a)



(b)

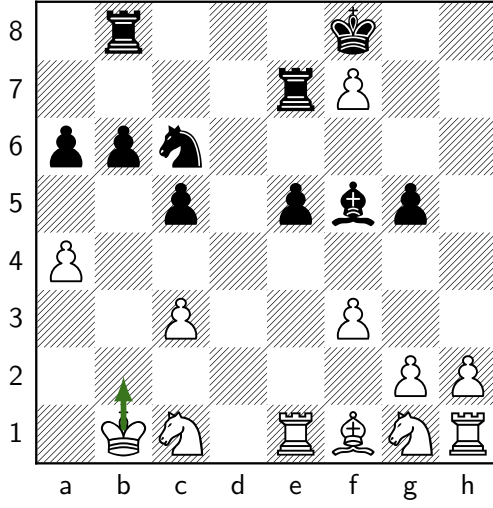


(c)

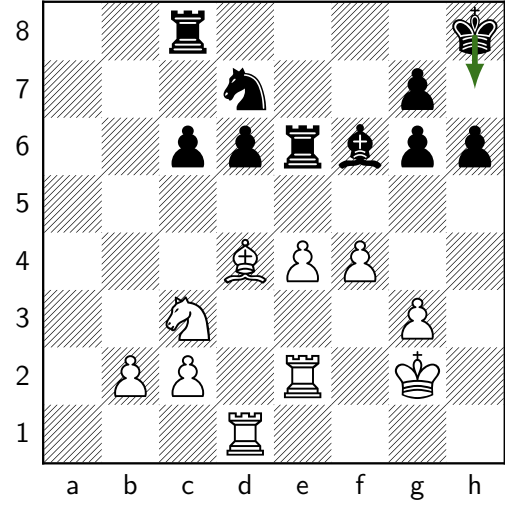


(d)

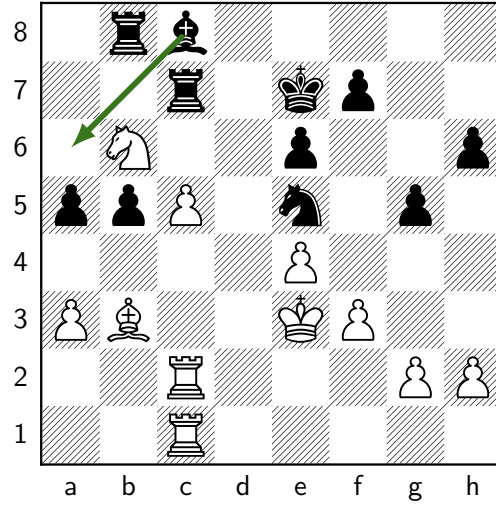
Figure 24: **Move 30, White: Qxf5**. (a) Our position; Melanie plays Qxf5 (h7f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#1057` (ply 58) [structural distance $d=112$, QAP obj=42]: the master played Qe2; under μ our Qh7 plays the role of that game’s Qd3; cross-type roles: our Nc1 as their Rb1, our Ng1 as their Rc2; transferred move h7f5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#3750` (ply 49) [structural distance $d=113$, QAP obj=18]: the master played Qd7; under μ our Qh7 plays the role of that game’s Qc2; cross-type roles: our Ng1 as their Ra8, our pb6 as their Re8; transferred move h7f5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#2613` (ply 43) [structural distance $d=113$, QAP obj=16]: the master played Rxd4; under μ our Qh7 plays the role of that game’s Re5 (a **cross-type** correspondence); cross-type roles: our Ng1 as their Rh8, our Nc1 as their Pa7; transferred move h7f5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qxf5 (summed QAP objective 76, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)



(b)



(c)

Figure 25: **Move 31, White: Kb2.** (a) Our position; Melanie plays Kb2 (b1b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite.2024-09_p0024#1196` (ply 61) [structural distance $d=84$, QAP obj=18]: the master played Kh7; under μ our *Kb1* plays the role of that game’s *Kh1*; cross-type roles: our *Nc1* as their *Rc8*, our *Pf3* as their *Nd7*; transferred move b1b2 (**endorses this move**). (c) Retrieved from `lichess_elite.2024-09_p0028#536` (ply 59) [structural distance $d=84$, QAP obj=15]: the master played Ba6; under μ our *Bf1* plays the role of that game’s *Bc1*; cross-type roles: our *Nc1* as their *Rb8*, our *Ng1* as their *Ph6*; transferred move f1d3. Synthesis: 1 retrieved analog transferred to Kb2 (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

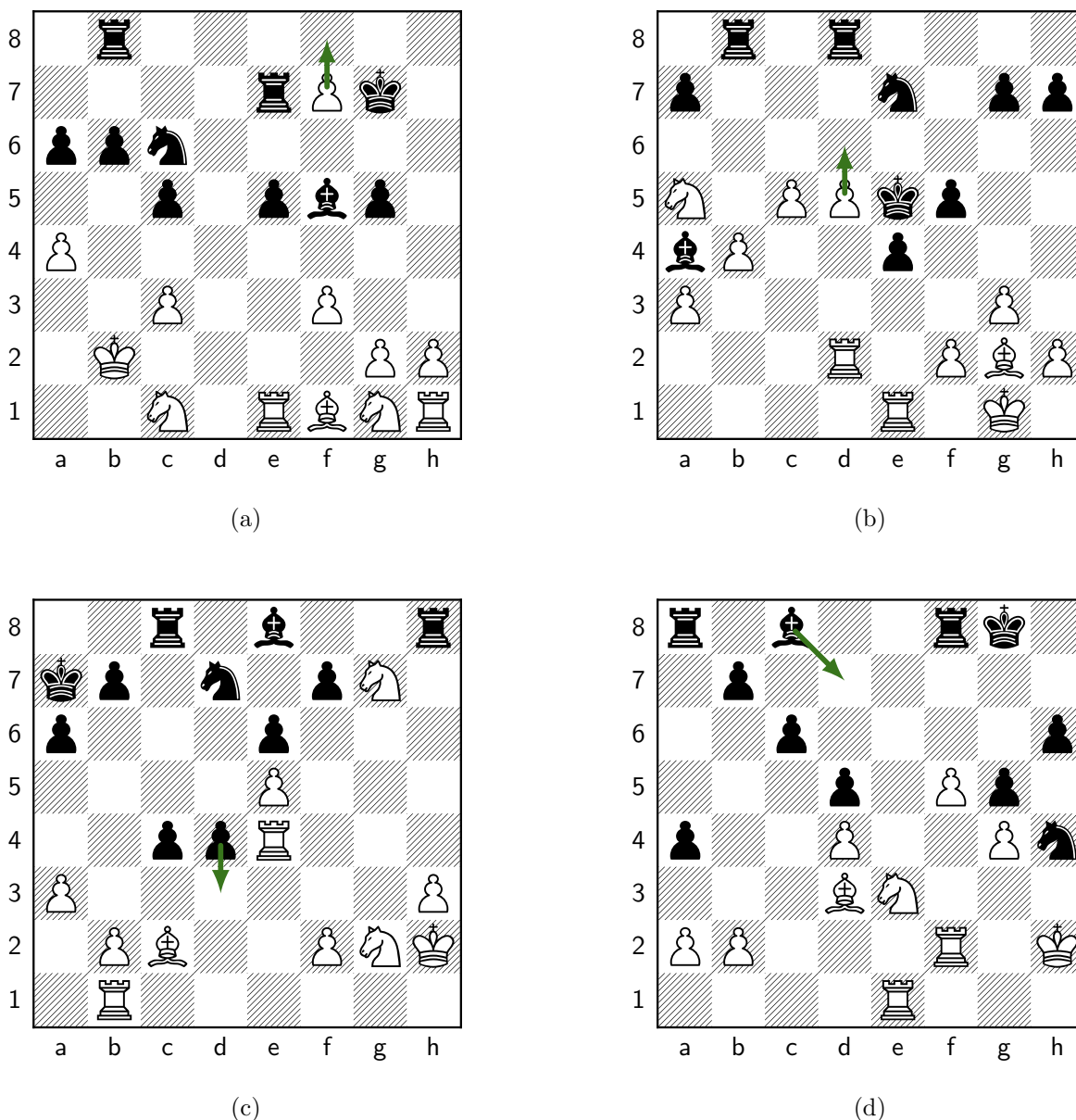
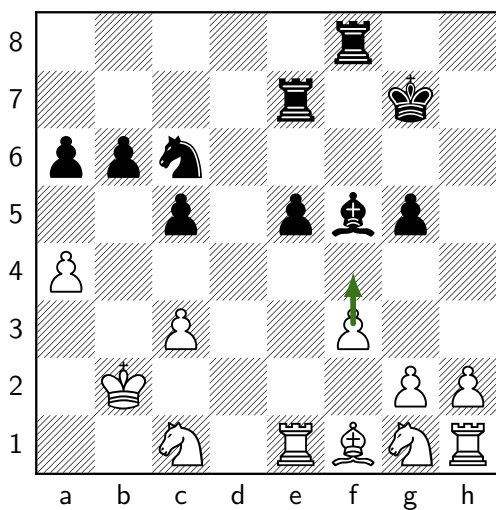
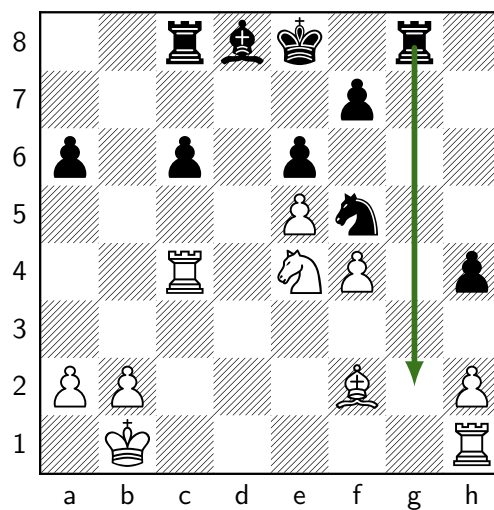


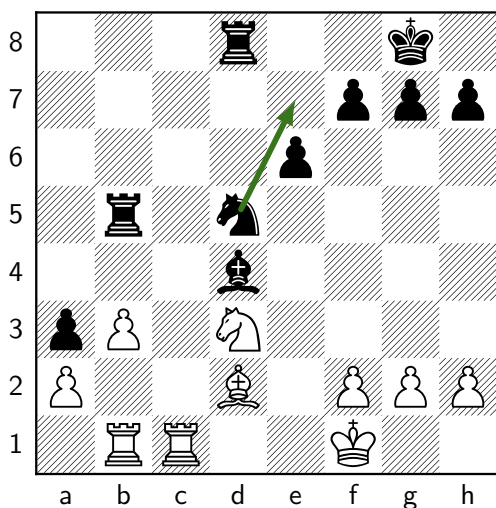
Figure 26: **Move 32, White: $f8=R$.** (a) Our position; Melanie plays $f8=R$ ($f7f8r$). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#238` (ply 68) [structural distance $d=86$, QAP obj=60]: the master played $d6$; under μ our $Pf7$ plays the role of that game’s $Pd5$; cross-type roles: our $Pc3$ as their $Rd2$, our $Re1$ as their $Bg2$; transferred move $f7f8r$ (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#491` (ply 73) [structural distance $d=84$, QAP obj=36]: the master played $d3$; under μ our $Pf7$ plays the role of that game’s $Pd5$; cross-type roles: our $Ng1$ as their $Rc8$, our $Pf7$ as their $Be8$; transferred move $f7f8r$ (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0016#3656` (ply 53) [structural distance $d=86$, QAP obj=39]: the master played $Bd7$; under μ our $Re1$ plays the role of that game’s $Bc1$ (a **cross-type** correspondence); cross-type roles: our $Nc1$ as their $Ra8$, our $Ng1$ as their $Pg5$; transferred move $e1e3$. Synthesis: 2 retrieved analogs transferred to $f8=R$ (summed QAP objective 96, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



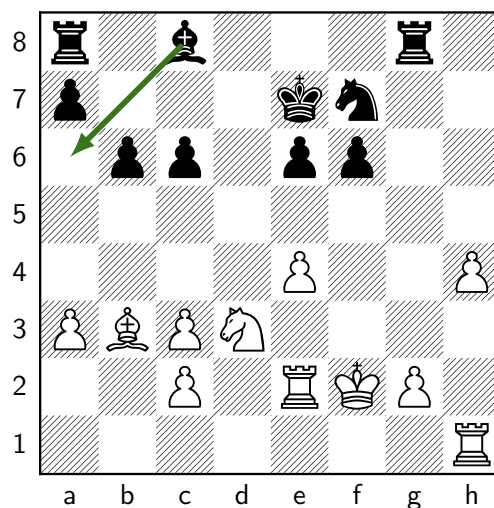
(a)



(b)



(c)



(d)

Figure 27: **Move 33, White: f4**. (a) Our position; Melanie plays f4 (f3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#406` (ply 49) [structural distance $d=59$, QAP obj=34]: the master played Rg2; under μ our Pf3 plays the role of that game’s Rg1 (a **cross-type** correspondence); cross-type roles: our nc6 as their Rc8, our Rh1 as their Pa6; transferred move f3f4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0025#3923` (ply 53) [structural distance $d=58$, QAP obj=33]: the master played Ne7; under μ our Pf3 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our nc6 as their Rd8, our Nc1 as their Ph7; transferred move f3f4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0026#1297` (ply 53) [structural distance $d=64$, QAP obj=41]: the master played Ba6; under μ our Pg2 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Pa4 as their Ra8, our Bf1 as their Rg8; transferred move g2g3. Synthesis: 2 retrieved analogs transferred to f4 (summed QAP objective 67, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

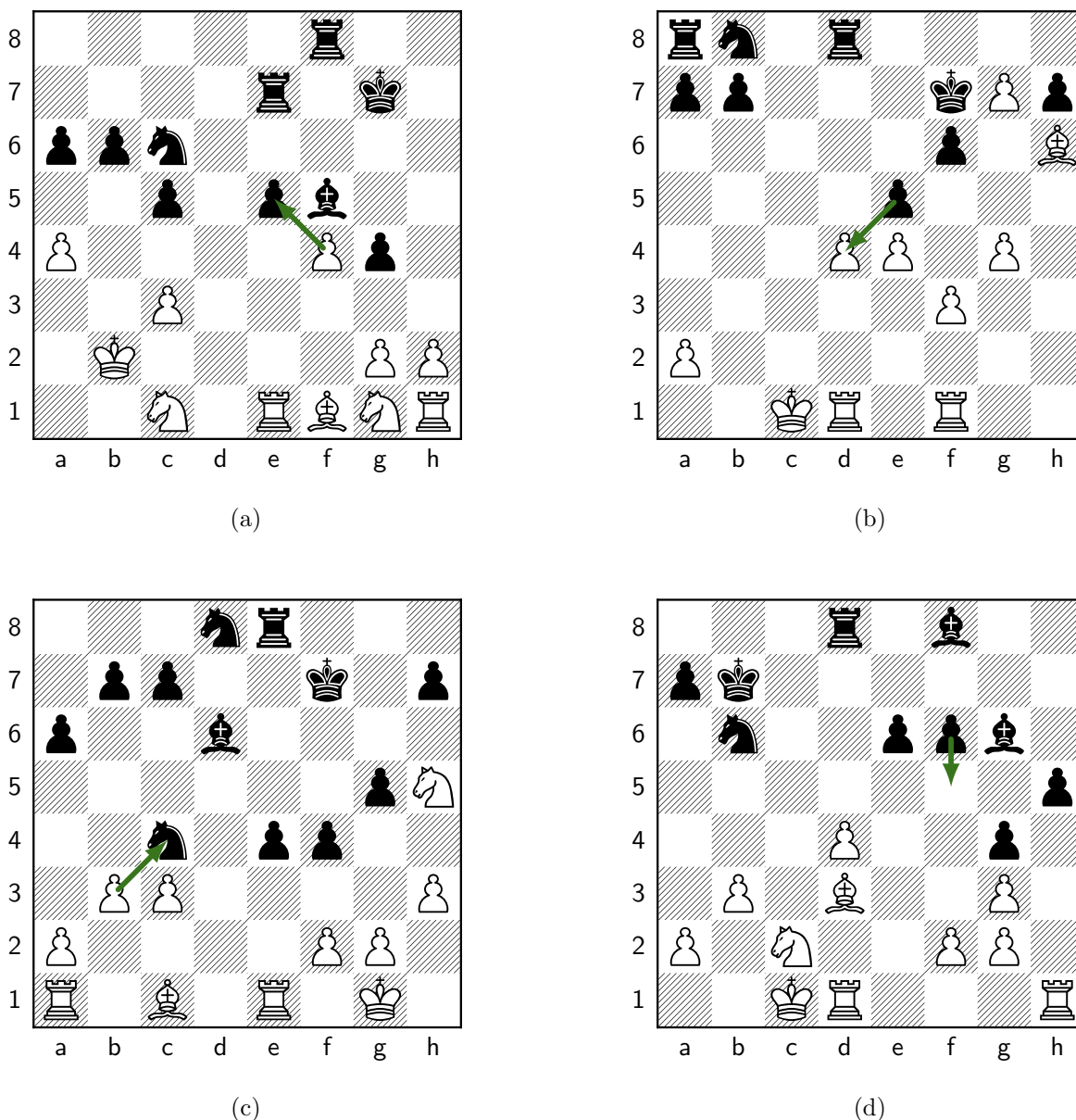
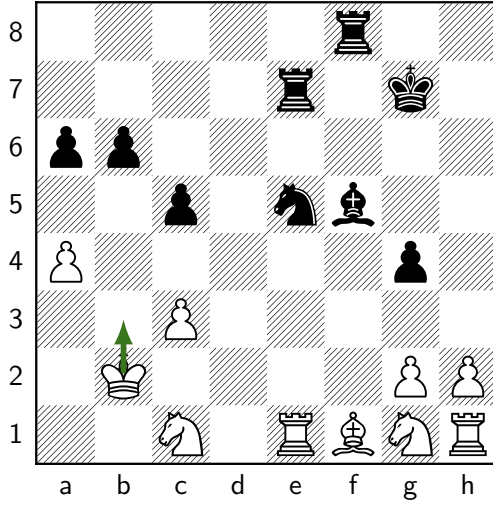
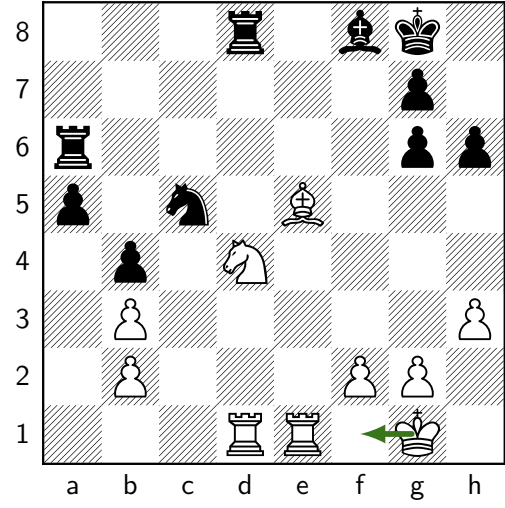


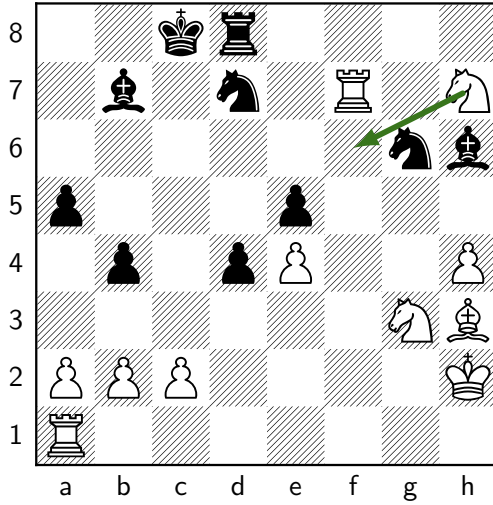
Figure 28: **Move 34, White: fxe5**. (a) Our position; Melanie plays **fxe5** (**f4e5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#3595` (ply 53) [structural distance $d=69$, QAP obj=97]: the master played **exd4**; under μ our *Pf4* plays the role of that game’s *Pe4*; cross-type roles: our *Pa4* as their *Ra8*, our *Bf1* as their *Nb8*; transferred move **f4e5** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#2253` (ply 50) [structural distance $d=74$, QAP obj=21]: the master played **bxc4**; under μ our *Pf4* plays the role of that game’s *Pb3*; cross-type roles: our *pa6* as their *Ra1*, our *Pc3* as their *Bc1*; transferred move **f4e5** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0025#3527` (ply 59) [structural distance $d=68$, QAP obj=65]: the master played **f5**; under μ our *Ng1* plays the role of that game’s *Pf3* (a **cross-type** correspondence); cross-type roles: our *pa6* as their *Nb6*, our *Nc1* as their *Pe6*; transferred move **g1f3**. Synthesis: 2 retrieved analogs transferred to **fxe5** (summed QAP objective 118, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



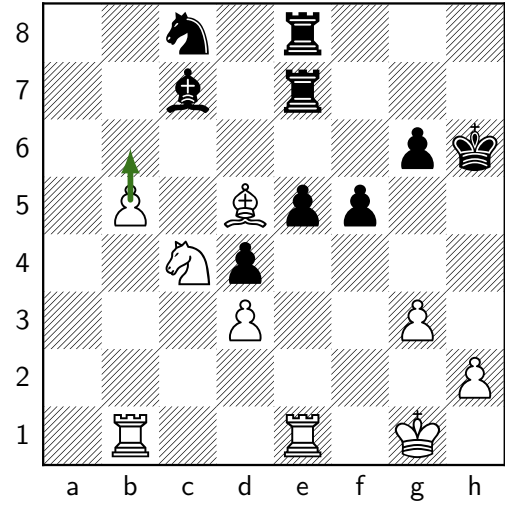
(a)



(b)

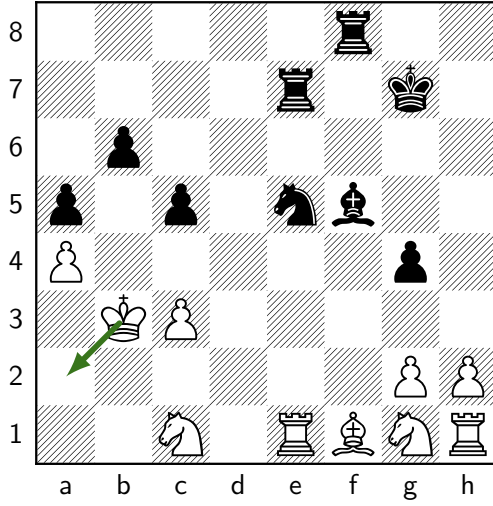


(c)

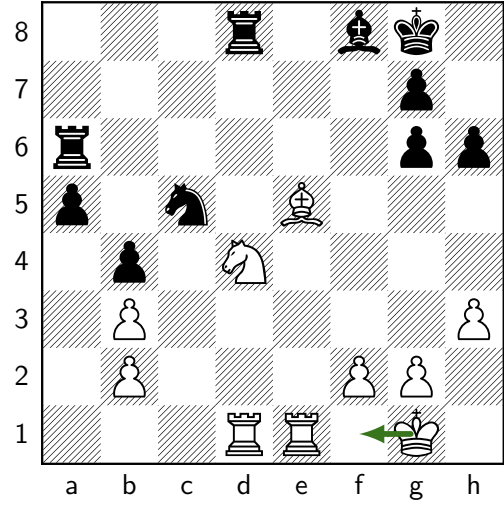


(d)

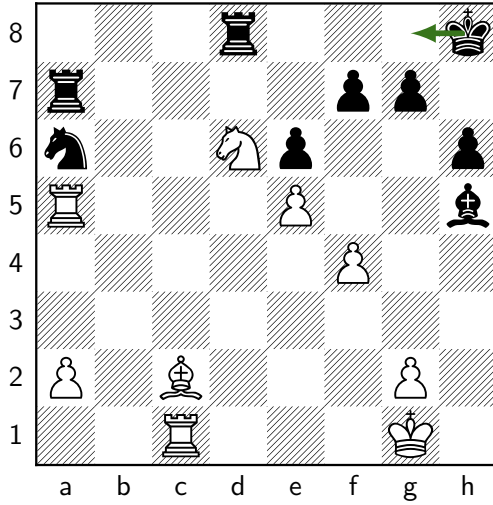
Figure 29: **Move 35, White: Kb3.** (a) Our position; Melanie plays Kb3 (b2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0023#3304` (ply 60) [structural distance $d=62$, QAP obj=50]: the master played Kf1; under μ our Kb2 plays the role of that game’s Kg1; cross-type roles: our bf5 as their Rd1, our Ng1 as their pb4; transferred move b2b3 (**endorses this move**). (c) Retrieved from `twic1500#277` (ply 62) [structural distance $d=66$, QAP obj=41]: the master played Nf6; under μ our Bf1 plays the role of that game’s Nh7 (a **cross-type** correspondence); cross-type roles: our Pc3 as their Ra1, our Nc1 as their Pa2; transferred move f1d3. (d) Retrieved from `lichess_elite_2024-09_p0003#1620` (ply 76) [structural distance $d=64$, QAP obj=30]: the master played b6; under μ our Pc3 plays the role of that game’s Pb5; cross-type roles: our Nc1 as their Rb1, our Bf1 as their Pg3; transferred move c3c4. Synthesis: 1 retrieved analog transferred to Kb3 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



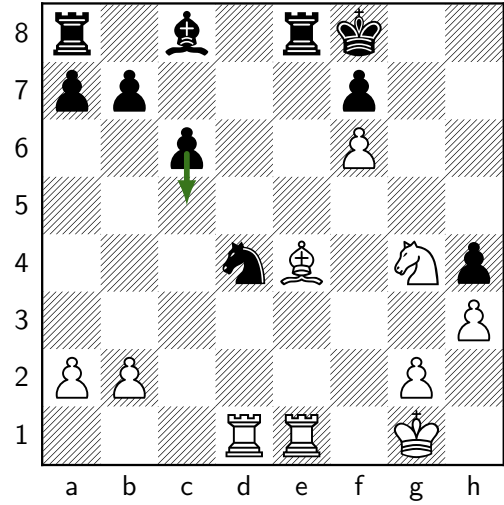
(a)



(b)

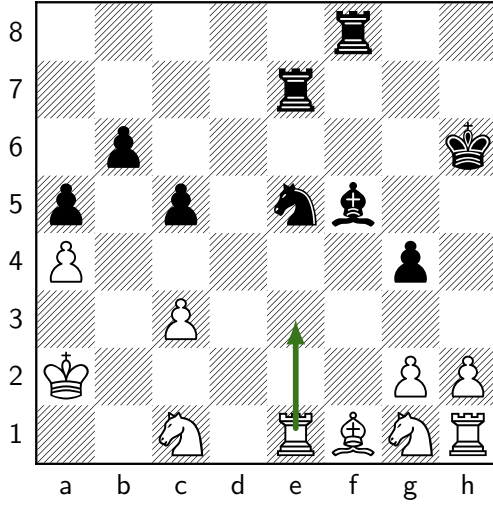


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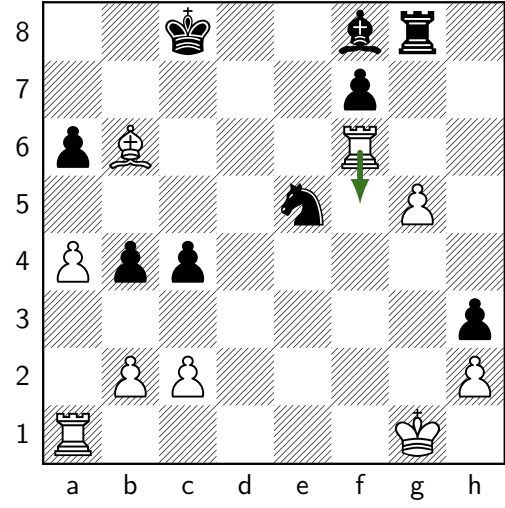


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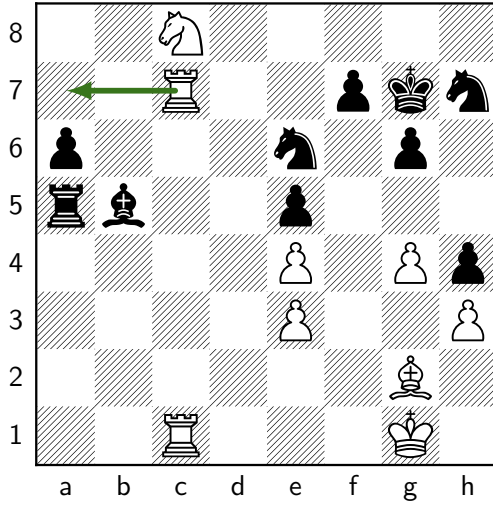
Figure 30: **Move 36, White: Ka2.** (a) Our position; Melanie plays Ka2 (**b3a2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0023#3304` (ply 60) [structural distance $d=68$, QAP obj=47]: the master played Kf1; under μ our *Kb3* plays the role of that game’s *Kg1*; cross-type roles: our *Nc1* as their *Re1*, our *Ng1* as their *Pb2*; transferred move **b3a2** (**endorses this move**). (c) Retrieved from `twic1500#400` (ply 65) [structural distance $d=71$, QAP obj=14]: the master played Kg8; under μ our *Kb3* plays the role of that game’s *Kh1*; cross-type roles: our *Nc1* as their *Rd8*, our *Ng1* as their *Ra7*; transferred move **b3a2** (**endorses this move**). (d) Retrieved from `twic1500#262` (ply 59) [structural distance $d=53$, QAP obj=40]: the master played c5; under μ our *Pc3* plays the role of that game’s *Pc3*; cross-type roles: our *Nc1* as their *Ra8*, our *Re1* as their *Bc8*; transferred move **c3c4**. Synthesis: 2 retrieved analogs transferred to Ka2 (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



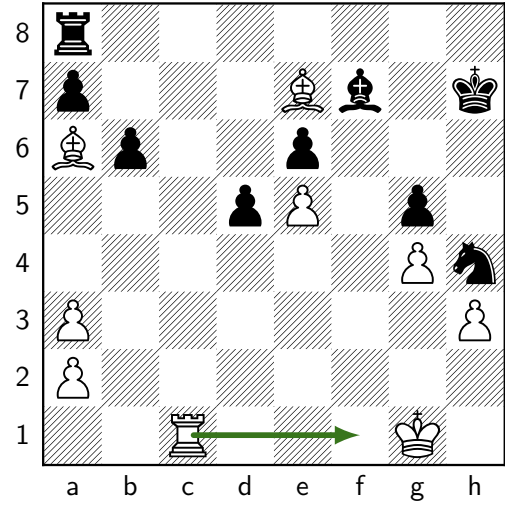
(a)



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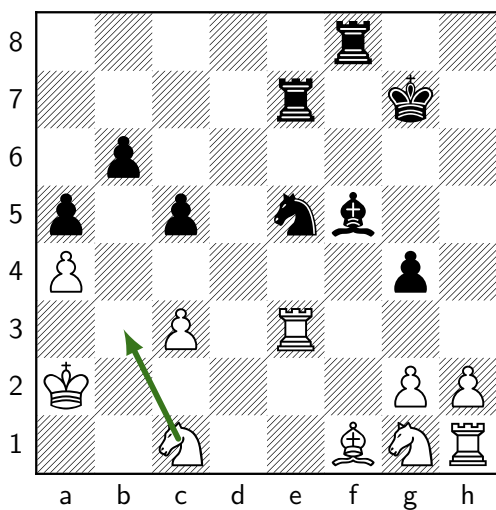


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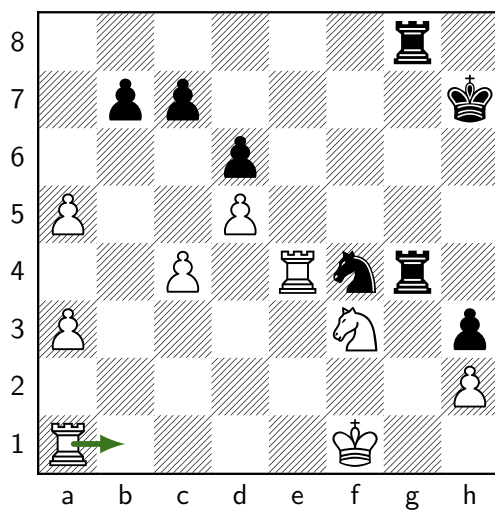


(d)

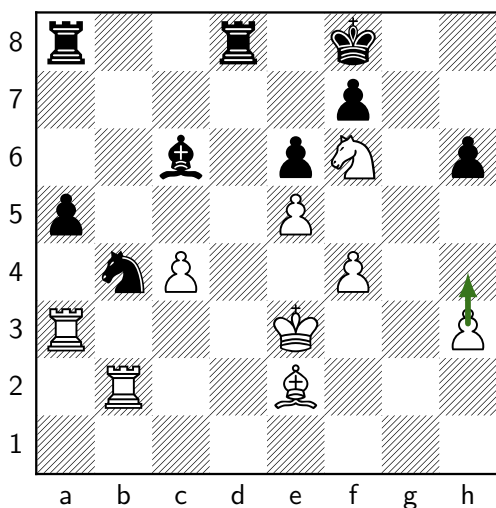
Figure 31: **Move 37, White: Re3**. (a) Our position; Melanie plays Re3 (e1e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#2916` (ply 66) [structural distance $d=63$, QAP obj=36]: the master played Rf5; under μ our Re1 plays the role of that game’s Rf6; cross-type roles: our pg4 as their Ra1, our Nc1 as their Pa4; transferred move e1e3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#701` (ply 84) [structural distance $d=64$, QAP obj=13]: the master played Ra7; under μ our Re1 plays the role of that game’s Rc7; cross-type roles: our Nc1 as their Rc1, our Pg2 as their Bg2; transferred move e1e3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0031#1629` (ply 58) [structural distance $d=70$, QAP obj=29]: the master played Rf1; under μ our Nc1 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Pa3, our Ng1 as their Ph3; transferred move c1b3. Synthesis: 2 retrieved analogs transferred to Re3 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



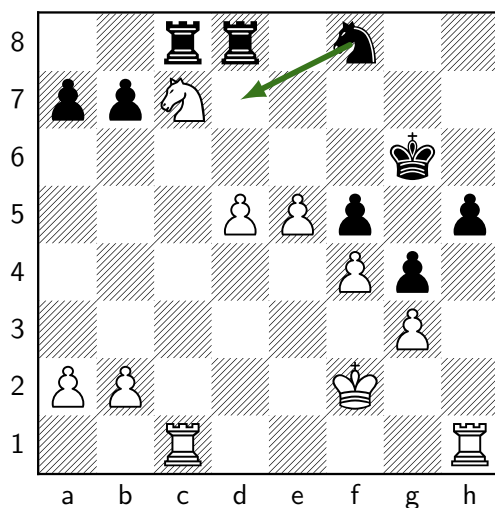
(a)



(b)

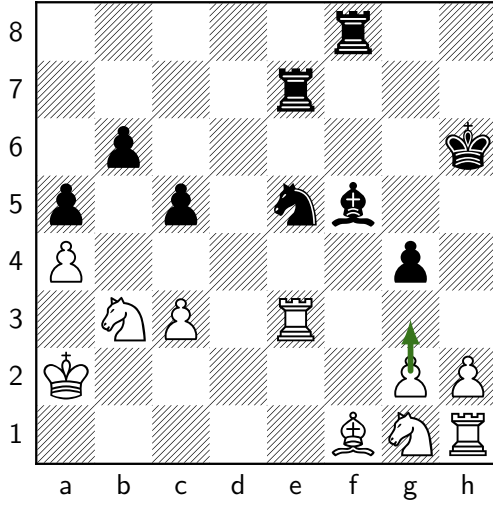


(c)

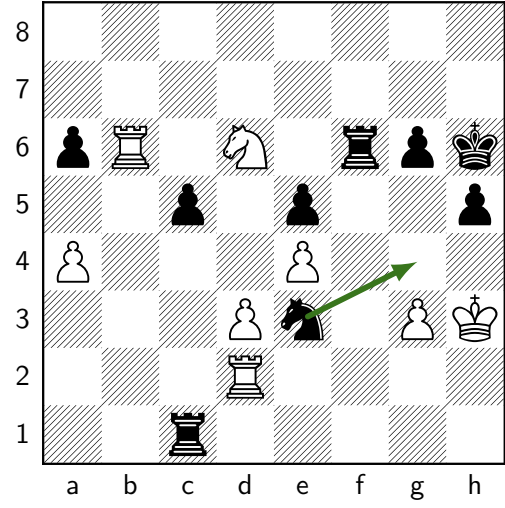


(d)

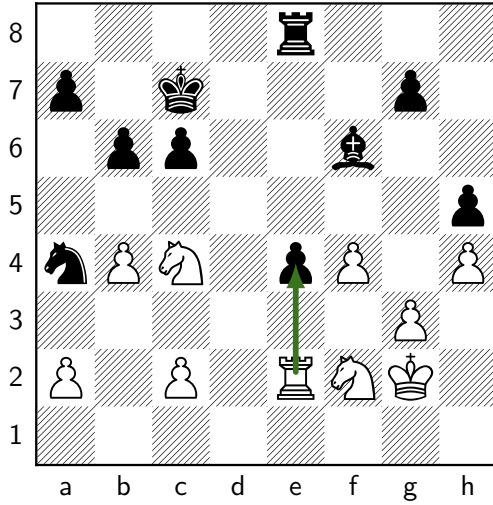
Figure 32: **Move 38, White: Nb3.** (a) Our position; Melanie plays Nb3 (c1b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0012#882` (ply 64) [structural distance $d=53$, QAP obj=30]: the master played Rb1; under μ our Nc1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Ng1 as their Pa3, our Pg2 as their Nf3; transferred move c1b3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#2972` (ply 88) [structural distance $d=62$, QAP obj=28]: the master played h4; under μ our Nc1 plays the role of that game’s Ph3 (a **cross-type** correspondence); cross-type roles: our Pc3 as their Be2, our Bf1 as their Pf4; transferred move c1b3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0028#3188` (ply 59) [structural distance $d=64$, QAP obj=28]: the master played Nd7; under μ our Nc1 plays the role of that game’s Nf1; cross-type roles: our Pc3 as their Rd8, our Ng1 as their Pa7; transferred move c1b3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nb3 (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



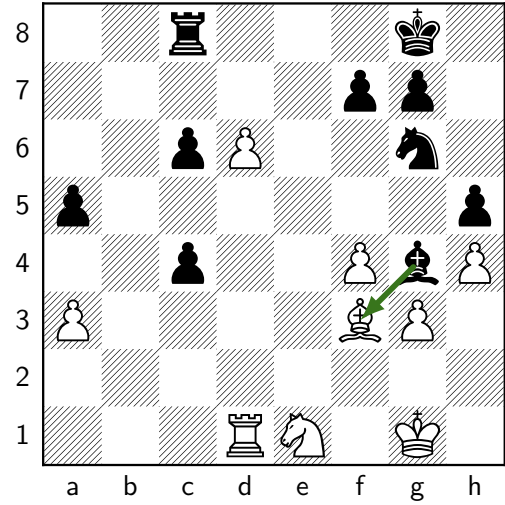
(a)



(b)

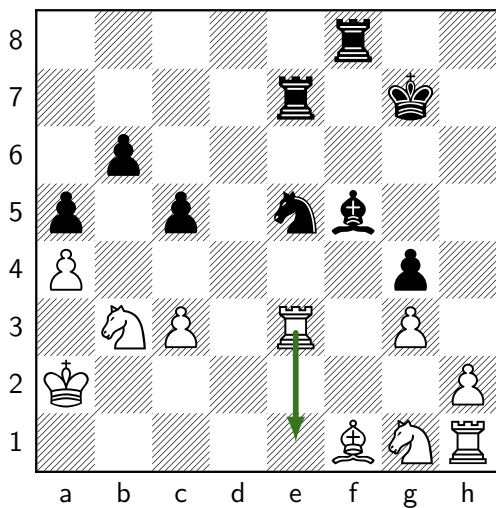


(c)

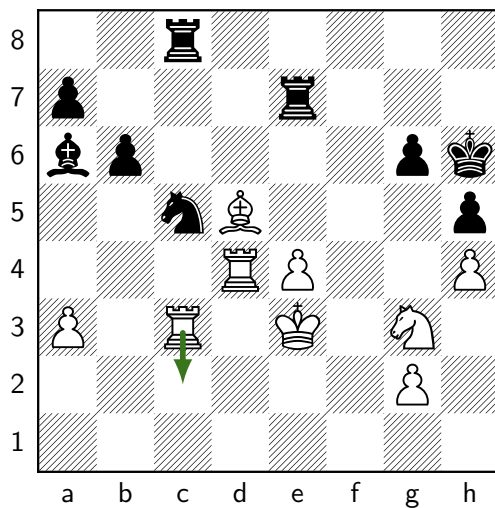


(d)

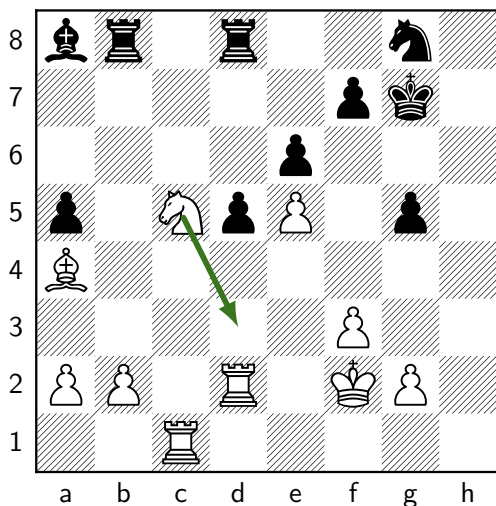
Figure 33: **Move 39, White: g3**. (a) Our position; Melanie plays g3 (g2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#1829` (ply 77) [structural distance $d=69$, QAP obj=54]: the master played Ng4; under μ our *Pg2* plays the role of that game’s *Ne6* (a **cross-type** correspondence); cross-type roles: our *Nb3* as their *rb6*, our *Ng1* as their *Pe5*; transferred move g2g3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#2156` (ply 74) [structural distance $d=68$, QAP obj=27]: the master played Rxe4; under μ our *Re3* plays the role of that game’s *Re2*; cross-type roles: our *Bf1* as their *Pa2*, our *Ng1* as their *Pc2*; transferred move e3e5. (d) Retrieved from `lichess_elite_2024-09_p0020#61` (ply 65) [structural distance $d=69$, QAP obj=25]: the master played Bxf3; under μ our *Re3* plays the role of that game’s *Bg5* (a **cross-type** correspondence); cross-type roles: our *Pc3* as their *Rc8*, our *bf5* as their *Pf7*; transferred move e3e5. Synthesis: 1 retrieved analog transferred to g3 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



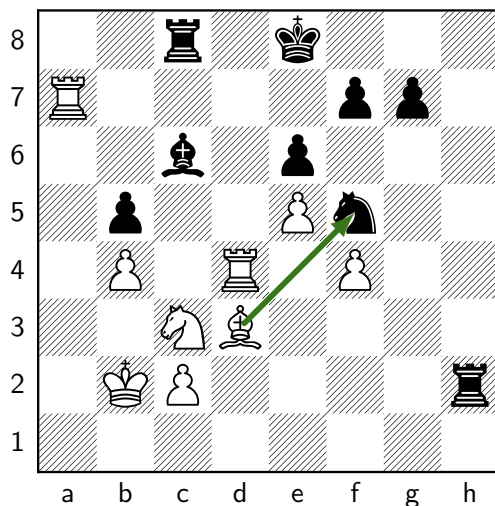
(a)



(b)

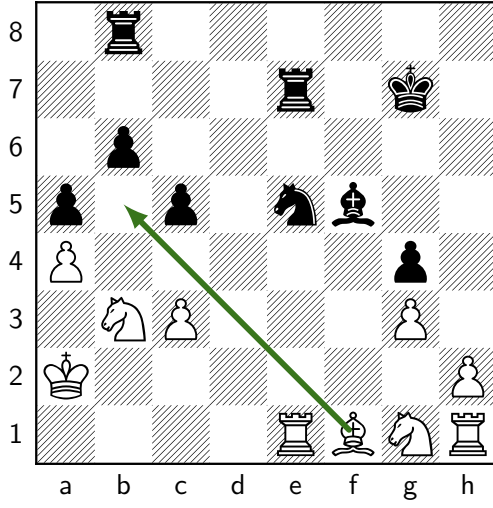


(c)

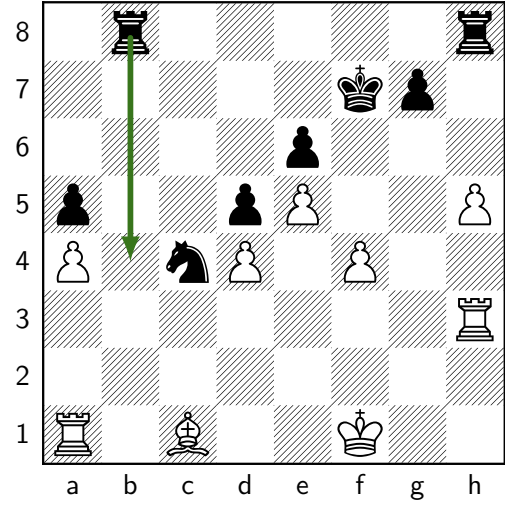


(d)

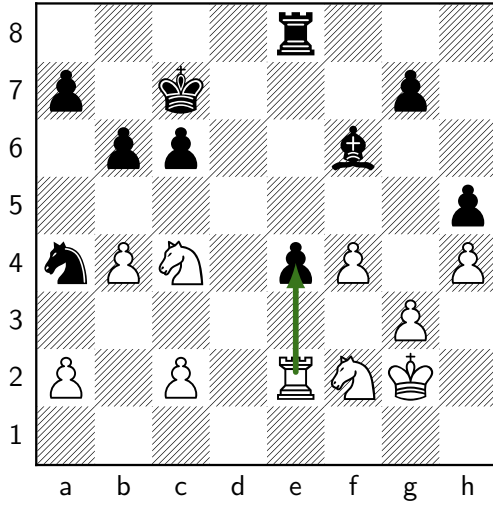
Figure 34: **Move 40, White: Re1**. (a) Our position; Melanie plays Re1 (e3e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#1982` (ply 92) [structural distance $d=68$, QAP obj=52]: the master played Rc2; under μ our Re3 plays the role of that game’s Rc3; cross-type roles: our Bf1 as their Pg2, our Ng1 as their Rd4; transferred move e3e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#3944` (ply 66) [structural distance $d=68$, QAP obj=20]: the master played Nd3; under μ our Nb3 plays the role of that game’s Nc5; cross-type roles: our Bf1 as their Rc1, our Ng1 as their Pf3; transferred move b3c1. (d) Retrieved from `lichess_elite_2024-09_p0022#1246` (ply 62) [structural distance $d=66$, QAP obj=18]: the master played Bxf5; under μ our Re3 plays the role of that game’s Bd3 (a **cross-type** correspondence); cross-type roles: our bf5 as their rh2, our Ng1 as their Pb4; transferred move e3e5. Synthesis: 1 retrieved analog transferred to Re1 (summed QAP objective 52, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



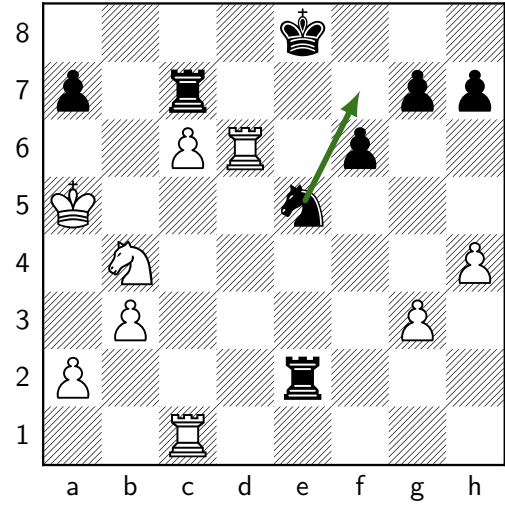
(a)



(b)

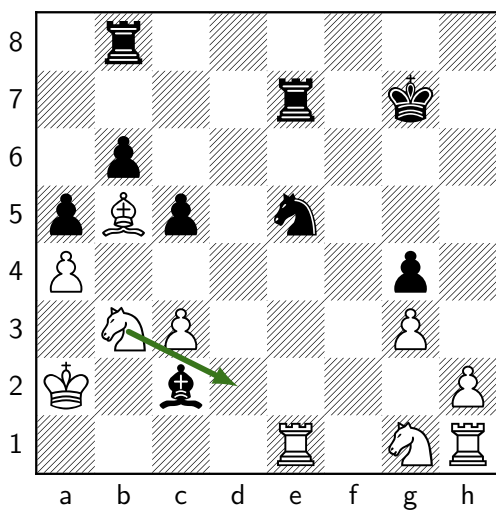


(c)

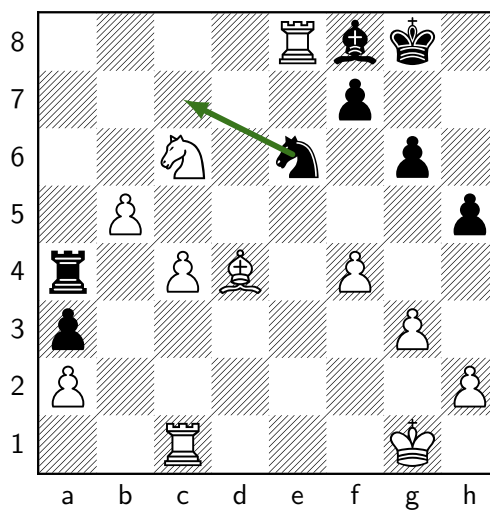


(d)

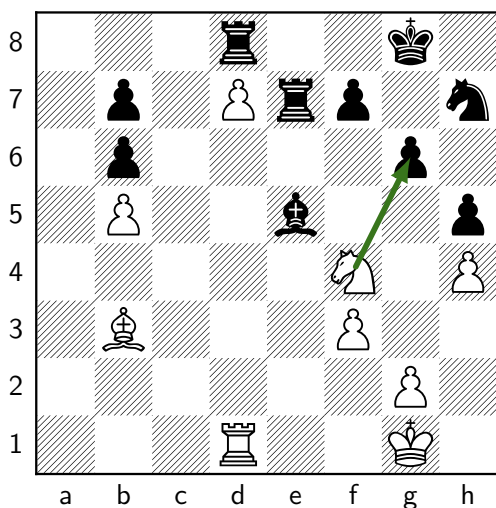
Figure 35: **Move 41, White: Bb5**. (a) Our position; Melanie plays Bb5 (f1b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#102` (ply 67) [structural distance $d=64$, QAP obj=39]: the master played Rb4; under μ our Bf1 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our ne5 as their ph5, our Ng1 as their pa4; transferred move f1b5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#2156` (ply 74) [structural distance $d=65$, QAP obj=51]: the master played Rxe4; under μ our Re1 plays the role of that game’s Re2; cross-type roles: our Bf1 as their Pa2, our bf5 as their Pc2; transferred move e1e5. (d) Retrieved from `lichess_elite_2025-11_p0000#1560` (ply 81) [structural distance $d=68$, QAP obj=46]: the master played Nf7; under μ our Re1 plays the role of that game’s Ne4 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Pa7, our Rh1 as their Ph7; transferred move e1a1. Synthesis: 1 retrieved analog transferred to Bb5 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



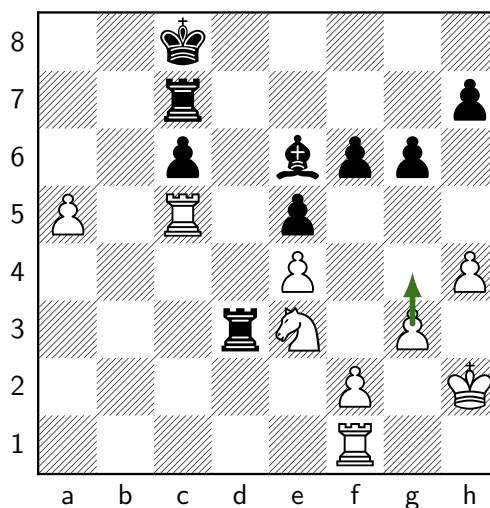
(a)



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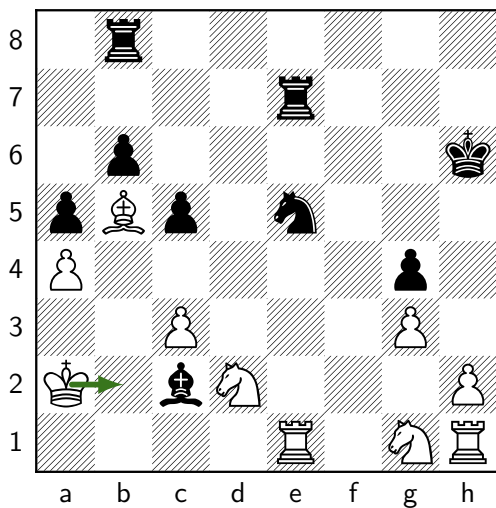


(c)

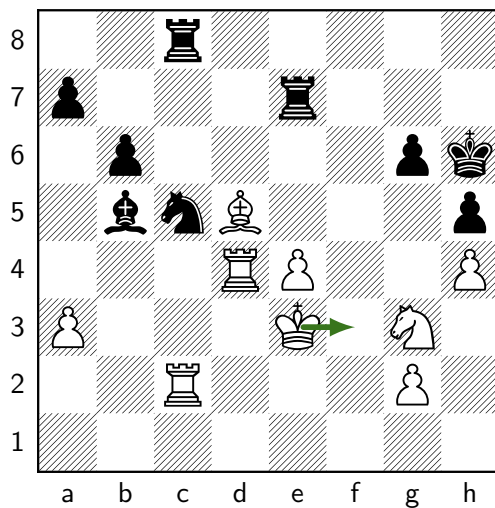


(d)

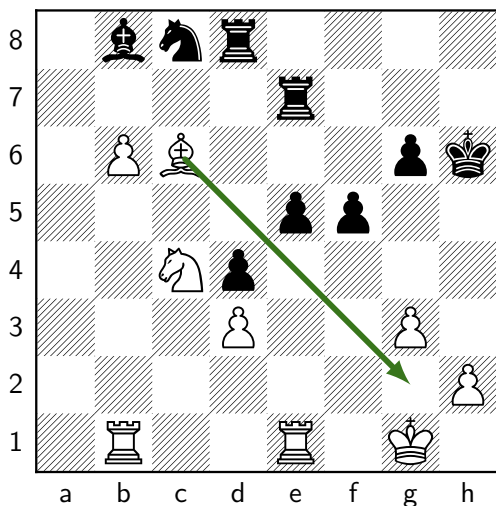
Figure 36: **Move 42, White: Nd2.** (a) Our position; Melanie plays Nd2 (b3d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#952` (ply 67) [structural distance $d=73$, QAP obj=44]: the master played Nc7; under μ our Nb3 plays the role of that game’s Ne3; cross-type roles: our bc2 as their re8, our Nb3 as their Bf8; transferred move b3d2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#3169` (ply 68) [structural distance $d=73$, QAP obj=65]: the master played Nxg6; under μ our Nb3 plays the role of that game’s Nf4; cross-type roles: our rb8 as their Pg2, our Ng1 as their pb7; transferred move b3a5. (d) Retrieved from `lichess_elite_2024-09_p0022#647` (ply 70) [structural distance $d=71$, QAP obj=41]: the master played g4; under μ our Bb5 plays the role of that game’s Pg3 (a **cross-type** correspondence); cross-type roles: our Ng1 as their Rf1, our bc2 as their rd3; transferred move b5f1. Synthesis: 1 retrieved analog transferred to Nd2 (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



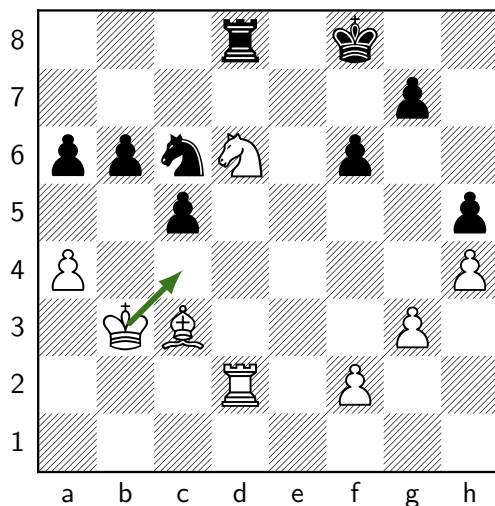
(a)



(b)

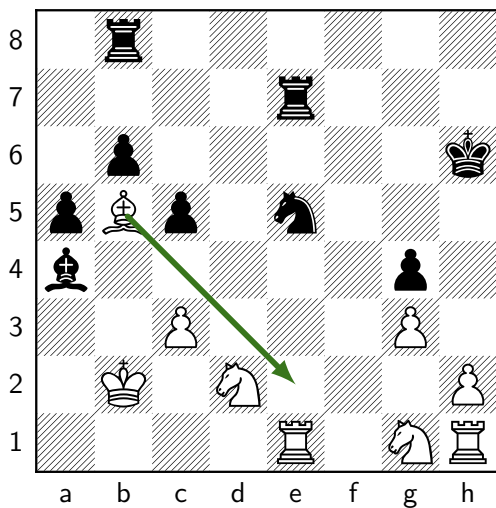


(c)

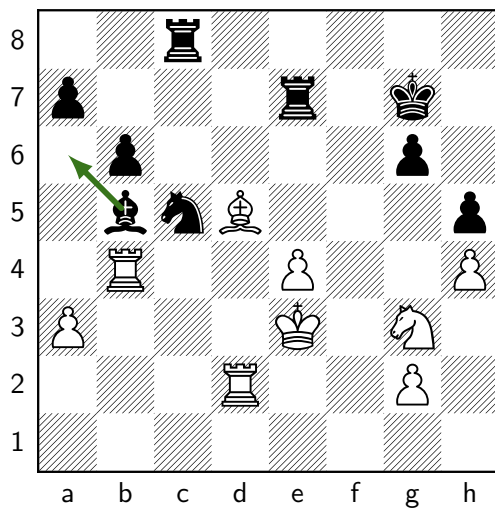


(d)

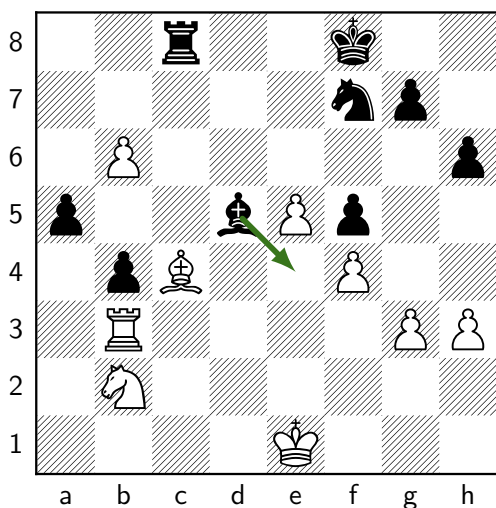
Figure 37: **Move 43, White: Kb2**. (a) Our position; Melanie plays Kb2 (a2b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#1982` (ply 94) [structural distance $d=67$, QAP obj=46]: the master played Kf3; under μ our Ka2 plays the role of that game’s Ke3; cross-type roles: our Rh1 as their Pg2, our Nd2 as their Pa3; transferred move a2b2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#1620` (ply 80) [structural distance $d=71$, QAP obj=42]: the master played Bg2; under μ our Bb5 plays the role of that game’s Bc6; cross-type roles: our Ng1 as their Rb1, our Nd2 as their Pd3; transferred move b5f1. (d) Retrieved from `lichess_elite_2024-09_p0025#352` (ply 72) [structural distance $d=65$, QAP obj=32]: the master played Kc4; under μ our Ka2 plays the role of that game’s Kb3; cross-type roles: our Nd2 as their pc5, our Rh1 as their nc6; transferred move a2a3. Synthesis: 2 retrieved analogs transferred to Kb2 (summed QAP objective 92, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



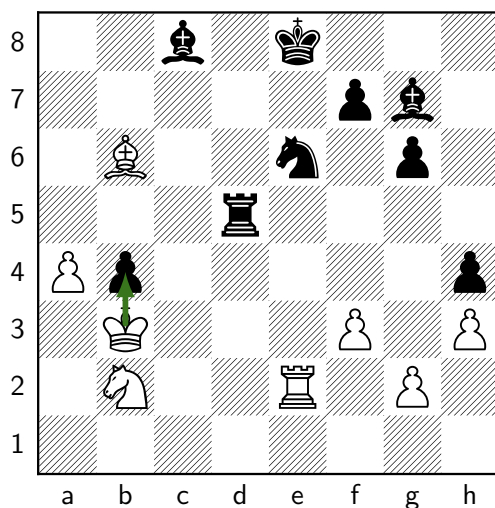
(a)



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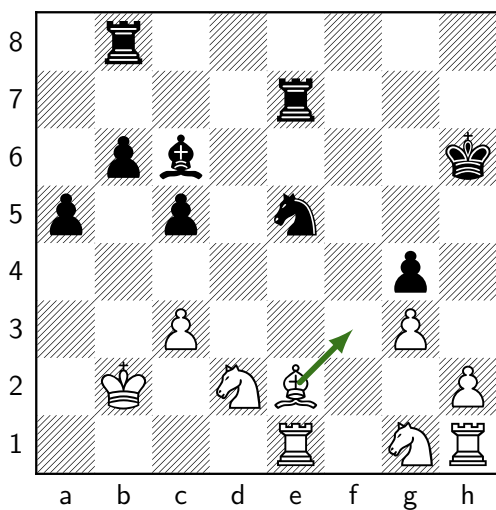


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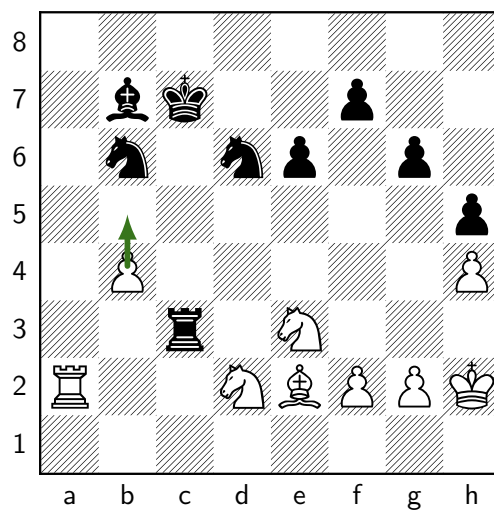


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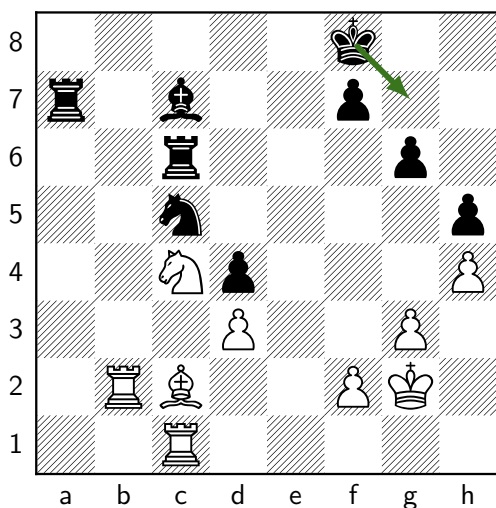
Figure 38: **Move 44, White: Be2.** (a) Our position; Melanie plays **Be2** (**b5e2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#1982` (ply 85) [structural distance $d=62$, QAP obj=33]: the master played **Ba6**; under μ our *Bb5* plays the role of that game’s *Bb4*; cross-type roles: our *Pc3* as their *Rc8*, our *Ng1* as their *Pa7*; transferred move **b5e2** (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#3305` (ply 87) [structural distance $d=61$, QAP obj=43]: the master played **Be4**; under μ our *Bb5* plays the role of that game’s *Bd4*; cross-type roles: our *Bb5* as their *Rc8*, our *re7* as their *Pa5*; transferred move **b5c6**. (d) Retrieved from `lichess_elite_2024-09_p0020#1181` (ply 88) [structural distance $d=64$, QAP obj=36]: the master played **Kxb4**; under μ our *Kb2* plays the role of that game’s *Kb3*; cross-type roles: our *ba4* as their *Ph3*, our *rb8* as their *pb4*; transferred move **b2a2**. Synthesis: 1 retrieved analog transferred to **Be2** (summed QAP objective 33, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



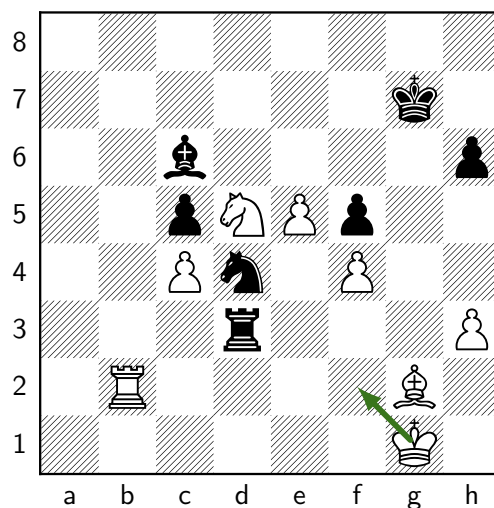
(a)



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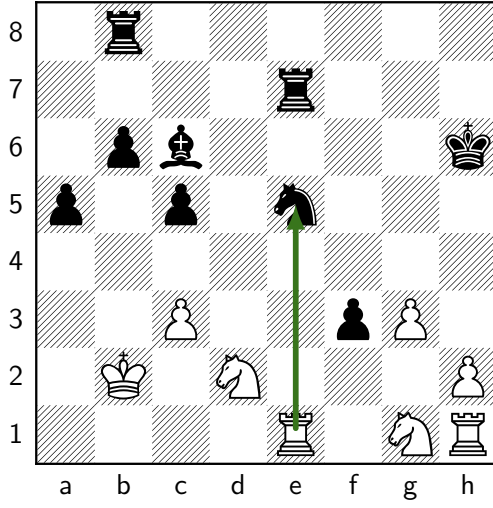


(c)

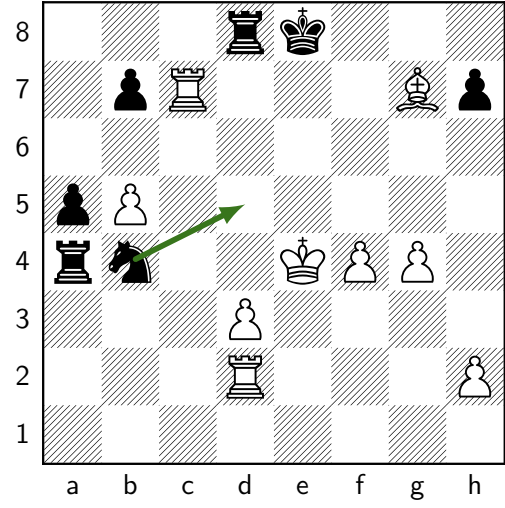


(d)

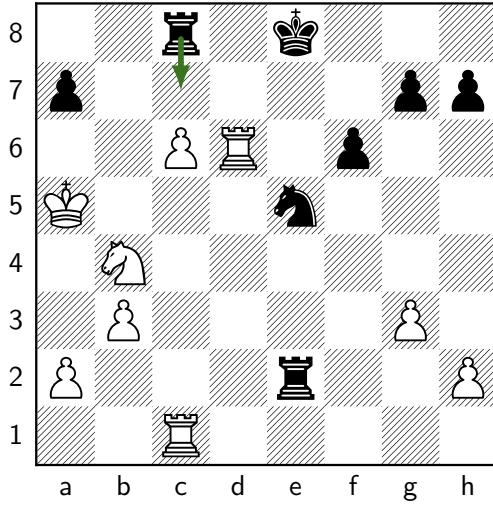
Figure 39: **Move 45, White: Bf3**. (a) Our position; Melanie plays Bf3 (e2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0020#2360` (ply 76) [structural distance $d=75$, QAP obj=21]: the master played b5; under μ our *Pc3* plays the role of that game’s *Pb4*; cross-type roles: our *Ng1* as their *Pg2*, our *bc6* as their *rc3*; transferred move c3c4. (c) Retrieved from `lichess_elite_2024-09_p0028#3916` (ply 85) [structural distance $d=36$, QAP obj=19]: the master played Kg7; under μ our *Kb2* plays the role of that game’s *Kf1*; cross-type roles: our *pa5* as their *Ra7*, our *bc6* as their *Nc5*; transferred move b2c2. (d) Retrieved from `lichess_elite_2024-09_p0022#2332` (ply 78) [structural distance $d=72$, QAP obj=14]: the master played Kf2; under μ our *Kb2* plays the role of that game’s *Kg1*; cross-type roles: our *Be2* as their *Rb2*, our *Pc3* as their *Bg2*; transferred move b2b3. Synthesis: no retrieved analog transferred to Bf3 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



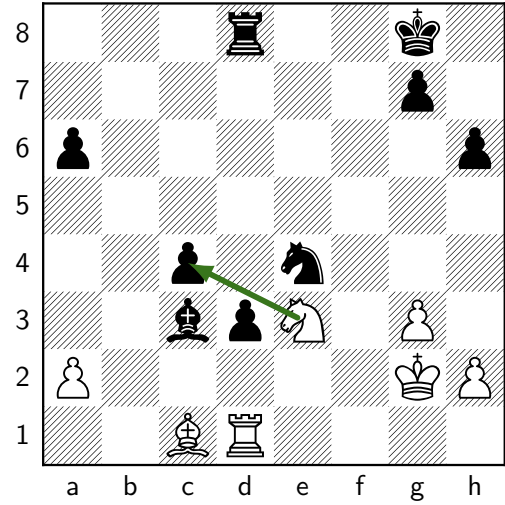
(a)



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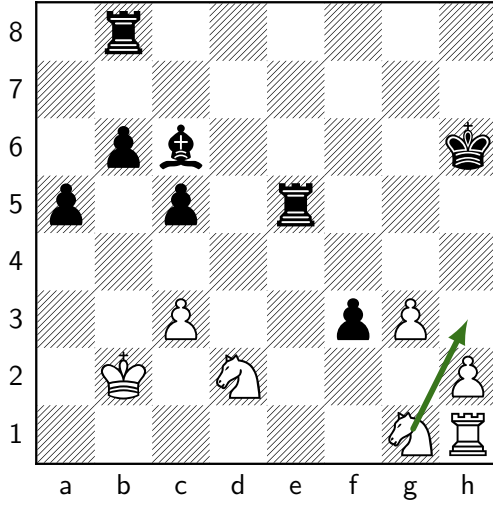


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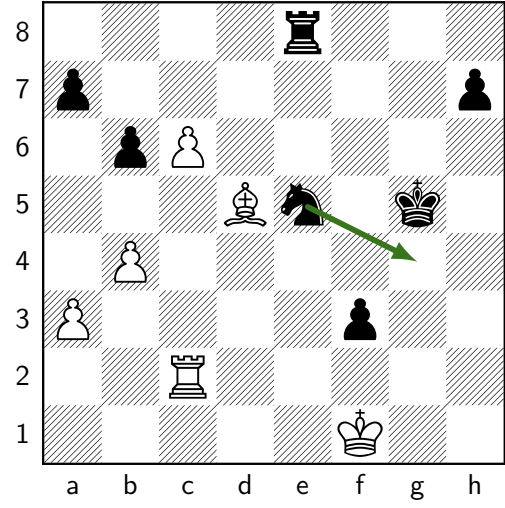


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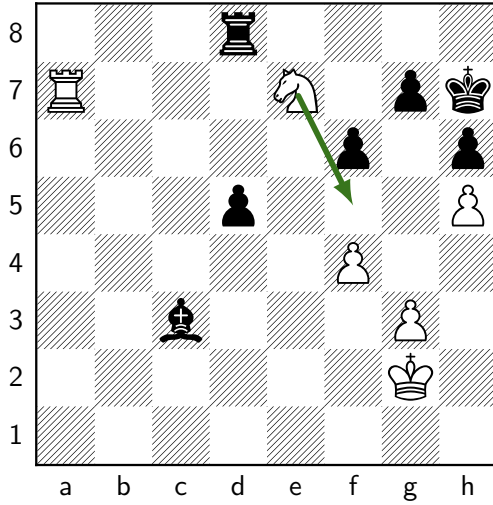
Figure 40: **Move 46, White: Rxe5**. (a) Our position; Melanie plays Rxe5 (e1e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#2483` (ply 79) [structural distance $d=59$, QAP obj=24]: the master played Nd5+; under μ our *Re1* plays the role of that game’s *Nb5* (a **cross-type** correspondence); cross-type roles: our *Ng1* as their *rc7*, our *pc5* as their *bg7*; transferred move e1e5 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#1560` (ply 79) [structural distance $d=57$, QAP obj=51]: the master played Rc7; under μ our *Re1* plays the role of that game’s *Rc1*; cross-type roles: our *Rh1* as their *Ph7*, our *ne5* as their *pc6*; transferred move e1d1. (d) Retrieved from `lichess_elite_2024-09_p0022#3677` (ply 86) [structural distance $d=58$, QAP obj=47]: the master played Nxc4; under μ our *Nd2* plays the role of that game’s *Ne3*; cross-type roles: our *Ph2* as their *Bc1*, our *re7* as their *bc3*; transferred move d2f1. Synthesis: 1 retrieved analog transferred to Rxe5 (summed QAP objective 24, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



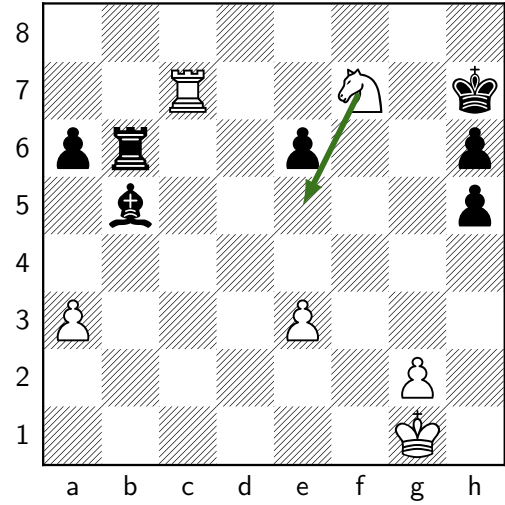
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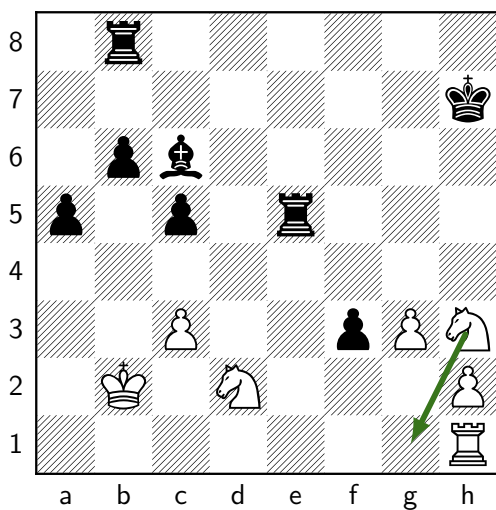


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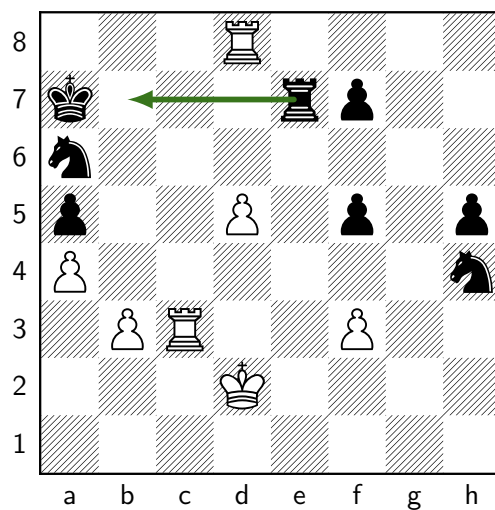


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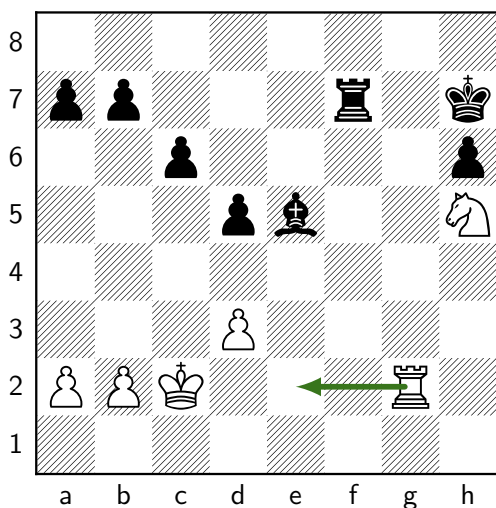
Figure 41: **Move 47, White: Nh3**. (a) Our position; Melanie plays Nh3 (g1h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#3371` (ply 85) [structural distance $d=38$, QAP obj=35]: the master played Ng4; under μ our Ng1 plays the role of that game’s Ne4; cross-type roles: our Nd2 as their bd5; transferred move g1h3 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#835` (ply 92) [structural distance $d=35$, QAP obj=22]: the master played Nf5; under μ our Ng1 plays the role of that game’s Ne7; cross-type roles: our Nd2 as their Pf4; transferred move g1h3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0010#567` (ply 80) [structural distance $d=38$, QAP obj=21]: the master played Ne5+; under μ our Ng1 plays the role of that game’s Nf7; cross-type roles: our pa5 as their bb5, our Nd2 as their ph5; transferred move g1h3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nh3 (summed QAP objective 122, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



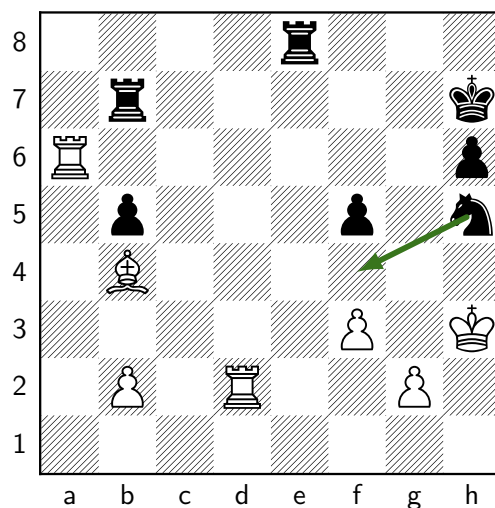
(a)



(b)

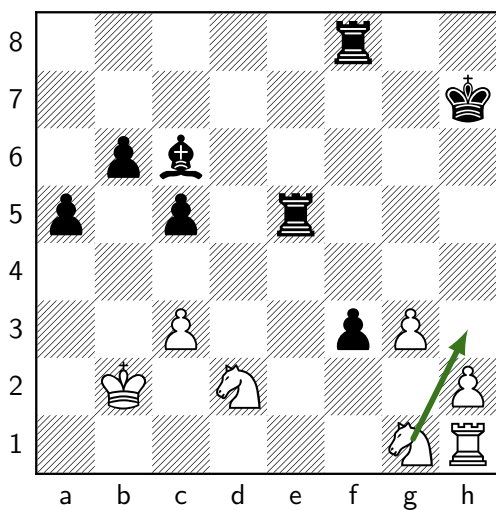


(c)

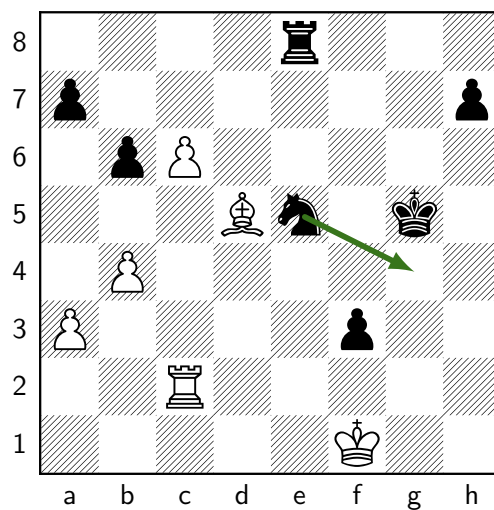


(d)

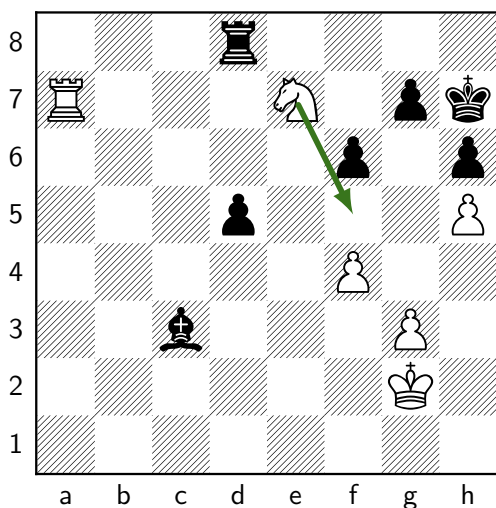
Figure 42: **Move 48, White: Ng1**. (a) Our position; Melanie plays Ng1 (h3g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0031#1409 (ply 93) [structural distance $d=32$, QAP obj=21]: the master played Rb7; under μ our Nh3 plays the role of that game’s Re2 (a **cross-type** correspondence); cross-type roles: our Ph2 as their rd8, our Rh1 as their Pf7; transferred move h3g1 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0013#2419 (ply 78) [structural distance $d=33$, QAP obj=14]: the master played Re2; under μ our Nh3 plays the role of that game’s Rg2 (a **cross-type** correspondence); cross-type roles: our Rh1 as their Pa2, our Nd2 as their be5; transferred move h3g1 (**endorses this move**). (d) Retrieved from lichess_elite_2025-11_p0000#1414 (ply 89) [structural distance $d=29$, QAP obj=8]: the master played Nf4+; under μ our Nh3 plays the role of that game’s Nh4; cross-type roles: our pa5 as their Re8, our Nd2 as their ra6; transferred move h3g1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ng1 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



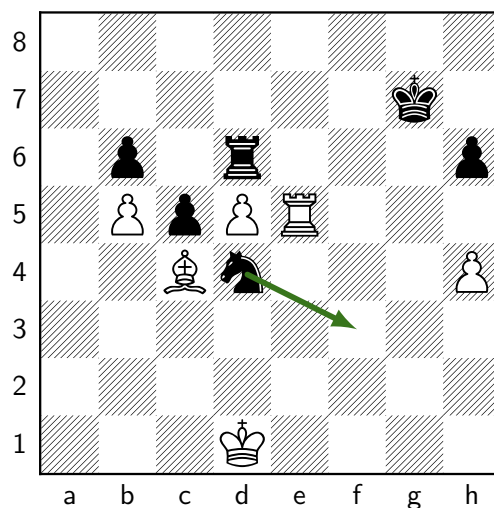
(a)



(b)

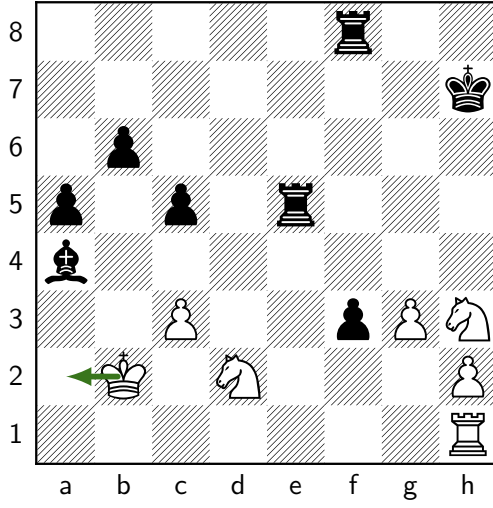


(c)

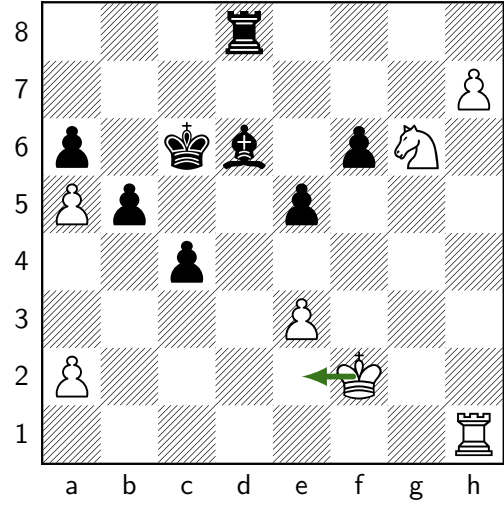


(d)

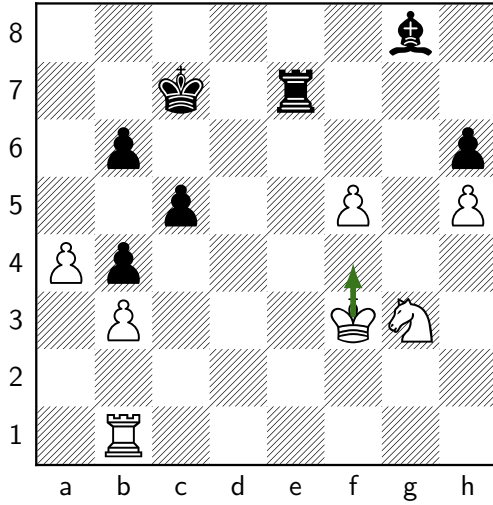
Figure 43: **Move 49, White: Nh3**. (a) Our position; Melanie plays Nh3 (g1h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2025-11_p0000#3371 (ply 85) [structural distance $d=37$, QAP obj=35]: the master played Ng4; under μ our Ng1 plays the role of that game’s Ne4; cross-type roles: our Nd2 as their bd5, our rf8 as their pb4; transferred move g1h3 (**endorses this move**). (c) Retrieved from lichess_elite_2025-11_p0000#835 (ply 96) [structural distance $d=36$, QAP obj=22]: the master played Nf5; under μ our Ng1 plays the role of that game’s Ne7; cross-type roles: our Nd2 as their Pf4; transferred move g1h3 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0018#2655 (ply 109) [structural distance $d=40$, QAP obj=17]: the master played Nf3; under μ our Ng1 plays the role of that game’s Nd5; cross-type roles: our Nd2 as their Rd6, our Rh1 as their Pc5; transferred move g1h3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Nh3 (summed QAP objective 100, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



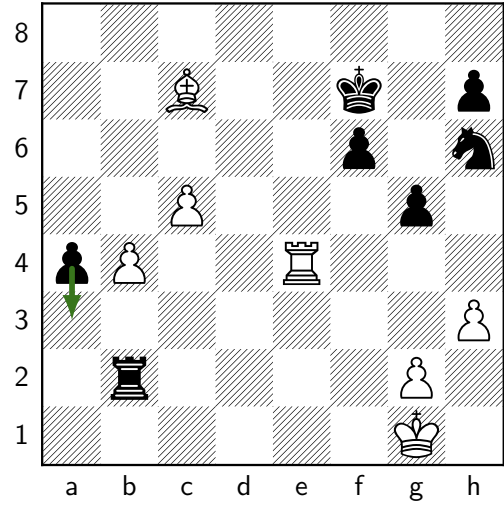
(a)



(b)

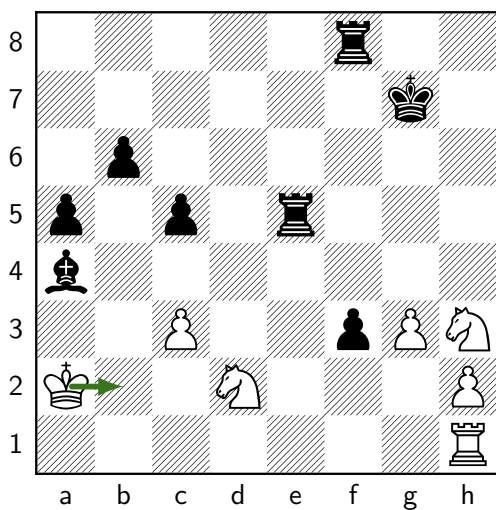


(c)

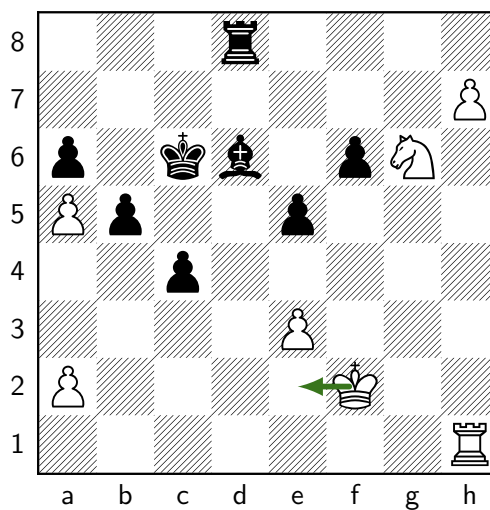


(d)

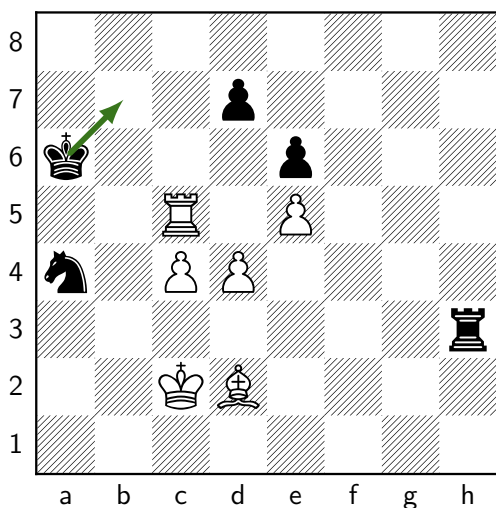
Figure 44: **Move 50, White: Ka2.** (a) Our position; Melanie plays Ka2 (b2a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#3426` (ply 88) [structural distance $d=33$, QAP obj=21]: the master played Ke2; under μ our Kb2 plays the role of that game’s Kf2; cross-type roles: our ba4 as their Pa2, our rf8 as their bd6; transferred move b2a2 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#2582` (ply 104) [structural distance $d=33$, QAP obj=11]: the master played Kf4; under μ our Kb2 plays the role of that game’s Kf3; cross-type roles: our pb6 as their Ng3, our Nh3 as their Pa4; transferred move b2a2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0010#1966` (ply 83) [structural distance $d=30$, QAP obj=22]: the master played a3; under μ our Rh1 plays the role of that game’s Pa5 (a **cross-type** correspondence); cross-type roles: our pc5 as their Nh6, our Nh3 as their pc5; transferred move h1f1. Synthesis: 2 retrieved analogs transferred to Ka2 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



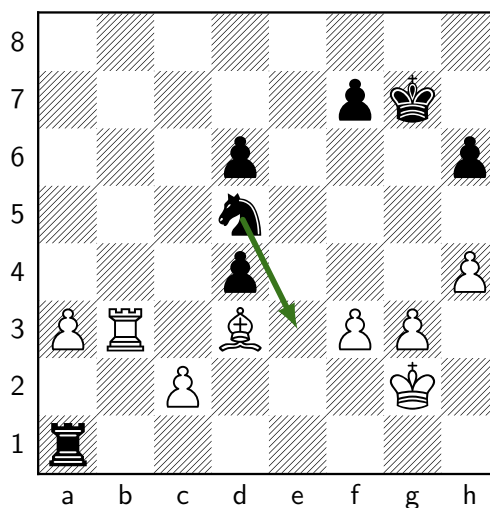
(a)



(b)

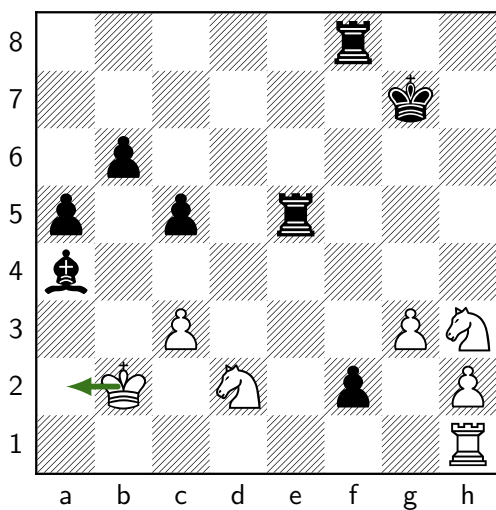


(c)

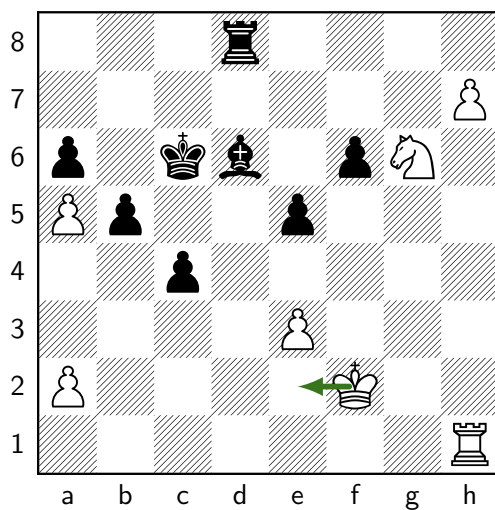


(d)

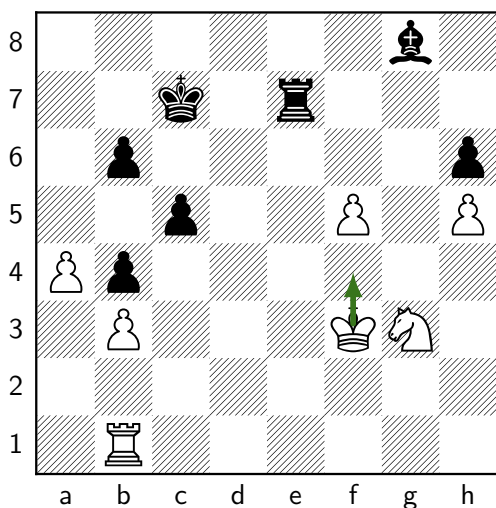
Figure 45: **Move 51, White: Kb2.** (a) Our position; Melanie plays Kb2 (a2b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#3426` (ply 88) [structural distance $d=31$, QAP obj=21]: the master played Ke2; under μ our Ka2 plays the role of that game’s Kf2; cross-type roles: our Nh3 as their Pe3, our rf8 as their bd6; transferred move a2b2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#2593` (ply 117) [structural distance $d=33$, QAP obj=4]: the master played Kb7; under μ our Ka2 plays the role of that game’s Ka3; cross-type roles: our Nd2 as their Pd7, our ba4 as their rc5; transferred move a2b2 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#261` (ply 97) [structural distance $d=36$, QAP obj=21]: the master played Ne3+; under μ our Nh3 plays the role of that game’s Nd4; cross-type roles: our Rh1 as their Pd4, our re5 as their bd3; transferred move h3f4. Synthesis: 3 retrieved analogs transferred to Kb2 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



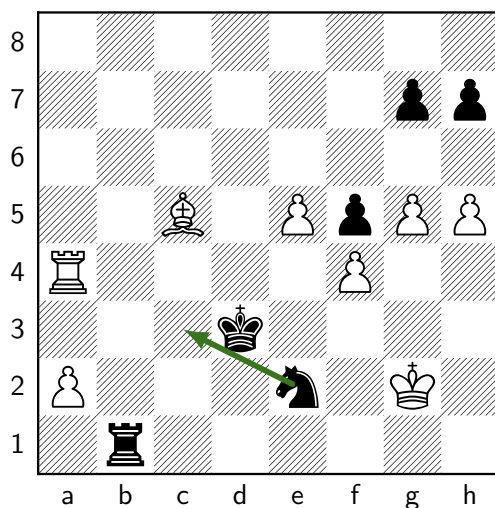
(a)



(b)

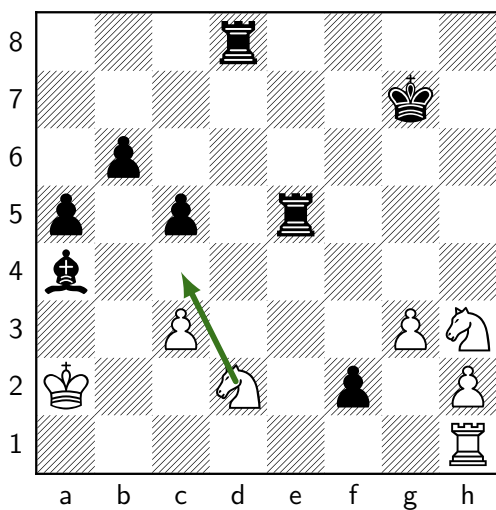


(c)

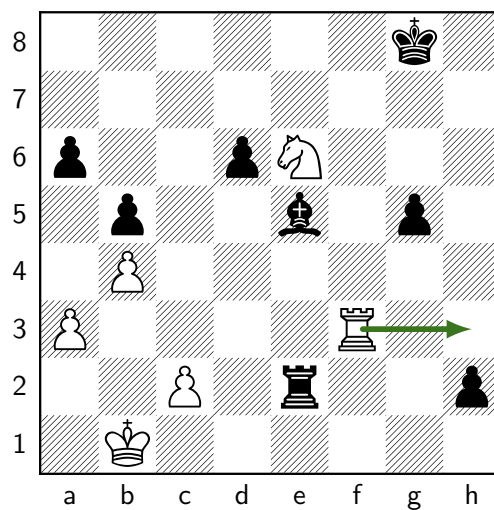


(d)

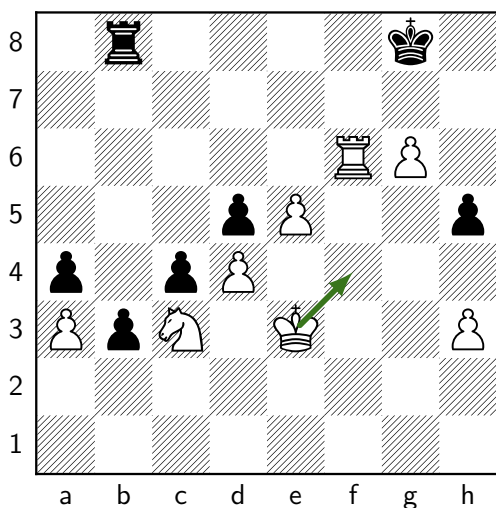
Figure 46: **Move 52, White: Ka2.** (a) Our position; Melanie plays Ka2 (b2a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#3426` (ply 88) [structural distance $d=31$, QAP obj=22]: the master played Kf2; under μ our Kb2 plays the role of that game’s Kf2; cross-type roles: our Nd2 as their Pe3, our ba4 as their Pa5; transferred move b2a2 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#2582` (ply 104) [structural distance $d=37$, QAP obj=10]: the master played Kf4; under μ our Kb2 plays the role of that game’s Kf3; cross-type roles: our pa5 as their Ng3, our Rh1 as their pb4; transferred move b2a2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0030#1309` (ply 103) [structural distance $d=33$, QAP obj=22]: the master played Nc3; under μ our Nh3 plays the role of that game’s Ne7; cross-type roles: our re5 as their bc5, our Nd2 as their pe5; transferred move h3f4. Synthesis: 3 retrieved analogs transferred to Ka2 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



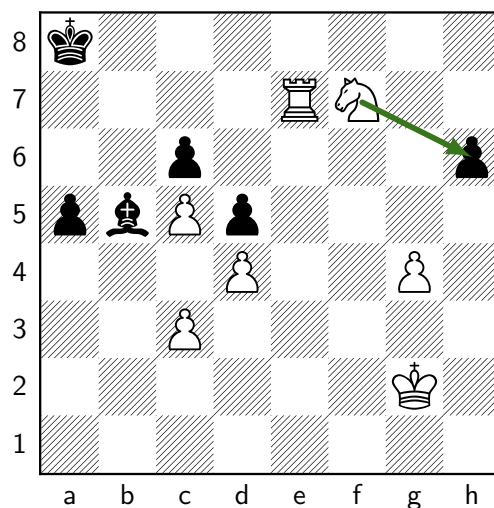
(a)



(b)

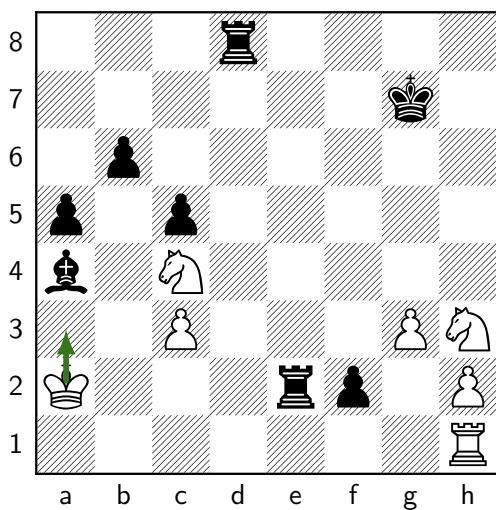


(c)

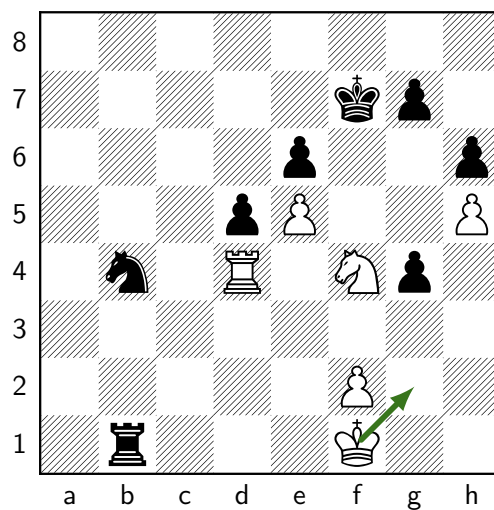


(d)

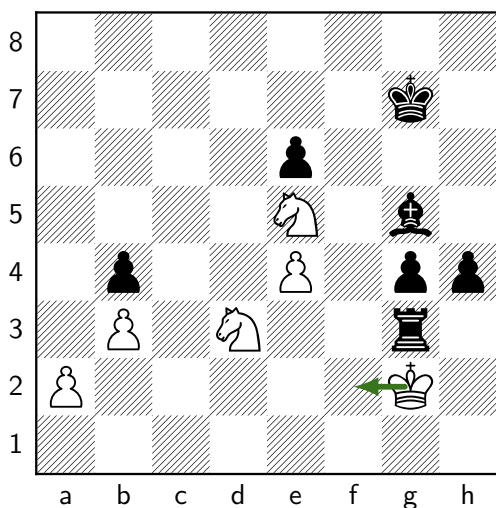
Figure 47: **Move 53, White: Nc4.** (a) Our position; Melanie plays Nc4 (d2c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#1640` (ply 96) [structural distance $d=34$, QAP obj=38]: the master played Rh3; under μ our Rh1 plays the role of that game’s Rf3; cross-type roles: our Nd2 as their Pc2, our ba4 as their ph2; transferred move h1c1. (c) Retrieved from `lichess_elite_2024-09_p0017#3791` (ply 94) [structural distance $d=35$, QAP obj=33]: the master played Kf4; under μ our Ka2 plays the role of that game’s Ke3; cross-type roles: our Rh1 as their Pa3, our Nd2 as their pb3; transferred move a2a1. (d) Retrieved from `lichess_elite_2024-09_p0012#3288` (ply 98) [structural distance $d=32$, QAP obj=21]: the master played Nxh6; under μ our Nh3 plays the role of that game’s Nf7; cross-type roles: our Rh1 as their pa5, our rd8 as their bb5; transferred move h3f2. Synthesis: no retrieved analog transferred to Nc4 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



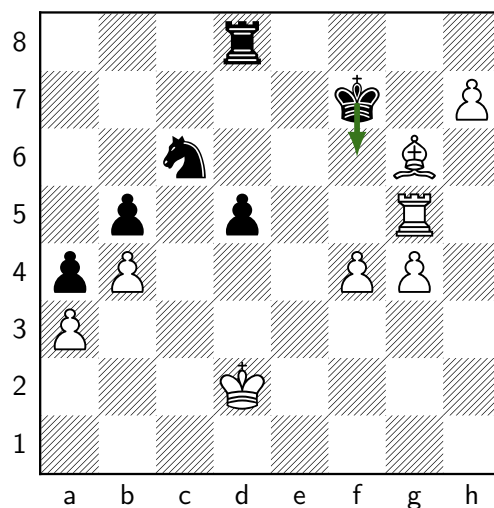
(a)



(b)

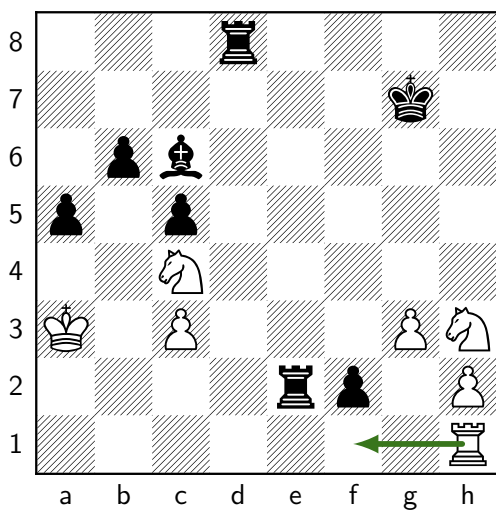


(c)

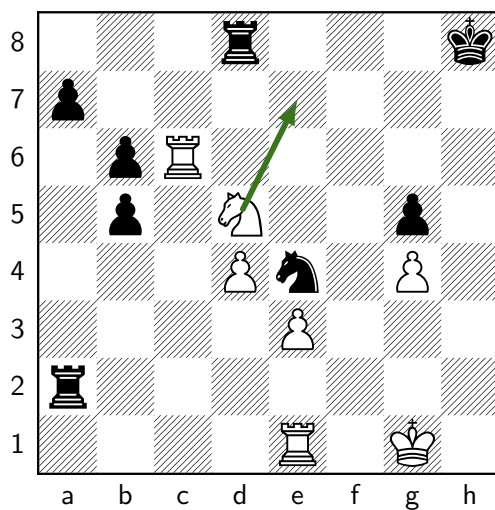


(d)

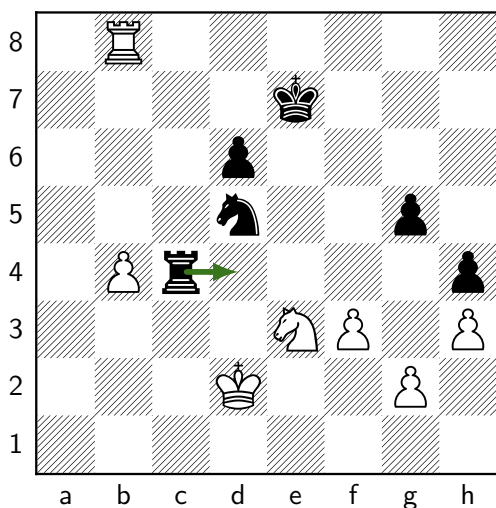
Figure 48: **Move 54, White: Ka3.** (a) Our position; Melanie plays Ka3 (a2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `twic1500#370` (ply 92) [structural distance $d=47$, QAP obj=71]: the master played Kg2; under μ our Ka2 plays the role of that game’s Kf1; cross-type roles: our pf2 as their nb4, our Nh3 as their Rd4; transferred move a2a3 (**endorses this move**). (c) Retrieved from `lichess_elite.2024-09_p0030#3175` (ply 108) [structural distance $d=47$, QAP obj=20]: the master played Kf2; under μ our Ka2 plays the role of that game’s Kg2; cross-type roles: our pf2 as their Nd3, our ba4 as their pg4; transferred move a2a3 (**endorses this move**). (d) Retrieved from `lichess_elite.2024-09_p0031#3142` (ply 121) [structural distance $d=41$, QAP obj=20]: the master played Kf6; under μ our Ka2 plays the role of that game’s Kf2; cross-type roles: our Pg3 as their Nc6, our re2 as their bg6; transferred move a2a3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ka3 (summed QAP objective 130, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



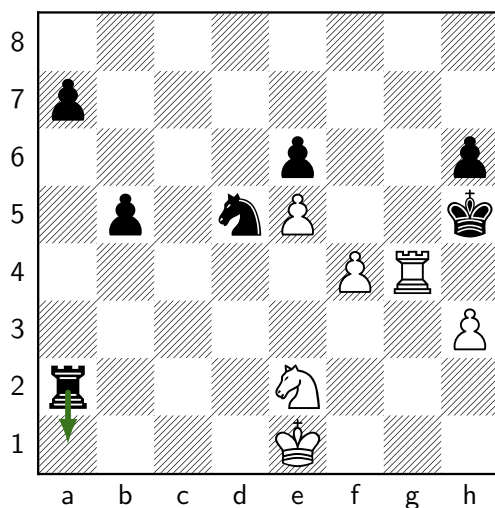
(a)



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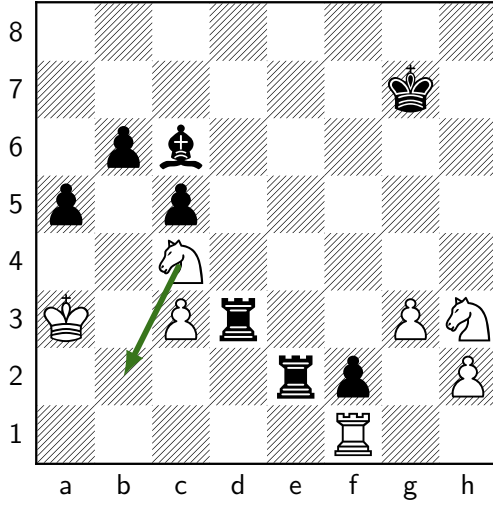


(c)

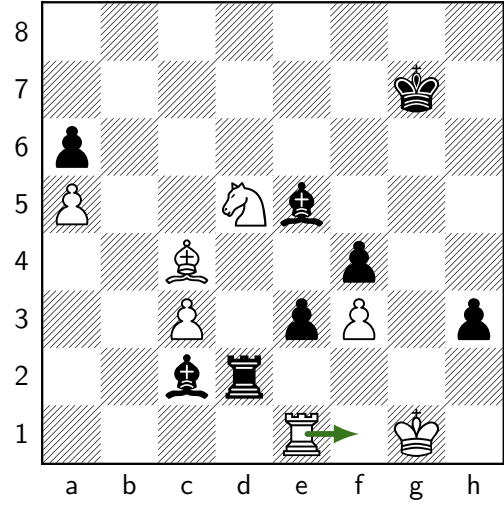


(d)

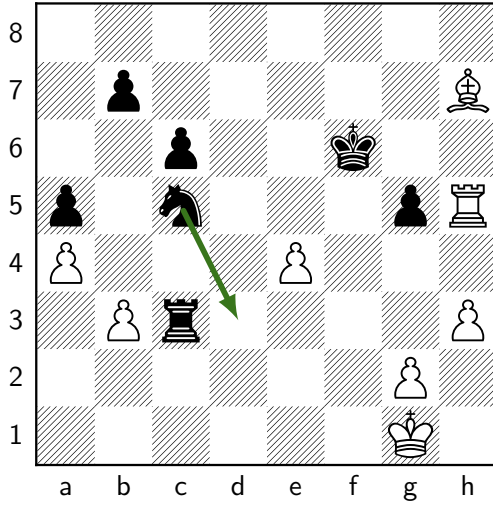
Figure 49: **Move 55, White: Rf1 (h1f1)**. (a) Our position; Melanie plays Rf1 (h1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#2692` (ply 98) [structural distance $d=44$, QAP obj=31]: the master played Ne7; under μ our *Rh1* plays the role of that game’s *Nd5* (a **cross-type** correspondence); cross-type roles: our *Nc4* as their *Re1*, our *Pc3* as their *ne4*; transferred move h1f1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#1753` (ply 97) [structural distance $d=44$, QAP obj=14]: the master played Rd4+; under μ our *Rh1* plays the role of that game’s *Rc5*; cross-type roles: our *pc5* as their *rb8*, our *Rh1* as their *Nd5*; transferred move h1f1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0031#313` (ply 95) [structural distance $d=52$, QAP obj=3]: the master played Ra1+; under μ our *Rh1* plays the role of that game’s *Ra7*; cross-type roles: our *Nc4* as their *Ph6*, our *Pc3* as their *ne2*; transferred move h1f1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rf1 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



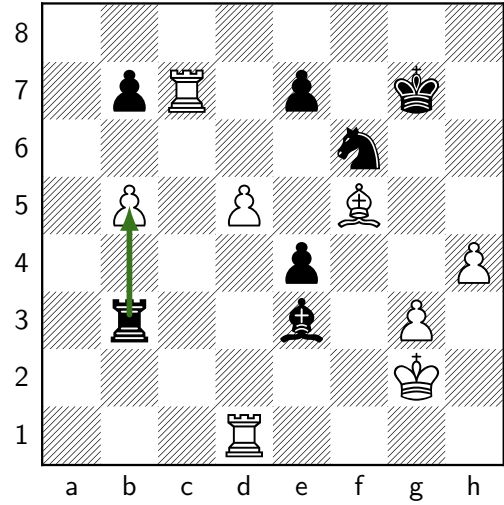
(a)



(b)

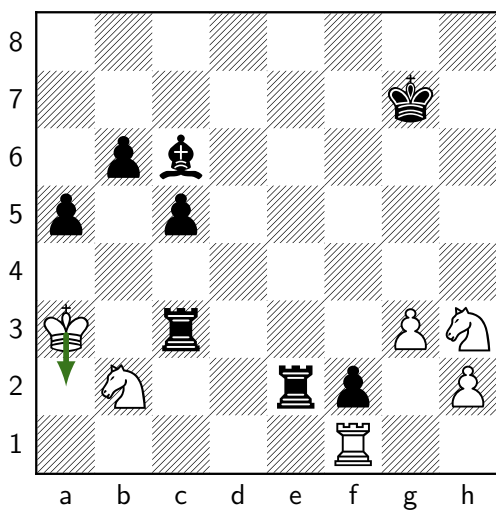


(c)

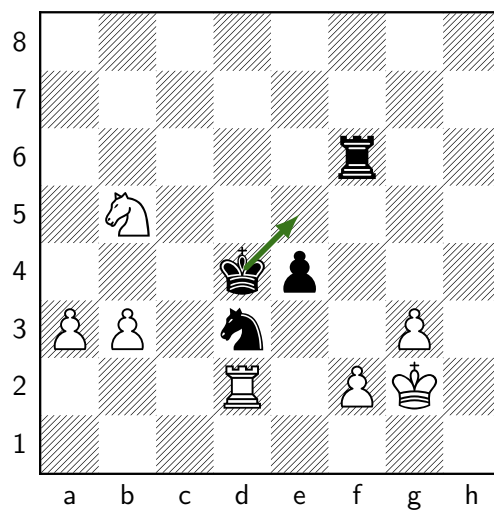


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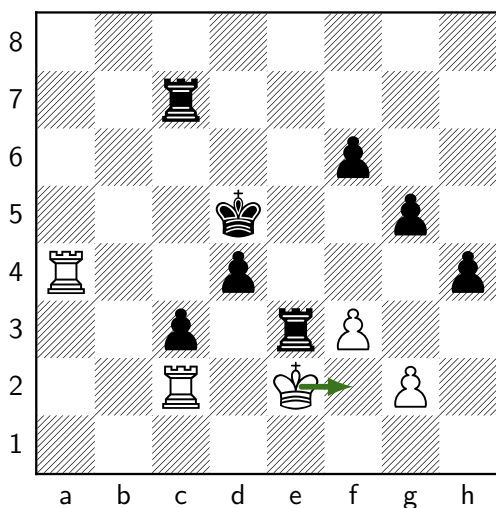
Figure 50: **Move 56, White: Nb2.** (a) Our position; Melanie plays Nb2 (c4b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#183` (ply 114) [structural distance $d=59$, QAP obj=53]: the master played Rf1; under μ our Nc4 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Pg3 as their bc2, our Nh3 as their Bc4; transferred move c4b2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#936` (ply 95) [structural distance $d=51$, QAP obj=89]: the master played Nd3; under μ our Nc4 plays the role of that game’s Nc4; cross-type roles: our Rf1 as their Pb7, our re2 as their bh7; transferred move c4d6. (d) Retrieved from `lichess_elite_2024-09_p0028#2919` (ply 99) [structural distance $d=54$, QAP obj=72]: the master played Rxb5; under μ our Nc4 plays the role of that game’s Rb6 (a **cross-type** correspondence); cross-type roles: our Rf1 as their bf5, our Nh3 as their Be3; transferred move c4b6. Synthesis: 1 retrieved analog transferred to Nb2 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



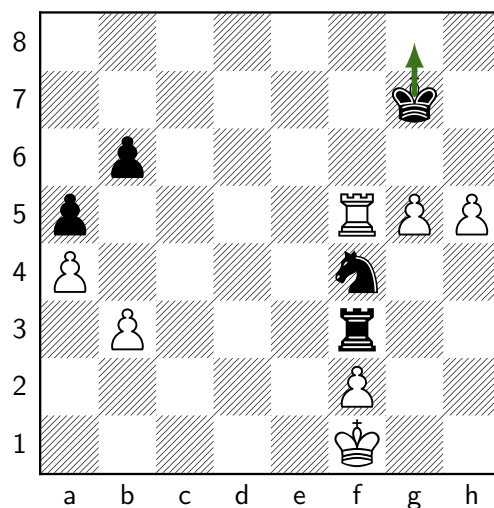
(a)



(b)



(c)



(d)

Figure 51: **Move 57, White: Ka2.** (a) Our position; Melanie plays Ka2 (a3a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#33337` (ply 99) [structural distance $d=47$, QAP obj=71]: the master played Ke5; under μ our Ka3 plays the role of that game’s Kd5; cross-type roles: our rc3 as their nb5, our Nb2 as their Pe4; transferred move a3a2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#2713` (ply 96) [structural distance $d=49$, QAP obj=37]: the master played Kf2; under μ our Ka3 plays the role of that game’s Ke2; cross-type roles: our pf2 as their Rc2, our Nb2 as their Pf3; transferred move a3a2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#1201` (ply 113) [structural distance $d=35$, QAP obj=5]: the master played Kg8; under μ our Ka3 plays the role of that game’s Kg2; cross-type roles: our Nb2 as their Pb6, our bc6 as their Pa5; transferred move a3a2 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Ka2 (summed QAP objective 123, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

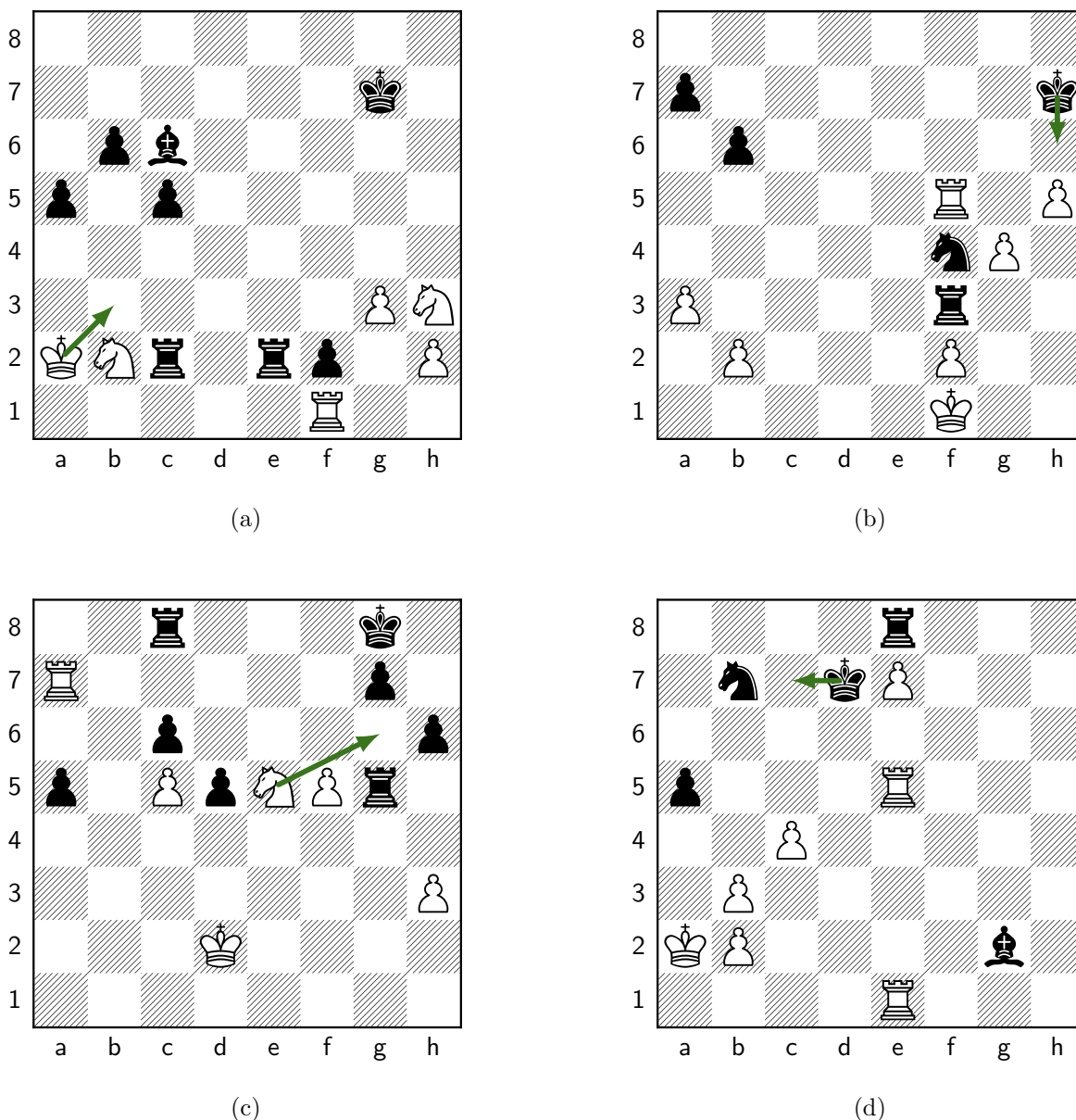


Figure 52: **Move 58, White: Kb3.** (a) Our position; Melanie plays Kb3 (a2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0022#1201 (ply 107) [structural distance $d=52$, QAP obj=47]: the master played Kh6; under μ our Ka2 plays the role of that game's Kh2; cross-type roles: our re2 as their pg4; transferred move a2b3 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0003#722 (ply 98) [structural distance $d=33$, QAP obj=15]: the master played Ng6; under μ our Nh3 plays the role of that game's Ne5; cross-type roles: our bc6 as their Ph3, our Rf1 as their pa5; transferred move h3g5. (d) Retrieved from lichess_elite_2024-09_p0019#2836 (ply 105) [structural distance $d=50$, QAP obj=14]: the master played Kc7; under μ our Ka2 plays the role of that game's Kd2; cross-type roles: our pa5 as their Nb7, our Nb2 as their pe7; transferred move a2a3. Synthesis: 1 retrieved analog transferred to Kb3 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

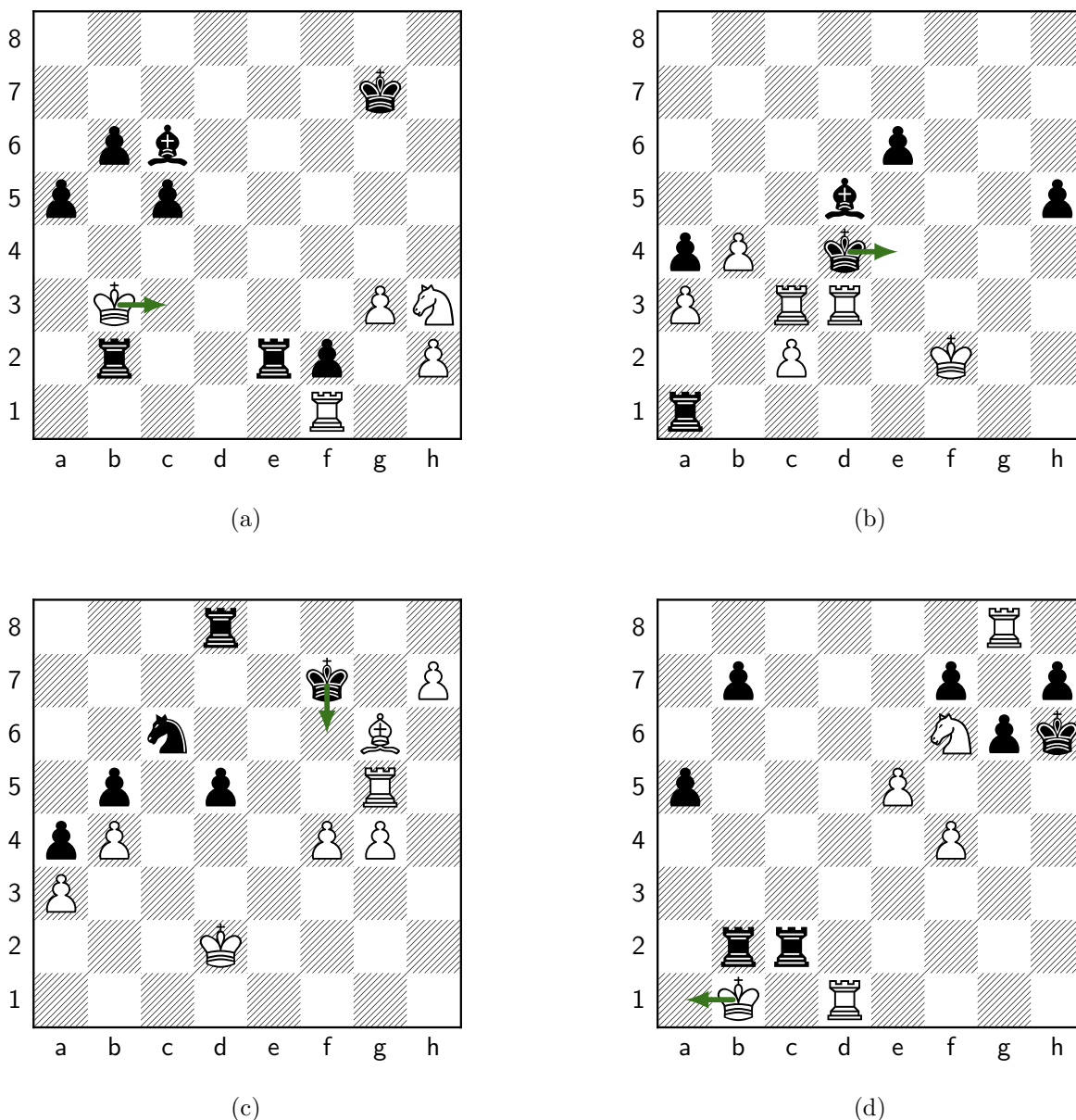


Figure 53: **Move 59, White: Kc3.** (a) Our position; Melanie plays Kc3 (b3c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0005#877 (ply 115) [structural distance $d=40$, QAP obj=71]: the master played Ke4; under μ our Kb3 plays the role of that game’s Kd5; cross-type roles: our Nh3 as their Pa4; transferred move b3c3 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0031#3142 (ply 121) [structural distance $d=38$, QAP obj=68]: the master played Kf6; under μ our Kb3 plays the role of that game’s Kf2; cross-type roles: our Nh3 as their Rd8, our Ph2 as their Nc6; transferred move b3c3 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0017#2447 (ply 110) [structural distance $d=37$, QAP obj=67]: the master played Ka1; under μ our Kb3 plays the role of that game’s Kb1; cross-type roles: our Rf1 as their Pf4, our pc5 as their Nf6; transferred move b3a3. Synthesis: 2 retrieved analogs transferred to Kc3 (summed QAP objective 139, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

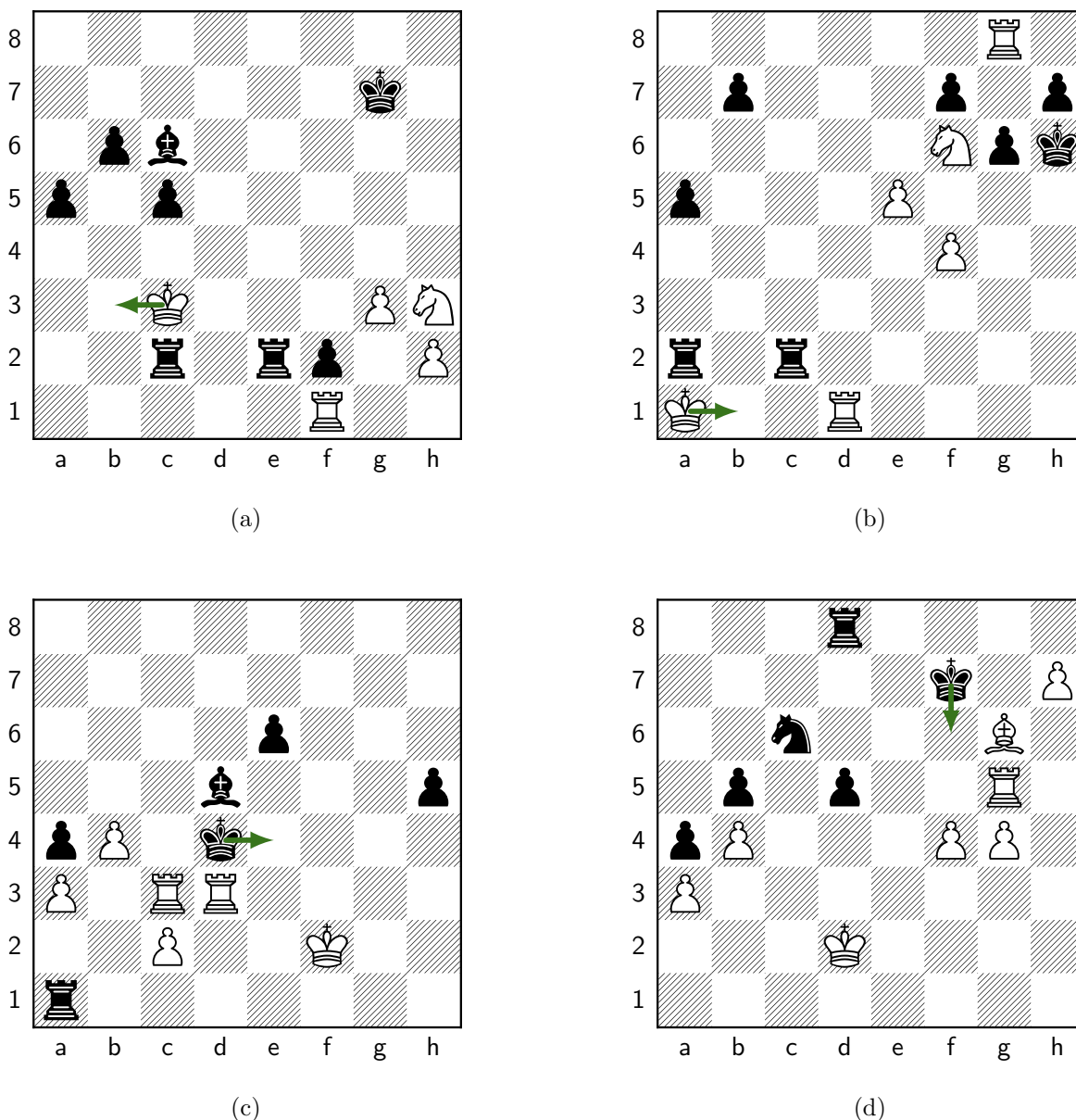


Figure 54: **Move 60, White: Kb3.** (a) Our position; Melanie plays Kb3 (c3b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#2447` (ply 112) [structural distance $d=34$, QAP obj=68]: the master played Kb1; under μ our *Kc3* plays the role of that game’s *Ka1*; cross-type roles: our *Rf1* as their *Pf4*, our *pc5* as their *Nf6*; transferred move c3b3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#877` (ply 115) [structural distance $d=40$, QAP obj=72]: the master played Ke4; under μ our *Kc3* plays the role of that game’s *Kd5*; cross-type roles: our *Pg3* as their *Bd5*, our *Nh3* as their *Pa4*; transferred move c3d3. (d) Retrieved from `lichess_elite_2024-09_p0031#3142` (ply 121) [structural distance $d=38$, QAP obj=69]: the master played Kf6; under μ our *Kc3* plays the role of that game’s *Kf2*; cross-type roles: our *Nh3* as their *Rd8*, our *Ph2* as their *Nc6*; transferred move c3d3. Synthesis: 2 retrieved analogs transferred to Kb3 (summed QAP objective 135, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

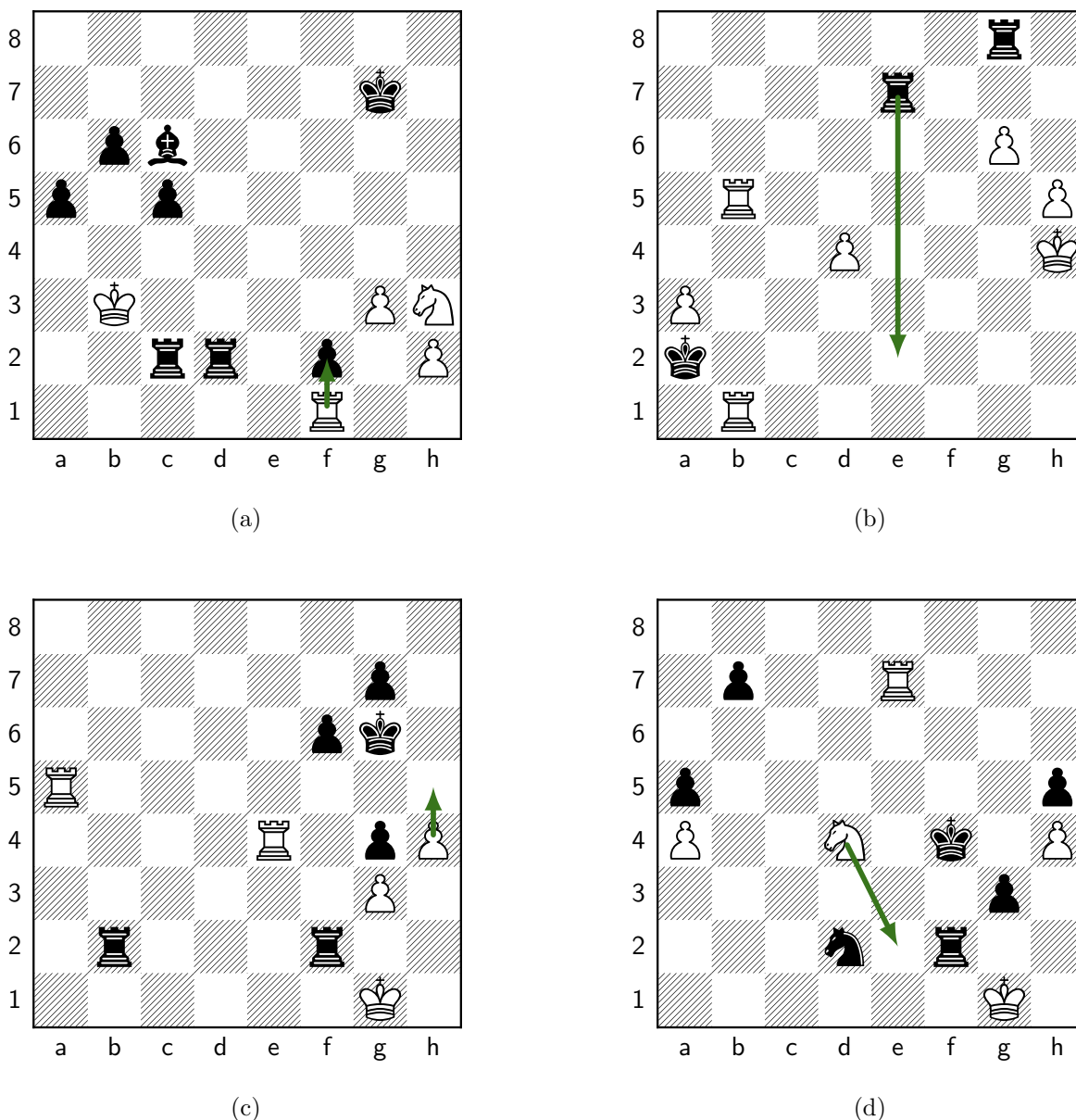
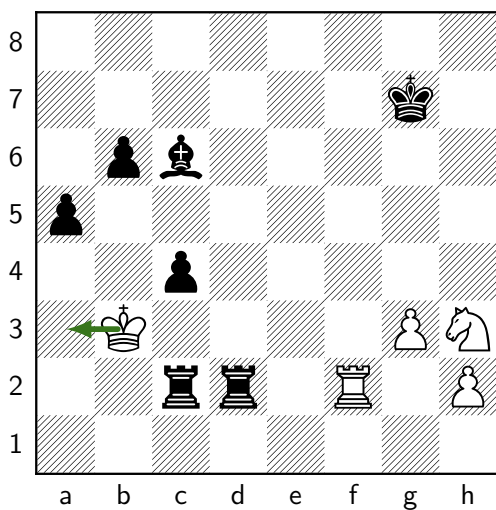
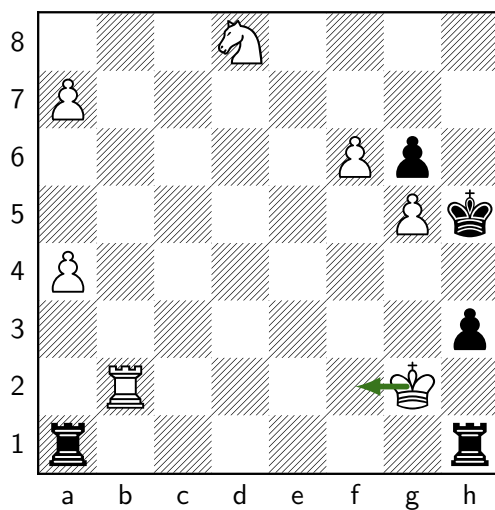


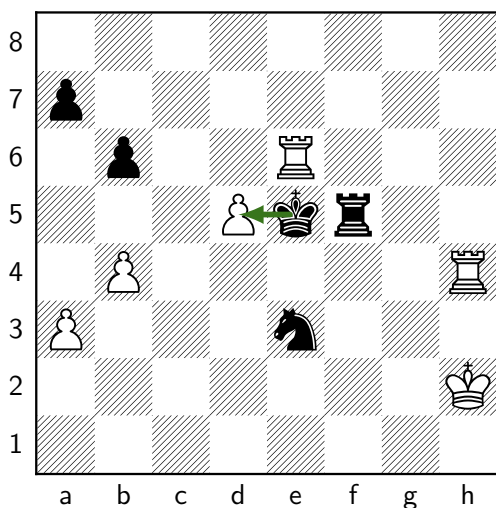
Figure 55: **Move 61, White: Rxf2.** (a) Our position; Melanie plays Rxf2 (f1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#3256` (ply 117) [structural distance $d=38$, QAP obj=23]: the master played Re2; under μ our Rf1 plays the role of that game’s Re2; cross-type roles: our Nh3 as their Rg8; transferred move f1f2 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#69` (ply 108) [structural distance $d=39$, QAP obj=21]: the master played h5+; under μ our Nh3 plays the role of that game’s Ph4 (a **cross-type** correspondence); cross-type roles: our Pg3 as their Re4; transferred move h3g1. (d) Retrieved from `lichess_elite_2024-09_p0023#1595` (ply 110) [structural distance $d=47$, QAP obj=20]: the master played Ne2+; under μ our Nh3 plays the role of that game’s Nd4; cross-type roles: our rd2 as their nd2; transferred move h3f2. Synthesis: 1 retrieved analog transferred to Rxf2 (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



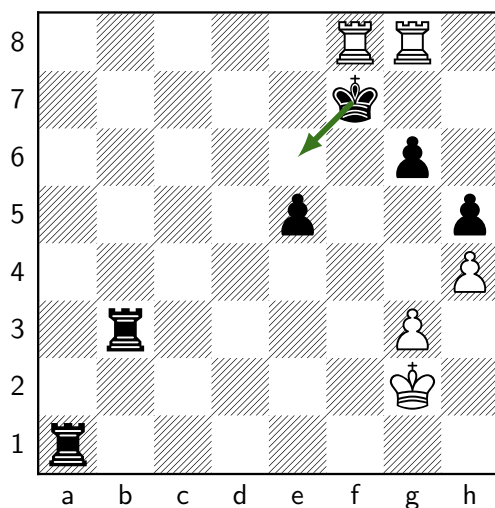
(a)



(b)



(c)



(d)

Figure 56: **Move 62, White: Ka3.** (a) Our position; Melanie plays Ka3 (b3a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#1046` (ply 116) [structural distance $d=32$, QAP obj=69]: the master played Kf2; under μ our Kb3 plays the role of that game’s Kg2; cross-type roles: our bc6 as their ra1, our Rf2 as their Pa4; transferred move b3a3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#1255` (ply 107) [structural distance $d=35$, QAP obj=68]: the master played Kxd5; under μ our Kb3 plays the role of that game’s Ke4; cross-type roles: our rd2 as their Pa7, our Rf2 as their Pb6; transferred move b3a3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0016#295` (ply 129) [structural distance $d=35$, QAP obj=68]: the master played Ke6; under μ our Kb3 plays the role of that game’s Kf2; cross-type roles: our pc4 as their rf8, our bc6 as their Pg6; transferred move b3a3 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Ka3 (summed QAP objective 677, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.